



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

COMMON SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

I/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

GROUP – A (CIVIL, CSE, IT) Branches

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1101	English *	4	3	1	--	4	30	70	100
B16 ENG 1102	Mathematics-I *	4	3	1	--	4	30	70	100
B16 ENG 1103	Mathematics-II *	4	3	1	--	4	30	70	100
B16 ENG 1104	Chemistry	4	3	1	--	4	30	70	100
B16 ENG 1106	Computer Programming Using C & Numerical Methods	4	3	1	--	4	30	70	100
B16 ENG 1108	History of Science and Technology	2	2	--	--	2	30	70	100
B16 ENG 1110	Chemistry Lab	2	--	--	3	3	50	50	100
B16 ENG 1112	Computer programming Using C & Numerical Methods Lab	2	--	--	3	3	50	50	100
B16 ENG 1114	Sports (Audit)	--	--	--	--	3	--	--	--
Total		26	17	5	6	31	280	520	800

Note: Add-On Course: Technical English *

*** Common to both Group - A and Group - B**

Code: B16 ENG 1101

ENGLISH
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Reading Skills

1. Addressing explicit and implicit meanings of a text on current topics.
2. Understanding the context.
3. Learning new words and phrases.
4. Using words and phrases in different contexts.

Writing Skills

1. Using the basic structure of a sentence.
2. Applying relevant writing formats to create paragraphs, essays, letters, emails, reports and presentations.
3. Retaining a logical flow while writing.
4. Planning and executing an assignment creatively.

Interactive Skills

1. Analyzing a topic of discussion and relating to it.
2. Participating in discussions and influencing them.
3. Communicating ideas effectively.
4. Presenting ideas coherently within a stipulated time.

Life Skills and Core Skills

1. Examining self-attributes and identifying areas that require improvement: self-diagnosis and self-motivation.
2. Adapting to a given situation and developing a functional approach to finding solutions: adaptability and problem solving.
3. Understanding the importance of helping others: community services and enthusiasm.

COURSE OUTCOMES:

1. The overall performance of the students will be enhanced after the course; they will be in a position to make presentations on topics of current interests – politics, famous personalities, science and technology, tourism, work and business environment, with increased public speaking skills.
2. Students will be able to read, listen, speak and write effectively in both academic and non-academic environment.
3. The students will be updated with certain real life situations, which they can handle when come face to face.

SYLLABUS

Listening Skills

Conversations: Life in a Hostel – Eating Away those Blues – Meeting Carl Jung – A Documentary on the Big Cat – A Consultant Interviewing Employees – A Conversation about a Business Idea.

Speaking Skills

Your Favorite Holiday Destination – Describe Yourself – Why we need to save our Tiger – A Dialogue – Your First Interview – Pair Work: Setting up a New Business.

Reading Skills

Reading Comprehension: Famous People – What is Personality, Personality based on Blood Groups – News Report, Magazine Article, Mobile Towers and Health – An Excerpt from a Short Story, An Excerpt from a Biography – Open Letter to Prime Minister, Business Dilemmas: An Email Exchange – A Review of IPL: The Inside Story, Mark Zuckerberg: World's Youngest Billionaire.

Writing Skills

Letter Writing, Essay Writing, Email Writing, Report Writing, Paragraph Writing, Drafting a Pamphlet, Argument Writing, Dialogue Writing.

Grammar

Types of Sentences, Articles, Prepositions, Gerunds and Infinitives, Conjunctions, Tense, Quantifiers, Punctuations, Correction of Sentences, Fill-in the Blanks.

Vocabulary

Synonyms, Antonyms, Idioms, One Word Substitution.

Life Skills and Core Skills

Self Awareness and Self Motivation – Communication, Adaptability – Motivation, Problem Solving – Personal Presentation Skills, Stress Management – Professionalism – Ethics.

TEXT BOOK:

1. Life Through Language: A Holistic Approach to Language Learning. Board of Editors, Pearson Publishers, India 2013.

REFERENCE BOOKS:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008
3. Know Your English (Volume 1 & 2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Maathukutty Monipally. Tata Mc Grahill P. 2009.

MATHEMATICS - I
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. How to find partial derivatives, Jacobians, how to change from one set of variables to another set of variables
2. Know Geometrical interpretation of partial derivatives
3. Learn applications to errors and approximations, maxima and minima
4. How to form ordinary differential equations (ODE) and solve equations of first order and first degree.
5. Applications of first order ODE to orthogonal trajectories, simple electrical circuits, Newton's law of cooling, Law of natural growth and decay
6. How to solve higher order linear differential equations with constant coefficients, Cauchy's linear equation, Legendre's linear equation and simultaneous first order ODEs
7. How to obtain Fourier Series for a periodic function of period 2π and Period $2C$ (C is a constant)
8. How to obtain Half range Cosine & Sine series and also Parseval's formula

COURSE OUTCOMES:

Students will be able to

1. Find partial derivatives, expand a function of more than one variable in a Taylor series and utilize them for errors and approximations, maxima and minima.
2. Solve a first order ODE and also find orthogonal trajectories and solve problems related to simple applications.
3. Solve a given higher order ODE, an equation with constant coefficients, a Cauchy's equation or a Legendre's equation.
4. Utilize knowledge of Fourier series for solving partial differential equations and also in understanding courses like Signals & Systems

SYLLABUS

Partial Differentiation

Functions of two or more variables – Partial derivatives – Homogeneous Functions – Euler's Theorem – Total Derivative – Change of Variables – Jacobians – Geometrical Interpretation: Tangent Plane and Normal to a Surface.

Applications of Partial Differentiation

Taylor's Theorem for functions of two variables – Errors and Approximations – Total Differential – Maxima and Minima of functions of two variables – Lagrange's Method of Undetermined Multipliers – Differentiation Under the Integral Sign – Liebnitz's Rules.

Ordinary Differential Equations of First Order and First Degree

Formation of ordinary differential equations (ODEs) – Solution of an ordinary differential equation – Equations of the First Order and First Degree – Linear Differential Equation – Bernoulli's Equation – Exact Differential Equations – Equations Reducible to exact equations.

Applications of Differential Equations of First Order

Orthogonal Trajectories – Simple electric (LR & CR) Circuits – Newton's Law of Cooling – Law of Natural growth and decay.

Linear Differential Equations of Higher Order

Solutions of Linear Ordinary Differential Equations With Constant Coefficients – Rules for finding Complimentary Function – Rules for finding the particular integral – Method of variation of parameters – Cauchy's linear equation – Legendre's Linear Equation – Simultaneous linear equations.

Fourier series

Introduction - Euler's Formulae - Conditions for a Fourier Expansion - Functions having points of discontinuity - Change of Interval - Odd and Even Functions - Expansions of Odd or Even Periodic Functions, Half-Range Series - Parseval's Formula.

TEXT BOOK:

1. Scope and Treatment as in "Higher Engineering Mathematics", by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

MATHEMATICS – II
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. The concept of rank, Normal form of a matrix, consistency of a system of linear algebraic equations
2. Eigen values and Eigen vectors of a matrix, the Cayley Hamilton theorem
3. Orthogonal reduction of a quadratic form.
4. Complex matrices and their properties
5. Laplace transform, existence, properties.
6. Inverse Laplace transform, convolution theorem, how to solve Ordinary Differential Equations (ODE) and simultaneous ODEs by the use of Laplace transforms.
7. Difference equations and their methods of solution.
8. Z-transform, its properties, important results and its use to solve difference equations.

COURSE OUTCOMES:

Students will be capable of

1. Utilizing the knowledge of matrices for solving linear simultaneous equations, find Eigen values and Eigen vectors and handle quadratic forms
2. Utilizing the knowledge of Laplace Transforms to find transforms of important functions that arise in applications and also solve ODE
3. Also utilizing the knowledge of Laplace Transforms in courses like Net Works, Signals & Systems and Control Systems
4. Utilizing the knowledge of difference equations and Z-transforms in understanding courses like Discrete Mathematical Structures and also Signals & Systems.

SYLLABUS

Matrices – I

Rank of a matrix - Normal Form – Solutions of Linear System of Equations- Consistency of Linear System of Equations – Rouche's Theorem(statement) - Direct Methods: Gauss Elimination Method, LU Factorization Method – Eigen Values and Eigen Vectors of a Matrix – Properties - Cayley – Hamilton Theorem – Inverse and Powers of a Matrix using Cayley – Hamilton Theorem.

Matrices – II

Diagonalization of a Matrix – Quadratic Forms – Reduction of Quadratic Form to Canonical Form – Nature of a Quadratic Form – Complex Matrices: Hermitian, Skew-Hermitian and Unitary Matrices and their Properties.

Laplace Transforms-I

Introduction – Existence Conditions – Transforms of Elementary Functions – Properties of Laplace Transforms – Transforms of Derivatives – Transforms of Integrals – Multiplication by 't' – Division by t – Evaluation of Integrals by Laplace Transforms – Laplace Transforms of Unit Step Function, Unit Impulse Function and Periodic Functions.

Laplace Transforms-II

Inverse Laplace Transform – different methods - Convolution Theorem – Applications of Laplace Transforms to Ordinary Differential Equations, Simultaneous Linear Differential Equations with Constant Coefficients.

Difference Equations

Definition - order and solution of a difference equation - Formation of difference equations - Linear difference equations - Rules for finding C.F. - Rules for finding P.I.- Simultaneous difference equations with constant coefficients. Application to deflection of a loaded string.

Z-transforms

Z-transforms – definition - some standard Z-transforms - Linear property, Damping rule - some standard results - shifting rules - initial and final value theorems - convolution theorem - Evaluation of inverse transforms - Applications of Z-transform to difference equation.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company

CHEMISTRY
(Common to CIVIL, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To develop full pledged laboratories for the analysis of engineering materials.
2. To develop full pledged lab for pollution, corrosion, polymers and electrochemical cells.
3. Extend the infrastructure facilities of the department for the project works of engineering students and also guide the project work under faculty supervision.
4. To develop full pledged lab for analysis of Water, Cement, Hydrogen Peroxide, pollutants and raw materials.

COURSE OUTCOMES:

1. Students learn in-depth about the topics of desalination of sea water, CNG, LPG Biogas, Semiconductors, Liquid crystals, Conducting polymers, fiber reinforced plastics, building materials.
2. Students understand the basic and advanced applied concepts.
3. Students learn to interrelate the theory and with the relevant experiment.
4. Students learn experimental techniques and understand the theory about experiments.

SYLLABUS**Water Chemistry**

Source of water- impurities- Hardness and its determination by EDTA method- Boiler troubles and their removal. Water softening methods- lime soda, zeolite and ion exchange. Municipal water treatment- Break point chlorination. Desalination of sea water – electro dialysis and reverse osmosis methods.

Building Materials

Portland cement: Manufacture-Chemistry involved in setting and hardening of cement – Cement concrete -RCC –Decay of concrete.
Refractories: Classification-Properties and Engineering applications.
Ceramics: Classification-Properties and uses.

Solid State Chemistry

Classification of solids-Types of crystals-properties-Imperfections in crystals. Band theory of solids. Chemistry of semi conductors –Intrinsic, Extrinsic, compound and defect. Organic semi conductors Purification of solids by Zone refining-Single crystal growth – epitaxial growth. Elementary ideas on Liquid crystals.

Corrosion Chemistry

Definition of corrosion- Types of corrosion-chemical & electrochemical corrosion –Pitting, stress corrosion, Galvanic corrosion, Water line corrosion Factors affecting corrosion – Prevention of corrosion- Cathodic protection. Corrosion inhibitors Protective coatings- Metallic coatings, electro plating, electroless plating, chemical conversion coatings- phosphate coatings, chromate coatings, anodizing. Organic coatings-Paints.

Polymers And Plastics

Definition-Types of polymerization – mechanism of free radical polymerization. plastics- Thermosetting and thermoplastic resins cellulose derivatives, vinyl resins, nylon 6,6, bakelite-Compounding of plastics-Fabrication of plastics. Fiber reinforced plastics – conducting polymers -Engineering applications of polymers.

Fuels And Lubricants

Solid fuels: Coal –Analysis of coal – Metallurgical coke- manufacture-Engineering applications.

Petroleum-refining-Knocking and Octane number of gasoline-Cetane number of diesel oil. Synthetic petrol – LPG-CNG–Applications. Rocket fuels – Propellants-classification.

LUBRICANTS: Principles of lubrications, classification of lubricants and properties of lubricants (any five)

TEXT BOOKS:

1. Engineering chemistry by Jain & Jain 15th edition Dhanpatrai publishing company.
2. Engineering chemistry by Dr. K.AnjiReddy and Dr. M.Sita rama reddy, silicon publications, 2015.

REFERENCE BOOKS:

1. A Text Book of Engineering Chemistry by S.S.Dara; S.Chand & Company Ltd.
2. Engineering Chemistry by B.Siva Sankar Mc.Graw Hill Education(India) Pvt.Ltd.
3. A Text Book of Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co.

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS
(Common to CIVIL, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To make the student familiar with programming in C and enable the student to implement the numerical methods described in this course using C as Programming Language.
2. To introduce the programming principles
3. To familiarize the variables construction, declarations, initializations etc.,
4. To introduce the Control statement
5. To introduce the arrays, structures, Unions, Files etc.,
6. To give the overall idea on Pointers
7. To give the idea about dynamic memory allocation
8. The nature of C language is emphasized in the wide variety of examples and applications.
9. To know about some popular programming languages.
10. To solve various numerical Methods
11. Student can write various programs by applying elementary concepts like arrays, structures, Unions, Files etc.,

COURSE OUTCOMES:

1. Student can understand basic terminology used in C programming.
2. Student can write programs by applying elementary algorithms to solve problems in C language.
3. Student can write, compile and debug programs in C language.
4. Student can Write programs to solve numerical methods
5. Student can be familiar with finite precision computation.

SYLLABUS

Introduction to C

Basic structure of C program, Constants, Variables and data types, Operators and Expressions, Arithmetic Precedence and associativity, Type Conversions. Managing Input and Output Operations, Formatted Input, Formatted Output.

Decision Making, Branching, Looping, Arrays & Strings: Decision making with if statement, Simple if statement, The if...else statement, Nesting of if...else statement, the else..if ladder, switch statement, the (?:) operator, the GOTO statement., The while statement, the do statement, The for statement, Jumps in Loops ,One, Two-dimensional Arrays, Character Arrays. Declaration and initialization of Strings, reading and writing of strings, String handling functions, Table of strings.

Functions

Definition of Functions, Return Values and their Types, Function Calls, Function Declaration, Category of Functions: No Arguments and no Return Values, Arguments but no Return Values, Arguments with Return Values, No Argument but Returns a Value, Functions that Return Multiple Values. Nesting of functions, recursion, passing arrays to functions, passing strings to functions, The scope, visibility and lifetime of variables. .

Pointers

Accessing the address of a variable, declaring pointer variables, initializing of pointer variables, accessing variables using pointers, chain of pointers, pointer expressions, pointers and arrays, pointers and character strings, array of pointers, pointers as function arguments, functions returning pointers, pointers to functions, pointers to structures-Program Applications

Structure and Unions

Defining a structure, declaring structure variables, accessing structure members, structure initialization, copying and comparing structure variables, arrays of structures, arrays within structures, structures within structures, structures and functions and unions, size of structures and bit-fields- Program applications.

File handling

Defining and opening a file, closing a file, Input/ Output operations on files, Error handling during I/O operations, random access to files and Command Line Arguments- Program Applications.

Numerical Methods

Solutions of Algebraic and Transcendental Equations: Bisection Method, Newton Raphson Method.

Interpolation - Newton's forward and backward Interpolation, Lagrange's Interpolation in unequal intervals.

Solutions of Linear Equations: Gauss Elimination Method, Jacobi and Gauss Seidel Methods. Numerical Integration: Trapezoidal rule, Simpson's 1/3 rule.

Solutions of Ordinary First Order Differential Equations: Euler's Method, Modified Euler's Method and Runge-Kutta Method.

TEXT BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt.Ltd.

REFERENCE BOOKS:

1. Let Us C, Yashwant Kanetkar, BPB Publications, 5th Edition.
2. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
3. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
4. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific

HISTORY OF SCIENCE AND TECHNOLOGY
(Common to CIVIL, CSE & IT)

Theory : 2 periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 2

COURSE OBJECTIVES:

1. To know the contributions of scientists for the development of society over a period of time.
2. To understand the Science and Technological developments that lead to human welfare.
3. To appreciate the Science and Technological contributions for the development of various sectors of the economy.
4. To identify the technological transfer versus economic progress of the countries.

COURSE OUTCOMES:

1. By the end of this course the students should be able to understand the contribution of Scientific and Technological developments for the benefit of society at large.

SYLLABUS**Historical Perspective of Science and Technology**

Nature and Definitions; Roots of Science – In Ancient Period and Modern Period (During the British Period); Science and Society; Role of Scientist in the Society.

Policies and Plans after Independence

Science and Technology Policy Resolutions; New Technology Fund; Technology Development (TIFAC); Programs aimed at Technological Self Reliance; Activities of Council of Scientific and Industrial Research.

Science and Technological Developments in Critical Areas

Space – The Indian Space Program: India's Geostationary Satellite Services – INSAT System And INSAT Services; **Defense Research and Technology** – Research Coordination, Research efforts and Development of technologies and Spin-off technologies for civilian use; **Nuclear Energy** – Effects of a nuclear explosion and India's safety measures.

Impact of Science and Technology in Major Areas

Ocean Development: Objectives of Ocean Development, Biological and Mineral resources, Marine Research and Capacity Building; **Biotechnology:** Meaning, Biotechnology techniques- Bioreactors, Cell fusion, Cell or Tissue Culture, DNA Fingerprinting, Cloning, Artificial Insemination and Embryo Transfer Technology and Stem Cell Technology; Application of Biotechnology – Medicine, Biocatalysts, Food Biotechnology, Fuel and Fodder and Development of Biosensors.

Technology Transfer and Development

Transfer of Technology – Types, Methods, Mechanisms, Process, Channels and Techniques; **Appropriate Technology** - Criteria and Selection of an Appropriate Technology; Barriers of Technological Change. .

TEXT BOOKS:

1. Kalpana Rajaram, Science and Technology in India, Published and Distributed by Spectrum Books (P) Ltd., New Delhi-58.
2. Srinivasan, M., Management of Science and Technology (Problems & Prospects), East – West Press (P) Ltd., New Delhi.

REFERENCE BOOK:

1. Dr. G.R. Kohli, History of Science and Technology and Environmental Movements in India, Surjeet Publications, New Delhi

CHEMISTRY LAB
(Common to CIVIL, CSE & IT)

Lab : 3 periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

LIST OF PRACTICAL EXPERIMENTS:

1. Estimation of Sodium hydroxide with HCl(Na_2CO_3 primary standard).
2. Estimation of Iron as Ferrous iron in an Ore Sample.
3. Estimation of Oxalic acid by a redox method
4. Estimation of Calcium in a sample of Portland cement.
5. Estimation of Volume strength of Hydrogen Peroxide.
6. Estimation of Mohr's salt by Potassium dichromate
7. Determination of Hardness of an underground water sample.
8. Estimation of Zinc by EDTA method.
9. Determination of Alkalinity of water sample.

DEMONSTRATION EXPERIMENTS:

10. Determination of Viscosity and Viscosity index of a lubricant.
11. Printed Circuit Board
12. Determination of dissolved oxygen in given water sample.
13. Potentiometric titrations.
14. P^{H} Determination by using P^{H} meter.

REFERENCE BOOKS:

1. Essentials Experimental Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co Pvt. Ltd.
2. Laboratory Manual on Engineering Chemistry by Dr. Sudha Rani, Dhanpatrai Publishing Company.
3. Engineering Chemistry Laboratory Manual by Dr. K.Anji Reddy, Tulip Publications.

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS LAB
(Common to CIVIL, CSE & IT)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

LIST OF PROGRAMMES:

1. Write a program to read x, y coordinates of 3 points and then calculate the area of a triangle formed by them and print the coordinates of the three points and the area of the triangle. What will be the output from your program if the three given points are in a straight line.
2. Write a program which generates 100 random numbers in the range of 1 to 100. Store them in an array and then print the array. Write 3 versions of the program using different loop constructs (eg. for, while and do-while).
3. Write a set of string manipulation functions eg. for getting a sub-string from a given position, copying one string to another, reversing a string and adding one string to another.
4. Write a program which determines the largest and the smallest number that can be stored in different data types like short, int, long, float and double. What happens when you add 1 to the largest possible integer number that can be stored?
5. Write a program which generates 100 random real numbers in the range of 10.0 to 20.0 and sort them in descending order.
6. Write a function for transporting a square matrix in place (in place means that you are not allowed to have full temporary matrix).
7. First use an editor to create a file with some integer numbers. Now write a program, which reads these numbers and determines their mean and standard deviation.
8. Implement bisection method to find the square root of a given number to a given accuracy.
9. Implement Newton Raphson Method to determine a root of polynomial equation.
10. Given a table of x and corresponding f(x) values, write a program which will determine f(x) value at an intermediate x value using Lagrange's Interpolation.
11. Write a function which will invert a matrix.
12. Implement Simpson's $1/3^{\text{rd}}$ rule for numerical integration.
13. Implement Trapezoidal rule for numerical integration.
14. Write a program to solve a set of linear algebraic equations.
15. Write a program to solve a differential equation using Runge-Kutta Method.

REFERENCE BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt.Ltd.

SPORTS
(Common to CIVIL, CSE & IT)

Contact Hrs. 3 Hrs.

This is an Audit Course. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD-ON COURSE
TECHNICAL ENGLISH
(Common to All Branches)

Total Hrs: 30

COURSE OBJECTIVES:

1. To improve the language proficiency of the students regarding the technical aspects.
2. To enhance the awareness about the formal speaking and writing skills in different contexts.
3. Motivating the students to improve their personal skills.

COURSE OUTCOMES:

1. Students improve their language skills in formal/ technical contexts.
2. They enhance their understanding of technical terms.
3. They improve their personal skills.

SYLLABUS

1. Defining Technical Writing.
2. Writing Skills
Business letters, E-mails
Report Writing
Writing Proposals & Business Quotations
Brochures, Newsletters
How to write an abstract & research paper
3. Speaking Skills
Technical Descriptions
PPT Presentations
Paper Presentations
4. Vocabulary
Technical Terms
5. Behavioral / Attitudinal Skills

TEXT BOOK:

1. Effective Technical Communication by Ashraf Rizvi: Tata McGraw Hill.

REFERENCE BOOKS:

1. Handbook for Technical Writing by David A. McMurrey and Joanne Buckley: Cengage Learning.
2. Technical Writing: Process and Product by Sharon J. Gerson and Steven M. Gerson: Pearson Publications

COMMON SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

I/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

GROUP – A (CIVIL, CSE & IT) Branches

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1201	Mathematics-III *	4	3	1	--	4	30	70	100
B16 ENG 1202	Physics	4	3	1	--	4	30	70	100
B16 ENG 1204	Engineering Graphics	4	2	--	3	5	30	70	100
B16 ENG 1206	Professional Ethics & Moral values	2	2	--	--	2	30	70	100
DS	Department Subject #	4	3	1	--	4	30	70	100
B16 ENG 1209	Physics Lab	2	--	--	3	3	50	50	100
B16 ENG 1211	Workshop	2	--	--	3	3	50	50	100
B16 ENG 1213	English Language Lab *	2	--	--	3	3	50	50	100
B16 ENG 1214	NCC/NSS (Audit)	--	--	--	--	3	--	--	--
Total		24	13	3	12	31	300	500	800

Note: Add-On Course: Technology Course-I *

*** Common to both Group-A and Group-B**

Civil: B16CE1208: Building Materials and Building Construction

CSE: B16CS1208: Probability, Statistics & Queueing Theory

IT: B16CS1208: Probability, Statistics & Queueing Theory

Code: B16 ENG 1201

MATHEMATICS – III
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. Equations of a line, shortest distance between skew lines
2. About sphere, cone and cylinder
3. Double integrals, change of order of integration
4. Beta function, Gamma function, error function
5. Applications of double integrals to Areas, Volumes, mass, centre of gravity, Moment of inertia
6. The concept of Fourier Integral, Fourier Transform, properties.
7. Parseval's formulae

COURSE OUTCOMES:

Students will be able to

1. Utilize knowledge of line, sphere etc. in his engineering subjects
2. Utilize the knowledge of Beta and Gamma functions and multiple integrals to evaluate the integrals they come across in their applications
3. Utilize the knowledge of Fourier Transform in courses like Signals and Systems and in the solution of partial differential equations at a later stage

SYLLABUS

Solid Geometry

Equations of a plane, Normal form, Intercept form, Equations of Straight Line – Conditions for a line to lie in a Plane – Coplanar lines – Shortest distance between two skew lines - Intersection of three Planes – Equations of Sphere – Tangent Plane to a Sphere –Cone – Cylinder.

Multiple Integrals-1

Double Integrals - Change of Order of Integration - Double Integrals in Polar Coordinates - Triple Integrals - Change of Variables.

Multiple Integrals-2

Beta Function - Gamma Function - Relation between Beta and Gamma Functions-Error Function - Area enclosed by plane curves - Volumes of solids - Area of a curved surface - Calculation of mass - Center of gravity of a plane lamina- Moment of inertia.

Fourier Transforms

Introduction – definition - Fourier integral - Sine and Cosine integrals - Complex form of Fourier integral - Fourier transform - Fourier Sine and Cosine transforms -Finite Fourier Sine and Cosine transforms - properties of Fourier transforms, Convolution theorem for Fourier transforms - Parseval's identity for Fourier transforms - Fourier transforms of derivatives of a function.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

PHYSICS
(Common to CIVIL, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To understand the basic and advanced concepts of classical and modern physics – Identify, analyze, formulate and solve a wide variety of real world problems.
2. To develop scientific reasoning and problem solving methods.
3. To bring awareness of latest technologies (physics based) – lasers and fiber optics.

COURSE OUTCOMES:

1. Students learn in depth about the topics of Lasers, fiber optics, quantum mechanical Theory and classical theories of thermodynamics and electromagnetism.
2. Students understand the classical and modern concepts.

SYLLABUS**Thermodynamics**

Introduction, Heat and Work, First Law of Thermodynamics and applications, Reversible and Irreversible Process, Carnot Cycle and Efficiency, Second Law of Thermodynamics, Carnot's Theorem, Entropy, Second Law in terms of entropy, Entropy and disorder, Third Law of Thermodynamics (Statement Only).

Electromagnetism

Effect of Magnetic Field on – moving charges, current in long straight wire and rectangular Current Loop, Hall Effect, Biot-Savart's Law, B near a Long Wire, B for a Circular Current Loop, Ampere's Law, B for a Solenoid, Faraday's Law of electromagnetic induction, Lenz's law, Inductance of a solenoid, L-R Circuit, Displacement Current, Maxwell's Equations (integral form) and their significance (without derivation).

Interference

Principle of Super Position – Young's Experiment – Coherence – Inference in thin transparent films, Newton's Rings, Michelson Interferometer and its applications.

Lasers

Introduction, spontaneous and stimulated emissions, requirements of laser device, Ruby Laser, Gas Laser (He-Ne Laser), Semiconductor diode Laser, Characteristics and applications of Lasers.

Optical Fibers

Introduction, Principles of light propagation in optical fiber, Acceptance angle, Numerical aperture, types of fiber, Applications of Optical Fibers, Optical Fiber in Communications, advantages.

Ultrasonics

Definition, Production of Ultrasonics by Magnetostriction and Piezoelectric methods, detection methods, acoustic grating, characteristics and applications of Ultrasonics.

Modern Physics

Introduction, de Broglie concept of matter waves, Properties of matter waves, experimental verification (Davisson-Germer experiment), Heisenberg uncertainty principle, Wave function and its physical significance, Schrodinger time independent wave equation, application to a particle in a box, Band theory of Solids, Kronig - Penney model (qualitative treatment), Origin of energy band formation in solids, Classification of materials into conductors, semi conductors and insulators .

Nanophase Materials

Definition, Synthesis – Synthesis methods, Condensation and Ball milling, Chemical vapour deposition method – sol-gel methods, Characterisation, analysis and applications of nano materials.

TEXT BOOKS:

1. Physics by David Halliday and Robert Resnick – Part I and Part II, Wiley Eastern India (pvt.) Ltd.
2. Engineering Physics by M.N. Avadhanulu & P.G. Kshirasagar; S. Chand & Company Ltd

REFERENCE BOOKS:

1. Modern Engineering Physics by A.S. Vasudeva, S. Chand & Company Ltd
2. University Physics by Young and Freedman, Addison-Wesley
3. Engineering Physics by R.K. Gaur and S.L. Gupta, Dhanpat Rai & CO

ENGINEERING GRAPHICS
(Common to CIVIL, CSE & IT)

Theory	: 2 Periods	Sessionals	: 30
Lab	: 3 Periods	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

COURSE OUTCOMES:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent sectional views of solids.
5. Develop the surfaces of regular solids.
6. Draw the isometric views of solids and combination of solids.

SYLLABUS

Introduction

Lines, Lettering and Dimensioning. Geometrical Constructions.

Curves

Conic sections: General construction of ellipse, parabola and hyperbola. Construction of involutes. Normal and Tangent.

Projections of Points

Principal or Reference Planes, Projections of a point situated in any one of the four quadrants

Projections of Straight Lines

Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of straight line inclined to both the reference planes:

Projections of Planes

Projection of Perpendicular planes: Perpendicular to both reference planes, perpendicular to one reference plane and parallel to other reference plane, perpendicular to one reference plane and inclined to other reference plane. Projection of Oblique planes. Introduction to Auxiliary Planes.

Projections of Solids

Types of solids: Polyhedra and Solids of revolution. Projection of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to other and axes inclined to both the reference planes.

Projections of Section of Solids

Section Planes: Parallel and inclined section planes, Sections and True shape of section, Sections of Solids: Prism, Pyramid, Cylinder and Cone .

Development of Surfaces

Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

Isometric Views

Introduction to Isometric projection, Isometric scale and Isometric view. Isometric views of simple planes. Isometric view of Prisms, Pyramids, cylinder and cone. Isometric view of a combination of solids.

TEXT BOOK:

1. Elementary Engineering Drawing by N.D.Bhatt, Charotar Publishing House.

REFERENCE BOOK:

1. Engineering Graphics by K.L. Narayana and P. Kannaiah, Tata Mc-Graw Hill.

PROFESSIONAL ETHICS AND MORAL VALUES
(Common to CIVIL, CSE & IT)

Theory	: 2 Periods	Sessionals	: 30
Exam	: 3 Hrs.	Ext. Marks	: 70
		Credits	: 2

COURSE OBJECTIVES:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

COURSE OUTCOME:

1. By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS**Ethics and Human Values**

Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

Engineering Ethics

Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics, Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

Engineering as Social Experimentation

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders, Role of engineers in promoting ethical climate, balanced outlook on law.

Safety Social Responsibility and Rights

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

Global Issues

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

TEXT BOOKS:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

REFERENCE BOOK:

1. Charles D, Fleddermann, “Engineering Ethics”, Pearson / PHI, New Jersey 2004 (Indian Reprint)

BUILDING MATERIALS AND BUILDING CONSTRUCTION
(For Civil Engineering)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To teach students about the Physical and Mechanical properties of construction materials and their respective testing procedure.
2. To teach students about the building materials available in market to be used for many components of building industry.
3. To teach students about the principles and methods to be followed in constructing various components of a building.
4. To teach students about the deterioration and repair of buildings.

COURSE OUTCOMES:

1. Learn and identify the relevant physical and mechanical properties pertaining to the construction industry.
2. Demonstrate the relevant BIS testing procedure to be carried out to ascertain the quality of building materials.
3. Develop ability to choose the modern construction material appropriate to the climate and functional aspects of the buildings.
4. Ability to supervise the construction technique to be followed in brick, stone and hollow block masonry, concreting, flooring, roofing, plastering and painting etc.
5. Learn about the causes of deterioration, crack pattern, and assessment of damages.
6. Learn about the construction techniques in repairing of buildings.

SYLLABUS**Bricks and Clay Products**

Bricks: Classification of Bricks, Manufacture of Bricks, general qualities of Bricks as per IS code, tests for good bricks as per IS code, including field tests, special forms of Bricks and their uses. Clay Products: Various types of tile manufacturing and their uses, Earth-ware, Terra-cotta, stone ware, porcelain, glazing of tiles etc.

Wood, Wood Based Products

Wood: cross section details of trees, their general properties, various types of defects in wood and timber, Methods of seasoning and their importance, various Mechanical Properties of timber, Decay of timber, Wood based Products: Veneers, Plywood and its types, Manufacturing of Plywood, plywood grades as per IS code, Laminated wood, merits of plywood and laminated wood, Lamin Boards, Block Boards, Batten board, Hard board, Particle boards and Composite boards.

Paints, Varnishes

Paints and Varnishes: Constituents and characteristics of paints, types of paint, their uses and preparation on different surfaces, painting defects, causes and remedies. Constituents of varnishes, uses of varnishes, different kinds of varnish, polishes. Painting of interior walls, exterior walls, wooden doors and windows – steel windows – various types of paints (chemistry of paints not included) including distempers; emulsion paints etc., Varnishes wood work finishing types. **Plastic:** definition, classification of plastic and modern

development of plastic, chemistry of plastic, polymerization, manufacturing process and their uses. **Glass:** properties of glass, types of glass and manufacturing. **Asbestos & Asphalt Bitumen & Tar :** Availability and uses of asbestos, properties of asbestos, various types of asbestos, difference between asphalt & bitumen, Types, uses and properties of Asphalt & Bitumen, composition of coal tar, wood tar, mineral tar and Naphtha.

Cements

Natural and artificial cements, types of various artificial cements and their uses. Wet and dry process of manufacturing ordinary Portland cement (OPC), Chemical and Physical analysis of OPC, various field and Laboratory Tests on OPC as per IS code. Storing of cement in the field and godowns

Foundations

Types of Foundations : Strip, Isolated, Strap, Combined Footings, Raft – Mat –Slab and Beam Raft, Box Type Raft, inverted arch foundations, SHELL foundations, Grillage foundations – Minimum depth of Foundation – Bearing capacity of soils **Masonry:** Different types of Stone Masonry – Plan, Elevation, Sections of Stone Masonry Works – Brick Masonry – Different Types of Bonds – Plan, elevation and Section of Brick Bonds upto Two-Brick wall thickness – Partition walls – Different types of Block Masonry – Hollow concrete Blocks – FAL- G Blocks, Hollow Clay Blocks.

Roofing

Mangalore tiled Roof, RCC roof, Madras Terrace, Hollow Tiled Roof, Asbestos Cement, Fibre glass, Aluminium G.I. Sheet roofing's. **Form Work, Scaffolding:** form work, Types of formwork, centering-scaffolding-types of scaffolding. **Trusses:** King Post & Queen Post Trusses – Steel roof Truss for 12m Span with details. **Wooden Doors And Windows:** Parallel – Glazed – Flush shutters, Plywood, Particle Board Shutters – Aluminum, PVC, Steel doors, windows and ventilators, various types of windows, Glazing – different varieties. **Stair Cases:** Stair cases or Stairway design (Architectural design or planning only) various types such as, straight flight – dog legged, quarter landing, open spiral, spiral stairs etc.

Concrete Technology and Mix Design

Polymer Concrete, Types of cement concretes, Ingredients and their characteristics, Cement concrete properties and relevant tests, storage, batching, mixing & transporting, placing & vibrating and curing. Concrete grades & mix designs upto M 20 as per IS code. Introduction to polymer concrete and it's uses.

TEXT BOOKS:

1. "Engineering Materials" [Material Science], by Rangwala 'Charotar Publications'
2. "Building Materials" by S.K. Duggal New Age International Publishers.
3. "Building Construction" by B.C. Punmia. Laxmi Publications.
4. "Concrete Technology Theory & Practice" by M.S.Shetty, 'S.Chand & Company Ltd.'
5. "Engineering Materials", 5th edition, By Surindra Singh,, Konark Publishers Pvt. Ltd., New Delhi.

REFERENCE BOOKS:

1. "Civil Engg. Materials", by Technical Teachers' Training Institute, Chandigarh, Tata-McGraw-Hill Publishing Company Ltd., New Delhi.
2. "Materials of construction", by R.C. Smith, McGraw-Hill Company, New York.
3. "Materials of construction", by D.N. Ghose, Tata-McGraw-Hill.
4. "Engineering Materials", By Sushil Kumar, Metropolitan Book Co., Pvt. Ltd, New Delhi
5. "Building Construction" Vol.II & III By W.B. McKay, E.L.B.S. and Longman, UK.
6. "Construction Technology" by R. Chudly Vols I & II 2nd Edition Longman, UK.

PROBABILITY, STATISTICS AND QUEUING THEORY
(Common to CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorials	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To discuss various definitions of probability and related theorems, understand the concepts of prior and posterior probabilities used in Baye's theorem and their use in decision making.
2. To illustrate the concept of a random variable and their properties
3. To learn the assumptions, p.m.f.'s /p.d.f.'s and analyse various statistical measures of a few discrete/continuous distributions.
4. To understand and compute the correlation coefficient, and estimation techniques from regression lines.
5. To develop a framework for testing of hypothesis in giving inferences about Population Parameter.
6. To study Queuing models and their Characteristics.

COURSE OUTCOMES:

At the end of the course a student able to

1. Handle the situation of uncertainty in decision making in our day-to-day life.
2. Identify the random variable as discrete/continuous and analyse it.
3. Predict the distribution suitable for the given data from its moments.
4. Measure the intensity of association between the variables and to fit a best suitable Curve for the given data.
5. Decide the test applicable for giving inference about Population Parameter based on Sample statistic.
6. Make business decisions about the resources needed to provide a service in day-to-day life applications including telecommunication, traffic engineering, computing and the design of factories, shops, offices and hospitals.

SYLLABUS**Probability:**

Various definitions of probability, Addition theorem, Conditional probability, Multiplication theorem, Baye's theorem of probability and Geometric probability.

Random Variables And Their Properties:

Definition, Distribution function, Properties of Distribution Function, Discrete Random Variable, Probability Mass Function, Discrete Distribution Function, Continuous Random Variable, Probability Density Function, Continuous Distribution Function. Mathematical Expectation of a Random Variable, Expected Value of function of a Random Variable, Addition Theorem and Multiplication Theorem of Expectation (**without proofs**), Statistical Measures like Mean, Variance, Moments and Covariance in terms of Expectations, Moment generating Function, Characteristic Function and Probability generating Function of a Random Variable.

Distributions:

Discrete Distributions:

Binomial distribution and Poisson distribution (Definition, Mean, Variance, m.g.f., Characteristic function, p.g.f., Fitting of distributions)

Continuous Distributions: Uniform distribution, its Mean and Variance. Normal Distribution and its Properties (without proofs), Standard Normal Variate, Importance of Normal distribution. Exponential Distribution, Definition, Mean Variance Memory less property of Exponential distribution

Bivariate Analysis and Curve Fitting:

CORRELATION: Definition, Karl Pearson's Coefficient of Correlation, Limits for correlation coefficient, Rank Correlation, Spearman's formula for rank correlation coefficient.

CURVE FITTING, Method of least Squares, Fitting of a Straight line, Fitting of a Parabola

REGRESSION ANALYSIS: Regression Lines, Regression Coefficients and their properties (without proofs)

Sampling Theory:

Sample, population, statistic, parameter, Sampling distribution, standard error, interval estimation. Testing of Hypothesis: Formulation of Null hypothesis, Alternative hypothesis, Critical region, Critical value, level of significance, Statistical Inference, Type-I-error, Type-II-error, One-tailed/Two-tailed test.

Large Sample Theory

Test of significance of single sample proportion, Test of significance for difference of proportions, Test of significance of single sample mean, Test of significance for difference of means.

Small Sample Theory: Degrees of freedom, Student's-t-distribution: definition, t-test for single mean, t-test for difference of means, Paired t-test for difference of means.

F-distribution: definition, F-test for equality of two population variances. Chi-square distribution definition, Chi-square test for goodness of fit, Chi-square test for single Population Variance.

Queuing Theory: Queue description, Birth and Death Process, Distribution of Inter-arrival times, Distribution of service times, Kendall's representation of a queueing model, Operating characteristics of a queueing model, steady-state solutions of $\{M/M/1: \infty/FCFS\}$ Model and $\{M/M/1; N/FCFS\}$ Model.

TEXT BOOK:

1. Probability, Statistics and Random Processes by T.Veerarajan, Tata Mc Graw Hill Pub.

REFERENCE BOOKS:

1. Probability & Statistics with Reliability, Queueing and Computer Applications by Kishore.S.Trivedi, Prentice Hall of India, 1999.
2. Fundamentals of Mathematical Statistics by S.C.Gupta and V.K.Kapoor, Sultan Chand & Sons Publishers.

PHYSICS LAB
(Common to CIVIL, CSE & IT)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

LIST OF EXPERIMENTS

1. Sonometer – verification of laws of transverse vibrations in stretched strings.
2. Melde's Experiment – Determination of the frequency of an electrically maintained tuning fork.
3. Newton's Rings – Determination of radius of curvature of a convex lens.
4. Diffraction Grating – Determination of Wavelengths of lines of mercury spectrum using spectrometer by normal incidence method.
5. Determination of Cauchy's constants of the material of the given prism using Spectrometer and mercury light.
6. Wedge Method – Determination of thickness of a paper by forming parallel interference fringes.
7. Variation of magnetic field along the axis of current carrying circular coil – Stewart and Gee's apparatus.
8. Carey Foster's bridge – Verification of laws of resistance.
9. Lee's Method – Determination of coefficient of thermal conductivity of a bad conductor.
10. Calibration of voltmeter using potentiometer.
11. Calibration of low range Ammeter using potentiometer.
12. Laser – Diffraction.

REFERENCE BOOK:

1. Advanced Practical Physics Vol I & II by SP Singh and MS Chauhan, Pragathi Prakasam Publications

WORKSHOP
(Common to CIVIL, CSE & IT)

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

COURSE OBJECTIVES:

1. To impart hands-on training on basic engineering trades.

COURSE OUTCOMES:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry

Bench Work, tools used in carpentry.

Jobs for Class work – half lap joint, mortise and tenon joint, half – lap dovetail joint, corner dovetail joint, central bridle joint.

Sheet Metal

Tools used in sheet metal work, Laying development of the sheet metal jobs, soldering.

Jobs for class works – Square tray, taper tray (sides), funnel, elbow pipe joint, 60° pipe joint.

Fitting

Tools used in fitting work, Different files, chisels, hammers and bench vice.

Jobs for class work – Square, hexagon, rectangular fit, circular fit and triangular fit.

REFERENCE BOOK:

1. Elements of workshop technology, Vol.1 by S.K. and A.K. Choudary, Media Promoters and Publishers Pvt. Ltd

ENGLISH LANGUAGE LAB
(Common to All Branches)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

The Language Lab focuses on the production and practices of sounds of language and familiarizes the students with use of English in everyday situations and contexts.

COURSE OBJECTIVES:

1. To make students recognize the sounds of English through Audio-Visual aids.
2. To help students build their confidence and help overcome their inhibitions and self consciousness while speaking in English. *The focus shall be on fluency.*
3. To familiarize the students with stress and intonation and enable them to speak English effectively.

COURSE OUTCOMES:

1. Students will be sensitized towards recognition of English sound pattern.
2. The fluency in speech will be enhanced.

SYLLABUS

1. English Sound Pattern – Letters
2. Sounds of English
3. Pronunciation
4. Stress and Intonation

Laboratory Practice Sessions:

1. Letters and Sounds, Worksheet-1
2. Interactions-1, Worksheet-2
3. The Sounds of English, Worksheet-3
4. Interactions-2, Worksheet-4
5. Pronouncing Words-Some Important Patterns, Worksheet-5
6. Interactions-3, Worksheet-2
7. Stress and Intonation, Worksheet-2

REFERENCE BOOKS:

1. Speak Well, Board of Editors, Orient Black Swan Publishers, Hyderabad, 2012.
2. Cambridge English Pronouncing Dictionary, Cambridge University Press, India, 2012.
3. A Textbook of English phonetics for Indian students by T. Balasubramanian, Macmillan publisher, 1981.

NCC/NSS
(Common to CIVIL, CSE & IT)

Contact Hrs. 3 Hrs.

This is an Audit Course. Every student should choose either NCC or NSS at the starting of the semester and pursue the same in that semester. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD ON COURSE
TECHNOLOGY COURSE – I
(Professional C Programming)
(Common to All Branches)

Total Hrs: 30

COURSE OBJECTIVES:

1. On successful completion of the course, students will be able to reinforce their learning of C, read code, and write good procedural code.

COURSE OUTCOMES:

1. Students will be able to solve a series of graduated problems
2. Students will be able to do projects in 'C' Language.

SYLLABUS

Introduction to Linux

Introduction to C with problem solving

Functions usage and application with solved problems

Introduction to pointers and applications with examples

Structures and Unions and file handling with program applications

Project using C

RESOURCES:

Talent Sprint Problem Bank, Project Euler



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

II/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

COMPUTER SCIENCE AND ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 CS 2101	Data Structures	4	3	1	--	4	30	70	100
B16 EC 2103	Elements of Electronics Engineering	4	3	1	--	4	30	70	100
B16 ENG 2102	Discrete Mathematical Structures	4	3	1	--	4	30	70	100
B16 CS 2102	Object Oriented Programming	4	3	1	--	4	30	70	100
B16 CS 2103	Digital Logic Design	4	3	1	--	4	30	70	100
B16 ENG 2103	Environmental Studies	2	3	1	--	4	30	70	100
B16 CS 2105	Data Structures Lab.	2	--	--	3	3	50	50	100
B16 CS 2106	Object Oriented Programming Lab.	2	--	--	3	3	50	50	100
B16 ENG 2104	English Proficiency	2	1	1	--	2	50	50	100
B16 ENG 2105	Industry Oriented Training.	1	--	--	2	2	50	--	50
Total		29	19	7	8	34	380	570	950

Code: B16 CS 2101

DATA STRUCTURES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Be familiar with basic techniques of algorithm analysis
2. Master the implementation of linked data structures such as linked lists and binary trees
3. Be familiar with advanced data structures such as balanced search trees, priority queues.
4. Be familiar with several sub-quadratic sorting algorithms including quick sort, merge sort and heap sort
5. Master analyzing problems and writing program solutions to problems using the above techniques.

Course Outcomes:

1. Describe how arrays, records, linked structures, stacks, queues, trees, and graphs are represented in memory and used by algorithms .
2. Describe common applications for arrays, records, linked structures, stacks, queues, trees, and graphs . Write programs that use arrays, records, linked structures, stacks, queues, trees, and graphs Demonstrate different methods for traversing trees [ABET (a)].
3. Compare alternative implementations of data structures with respect to performance [ABET (a, b, c)].
4. Compare and contrast the benefits of dynamic and static data structures implementations [ABET (a, b, c)].
5. Describe the concept of recursion, give examples of its use, describe how it can be implemented using a stack [ABET (a, c)].
6. Discuss the computational efficiency of the principal algorithms for sorting, searching.

SYLLABUS

Basic Concepts

System Life Cycle, Algorithm Specification, Recursive Algorithms, Data Abstraction, Performance Analysis, Space Complexity, Time Complexity, Asymptotic Notation, Comparing Time Complexities.

Arrays and Structures

Array as an Abstract Data Type, Polynomial Abstract Data Type, Introduction to Sparse Matrix, Sparse Matrix Abstract Data Type, Transposing a Sparse Matrix, Sparse Matrix Multiplication, Representation of Multidimensional Arrays, Structures and Unions, Internal Implementation of Structures, Self-Referential Structures.

Recursion, Simple Searching and Sorting Techniques

Recursive functions, Introduction to Searching, Sequential Search, Binary Search, Interpolation Search, Selection Sort, Bubble Sort, Insertion Sort, Quick Sort, Introduction to Merge Sort, Iterative Merge Sort, Recursive Merge Sort, Shell Sort.

Stacks and Queues

Stack Abstract Data Type, Queue Abstract Data Type, Stacks and Queues using arrays, Mazing Problem, Introduction to Evaluation of Expressions, Evaluating Postfix Expressions, Infix to Postfix and Prefix conversion, Multiple Stacks and Queues, Circular Queues using arrays.

Linked Lists

Pointers, Dynamically Allocated Storage using pointers, Singly Linked Lists, Dynamically Linked Stacks and Queues, Polynomials, Representing Polynomials as Singly Linked Lists, Adding Polynomials, Erasing Polynomials, Polynomials as Circularly Linked Lists, Additional List Operations, Operations for Singly Linked Lists, Operations for Doubly Linked Lists, Radix Sort.

Trees

Representation of Trees, Binary Trees Abstract Data Type, Properties of Binary Trees, Binary Tree Representations, Binary Tree Traversals, Additional Binary Tree Operations, Threaded Binary Trees, Heap Abstract Data Type, Priority Queues, Insertion into a max heap, Deletion from a max heap, Heap Sort, Introduction to Binary Search Trees, Searching a Binary Search Tree, Inserting an Element into a Binary Search Tree, Deleting an Element from a Binary Search Tree, Height of a Binary Search Tree, Counting Binary Trees.

Graphs

Graph Abstract Data Type, Definitions, Graph Representations, Elementary Graph Operations, Depth First Search, Breadth First Search, Connected Components, Spanning Trees, Minimum Cost Spanning Trees, Prim's and Kruskal's Algorithms, Shortest Paths and Transitive Closure, Single Source All Destination - Dijkstra's Algorithm, All Pairs Shortest Paths - Floyd's Algorithm, Transitive Closure using Warshall's Algorithm.

Text Books:

1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahani and Anderson-Freed, Universities Press, 2008.

Reference Books:

1. Data Structures using C by Aaron M. Tenenbaum, Y. Langsam and M.J. Augenstein, Pearson Education, 2009.
2. Data Structures with C by Seymour Lipschutz, Schaum Outline series, 2010.
3. Data Structures using C by R. Krishna Moorthy G. Indirani Kumaravel, TMH, New Delhi, 2008.

ELEMENTS OF ELECTRONICS ENGINEERING (Common to CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To give the exposure to the students on semiconductor physics of the intrinsic and extrinsic semiconductors.
2. To give the exposure to the students on the basics of semiconductor diodes, special purpose diodes like Zener diode, Photo diode, LED and tunnel diode.
3. To give the exposure to the students on rectifies circuits using diodes.
4. To give the exposure to the students on basics of BJT, JFET and MOSFET and biasing of BJT and FETs
5. To give the exposure to the students on the analysis of transistor at low and high frequencies.

Course Outcomes:

After completion of the course the students will be able to

1. Understand the physical structure, principles of operation, electrical characteristics and circuit models of diodes, BJTs and FETs
2. Use this knowledge to analyze and design basic electronic application circuits.
3. Extend the understanding of how electronic circuits and their functions fit into larger electronic systems.

SYLLABUS

Transport phenomenon in semiconductors :

Intrinsic and Extrinsic semiconductors, Charge densities in semiconductors, Drift and Diffusion currents, Hall effect, Mass action law.

P-N Junction Diode :

Basic operation and V-I characteristics of semiconductor diode, diode current equation, Zener diode, LED, Photo diode and tunnel diode (*Introductory treatment only*)

Diode Rectifiers :

Half wave and full wave rectifiers with and without filters, Bridge Rectifier expressions – Ripple factor, Efficiency, capacitor filters

Bipolar Junction Transistor :

Introduction, construction, basic operation, modes of operation-Active, cutoff and saturation, Transistor circuit configurations- CE, CB, CC – input and output Characteristics in various configurations (*Introductory treatment only*)

Transistor Biasing and Thermal Stabilization :

Transistor Biasing, Thermal runaway, Stabilization, Different methods of Biasing- Fixed bias, Collector feedback bias, self bias, Bias Compensation.

Transistor Amplifiers :

CE, CB, CC amplifier h-parameter model for Transistor amplifier

Field Effect Transistors :

Junction Field effect Transistors (JFET)- JFET characteristics, JFET parameters, JFET biasing, MOSFETS- Depletion and Enhancement MOSFET.

Text Books :

1. Electronic Device and Circuits by Sanjeev Guptha

Reference Books :

1. Integrated Electronics – Millman & Halkias

DISCRETE MATHEMATICAL STRUCTURES
(Common to CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To understand mathematical arguments using logical connectives and quantifiers and verify the validity arguments using propositional, predicate logic & truth tables.
2. To know about the concepts of counting techniques
3. To know how to solve recurrence relations.
4. To understand various types of relations and discuss various properties of the relations.
5. To understand the concepts in graphs and trees.
6. To learn about Groups, Lattices and their properties, Boolean algebra and its importance in computer field and minimization of Boolean expressions.

Course Outcomes:

At the end of the course students will be able to

1. Rewrite the mathematical arguments using logical connectives and quantifiers and verify the validity of the arguments using propositional and predicate logic.
2. Solve different counting problems.
3. Solve the recurrence relations which occur in many fields.
4. Identify and give examples of various types of relations and describe various properties of relations.
5. Determine isomorphism of graphs and utilize the concepts in graphs & trees in their fields.
6. Understand the importance of Groups, lattice structures and their diagrammatic representations and also the importance of Boolean algebra in computer science.

SYLLABUS

Introduction :

Sets – Operations on sets – Logic : Logical inferences, Methods of proof of an implications– First order logic and other proof methods-Rules of inference for quantified propositions-mathematical induction.

Elementary Combinatorics & Recurrence relations :

Basics of counting – Combinations and Permutations – their enumeration with and without repetition - Principle of Inclusion and Exclusion and its applications, Generating functions of sequences - Calculating their coefficients-Recurrence relations-solving recurrence relations-method of characteristic roots-Non-homogeneous recurrence relations and their solutions.

Relations and Diagraphs :

Relations and directed graphs-Special properties of binary relations-equivalence relations-Ordering relations-operations on relations-Paths and closures-Directed graphs and Adjacency matrices.

Graphs Theory :

Basic concepts – Isomorphism – sub graphs - planar graphs - Euler's formula -Multi graphs and Euler Circuits - Hamiltonian graphs – Graph coloring and Chromatic number – Four color theorem - Trees and their properties – definitions of different tree structures.

Groups :

Definitions of Binary operation, Algebraic Structure, Semi-group, Monoid, Group and Abelian group.

Lattices :

Lattices and Properties of lattices – lattices as partially ordered sets – sublattices – Direct product and Homomorphisms - Isomorphisms – Modular lattices Distributive lattices – Complemented lattices.

Boolean Algebra :

Definition – Sub algebra – Direct product – Homomorphisms – Isomorphisms – Boolean functions – Representation of Boolean functions – Minimizations of Boolean functions using K-maps.

Text Books :

1. For the first four topics, scope and treatment as in “Discrete Mathematics for computer scientists and mathematicians” by Joe. L .Mott, Abraham Kandel & T.P. Baker, Prentice Hal of India Ltd, New Delhi.
2. For the topics five and six, scope and treatment as in “Discrete mathematical structures with applications to computer science” by J.P. Trembly & R. Manohar, Tata McGraw-Hill Publishing company, New Delhi.

Reference Books :

1. “Discrete mathematics and its applications” by Keneth. H. Rosen, Tata McGraw-Hill Publishing Company, New Delhi.
2. “Discrete Mathematics” by Richard Johnson Baug, Pearson Education, New Delhi.
3. “Discrete and Combinatorial Mathematics” by Ralph. G. Grimaldi, Pearson Education, New Delhi.

OBJECT ORIENTED PROGRAMMING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

On completing this course student will be able to

1. Understand the syntax and principles of Object oriented programming language, and to programs using control statements, classes and interfaces.
2. Design and development of secure and extendable C++ applications.
3. Understanding the concepts of oops, different predefined classes and packages
Understand the concepts of polymorphism

Course Outcomes:

1. Students will be able to handle I/O streams and Run time errors.
2. Students will be able to construct applications and Identify where data structures are appearing in them

SYLLABUS**Part 1: C++****Basics of Object Oriented Programming:**

Object Oriented Paradigm, Principles of OOP, benefits of OOP, data types, declarations, expressions and operator precedence, functions, scope of variables.

Introduction to C++:

Classes and objects, Constructors & Destructors, constructor with dynamic allocation, explicit constructor, Operator Overloading through Unary, Binary, Assignment and Stream operators & type conversions.

Inheritance and Manipulating Strings:

Derived classes, syntax of derived classes, making private members inheritable, single, multilevel, multiple, hierarchical, hybrid inheritance, Virtual base Class, abstract classes, Creating String Objects, Manipulating String Objects, Relational Operations, Accessing String Characteristics.

Polymorphism:

Pointers, virtual functions and polymorphism- pointers to objects, this pointer, pointers to derived classes, virtual and pure virtual functions, Dynamic polymorphism, Virtual destructor, Virtual Base Class, Dynamic Casting, Cross Casting, Down Casting.

Templates, Exception handling, Streams and Files in C++:

Class templates, Function templates, member function templates, exception handling, managing console I/O operations, Stream Classes, Formatted and Unformatted i/o operations, managing output with manipulators, working with files.

Part 2: JAVA

Introduction to JAVA:

Introduction, Classes and Objects, Inheritance, Arrays, strings and Vectors, Exception Handling, Managing I/O files in Java.

Packages and Interface, and Multi threading: Packages, Interfaces, creating, threads, thread states, thread methods, exceptions, priority in threads, synchronization, Runnable interface, life cycle of an Applet.

Text Books:

1. Object oriented Programming using C++: E. Balagurusamy, PHI.
2. Programming with JAVA- A primer: E. Balagurusamy, PHI.

Reference Books:

1. JAVA 2.0- Complete Reference: Herbert Schildt & F. Naughton.
2. Introduction to JAVA PROGRAMMING by Y.Daniel Liang (PHI)
3. Object Oriented Programming in C++: N. Barkakati, PHI
4. Object Oriented Programming through C++ by Robot Laphore.
5. Object Oriented Analysis and Design by Andrew Haigh – (Tata Mcgrah Hjill.)

DIGITAL LOGIC DESIGN

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To introduce the basic principles for design of combinational circuit and sequential circuits.
2. To learn simple digital circuits in preparation for computer engineering.

Course Outcomes:

A student who successfully fulfills the course requirements will have demonstrated:

1. An ability to define different number systems, binary addition and subtraction, 2's complement representation and operations with this representation.
2. An ability to understand the different Boolean algebra theorems and apply them for logic functions.
3. An ability to define the Karnaugh map for a few variables and perform an algorithmic reduction of logic functions.
4. An ability to define the following combinational circuits: multiplexer, de-multiplexers encoders/decoders, comparators, arithmetic-logic units; and to be able to build simple circuits.
5. An ability to understand asynchronous and synchronous sequential circuits, like counters and shift registers.
6. An ability to understand memories like RAM and ROM, Programmable Logic Array and Programmable Array Logic.

SYLLABUS**Binary Systems:**

Digital Systems. Binary Numbers. Number Base Conversions. Octal and Hexadecimal Numbers. Complements. Signed Binary Numbers. Binary Codes. Binary Storage and Registers. Binary Logic.

Boolean Algebra and Logic Gates:

Basic Definitions. Axiomatic Definition of Boolean Algebra. Basic Theorems and Properties of Boolean Algebra. Boolean Functions. Canonical and Standard Forms. Other Logic Operations. Digital Logic Gates. Integrated Circuits.

Combinational Logic Design, Gate-Level Minimization:

The Map Method. Four- Variable Map. Five-Variable Map. Product of Sums Simplification. Don't-Care Conditions. NAND and NOR Implementation. Other Two- Level Implementations. Exclusive-OR Function. Hardware Description Language(HDL).

Combinational Logic:

Combinational Circuits. Analysis Procedure. Design Procedure. Binary Adder-Subtractor. Decimal Adder. Binary Multiplier. Magnitude Comparator. Decoders. Encoders. Multiplexers. HDL For Combinational Circuits.

Sequential Logic Design, Synchronous Sequential Logic:

Sequential Circuits .Latches .Flip-Flops. Analysis of Clocked Sequential Circuits. HDL For Sequential Circuits. State Reduction Assignment. Design Procedure.

Registers and Counters:

Registers. Shift Registers. Ripple Counters. Synchronous Counters. Other Counters. HDL for Registers and Counters.

Memory and Programmable Logic:

Introduction. Random-Access Memory. Memory Decoding, Error Detection and Correction. Read-Only Memory. Programmable Logic Array. Programmable Array Logic. Sequential Programmable Devices.

Text Books:

1. Digital Design, 3rd Edition, M. Morris Mano, Pearson Education.

Reference Books:

1. Digital Logic Design Principles, Norman Balabanian & Bradley Carlson, John Wiley & Sons (Asia) Pvt. Ltd., 2002.
2. Fundamentals of Digital Logic with VH.

ENVIRONMENTAL STUDIES
(Common to CIV, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs	Credits	: 2

Course Objectives:

Students learn

1. To develop an awareness and sensitivity to the total environment and its related problems.
2. To participate actively participation in environmental protection and improvement.
3. To develop skills for active identification and development of solutions to environmental problems
4. To evaluate environment programmes in terms of social, economic, ecological and aesthetic factors.
5. To Create a “CONCERN AND RESPECT FOR THE ENVIRONMENT”

Course Outcomes:

Students will be able to

1. Get awareness among the students about the nature and natural ecosystems.
2. Learn sustainable utilization of natural resources like water, land, minerals, air.
3. Learn resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation.
4. Safe guard against industrial accidents particularly nuclear accidents.
5. Learn Constitutional provisions for the protection of natural resources.

SYLLABUS**Global Environmental Crisis:**

Environmental Studies - Definition, Scope and importance, Need for public awareness.
Global Environmental Crisis

Ecosystems:

Basic concepts, Forest Ecosystems, Grassland Ecosystems and Desert Ecosystems, Aquatic Ecosystems

Biodiversity:

Introduction to Biodiversity, Value of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of bio-diversity.

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people, **Water resources:** Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems

Energy Resources: Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and global warming, acid rain and Ozone layer depletion.

Environmental Problems in India:

Drinking water, Sanitation and Public health, population growth and environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

Text Books:

1. Environmental Studies (From Crisis to Cure) by R. Rajagopalan, Oxford university Press, 2008
2. Environmental Studies by Anubha Kaushik & C.P. Kauskik, New Age International (P) Ltd, New Delhi, 2006

Reference Books:

1. Environmental Sciences by G.Tyler Miller, JR,10th ed, Thomson publishers, 2004

DATA STRUCTURES LAB

Lab : 3 Periods	Sessionals : 50
Exam : 3 Hrs.	Ext. Marks : 50
	Credits : 2

Course Objectives:

1. To implement stacks and queues using arrays and linked lists.
2. To develop programs for searching and sorting algorithms.
3. To write programs using concepts of various trees.
4. To implement programs using graphs.

Course Outcomes:

1. Student will be able to write programs to implement stacks and queues.
2. Ability to implement various searching and sorting techniques.
3. Ability to implement programs using trees and graphs.

SYLLABUS

Implement the following programs using C-Language.

1. Write a program for sorting a list using Bubble sort and then apply binary search.
2. Write a program to implement the operations on stacks.
3. Write a program to implement the operations on circular queues.
4. Write a program for evaluating a given postfix expression using stack.
5. Write a program for converting a given infix expression to postfix form using stack.
6. Write a program for implementing the mazing problem.
7. Write a program for the representation of polynomials using linked list and for the addition of two such polynomials
8. Write a program for quick sort
9. Write a program for Merge sort.
10. Write a program for Heap sort
11. Write a program to create a binary search tree and for implementing the in order, preorder, post order traversal using recursion
12. Write a program for finding the transitive closure of a digraph
13. Write a program for finding the shortest path from a given source to any vertex in a digraph using Dijkstra's algorithm
14. a) Write a program for finding the Depth First Search of a graph.
b) Write a program for finding the Breadth First Search of a graph

Reference Books:

1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahani and Anderson-Freed, Universities Press, 2008.

OBJECT ORIENTED PROGRAMMING LAB

Lab : 3 Periods	Sessionals : 50
Exam : 3 Hrs.	Ext. Marks : 50
	Credits : 2

Course Objectives:

1. To develop programs using basic OOPS concepts such as classes and objects.
2. To implement programs using Inheritance concepts.
3. To implement programs using Exception handling.
4. To develop programs using operator overloading concepts.

Course Outcomes:

1. Student will be able to use OOPs concepts.
2. Ability to apply Inheritance concepts to several problems.
3. Ability to use Exception Handling concepts.

SYLLABUS**Part I: UML**

Take a own scenario and draw the UML Diagrams (Structural, Behavioral and Interactive).

Part II: Implement the following programs using C++

1. Write a Program that implements stack operations using classes and objects.
2. Write a Program performing complex number addition using friend functions.
3. Write a Program for complex number addition using operator overloading.
4. Write a Program to perform string operations by overloading operators.
5. Write a Program on hierarchical inheritance showing public, private and protected inheritances.
6. Write a Program for computation of student's result using hybrid inheritance.
7. Write a Program implementing bubble-sort using templates.
8. Write a Program on virtual functions.
9. Write a Program for Templates.
10. Write a Program for copying one file to another file using streams.
11. Write a Program for writing and reading a class object to a file.
12. Write program to implement
 - a. One catch block and all Exceptions
 - b. using Multiple Catch blocks.
13. Write a program to implement the finally block.
14. Write a program to implement pointers to a derived class and virtual base classes.
15. Write a program to implement conversion of objects between different classes using conversion functions.
16. Write a program to implement function overloading- with various data types, with different number of arguments.

17. Write a program to evaluate mixed mode expressions and implicit type conversions.
18. Write a program to show that there is ambiguity in Multiple Inheritance.
19. Write a program to implement a virtual destructor.
20. Write a program to mimic a bank management system (user logins, requests for withdraw /credit, system verifies whether enough balance is available, update the account summary, etc.)

Part III: Implement the following programs using Java

1. Write a program to give an example for command line arguments.
2. Write a program to sort given list of numbers.
3. Write a program to implement linear search and binary search.
4. Write a program for this operator and super keyword.
5. Write a program that gives demonstration of static variables and methods.
6. Write a program that illustrates the simple inheritance, multilevel inheritance.
7. Write a program that demonstrates the difference between method overloading and overriding.
8. Write a program that demonstrates the difference between method overloading and constructor overloading.
9. Write a program that describes the exception handling mechanism.
10. Write a program that uses try and catch blocks and check whether the given array size is negative or not.
11. Write a program that describes the user defined exception.
12. Write a program that illustrates the creation of threads by using runnable class.
13. Write a program that illustrates the creation of threads
14. Write a program that illustrates the multiple inheritances by using interfaces.
15. Write a program that describes the life cycle of an applet.
16. Write a program that displays the number of characters, lines and words in a text file.
17. Write a program on packages.
18. Write a program to copy contents of a file into another file using File streams.
19. Write a Program for handling Array Index out of Bounds Exception and Divide-by-zero Exception.
20. Write a Program for interfaces.
21. Write a Program on Threads.
22. Write a Program for Constructors.
23. Write a Program for Wrapper Classes.
24. Write a Program for Thread Priority.
25. Write a program for Producer Consumer Problem.

Reference Books:

1. Object Oriented Programming with C++, 6e, E. Balagurusamy, TMH, India, 2013.
2. Programming with JAVA-A Primer, 5e, E. Balagurusamy, TMH, India, 2014.
3. The Unified Modeling Language user Guide, Grady Booch, James Rumbaugh and Ivar Jacobson, Addison-Wesley, 1999.

ENGLISH PROFICIENCY
(Common to All Branches)

Theory	: 1 Period	Sessionals	: 50
Tutorial	: 1 Period	Ext. Marks	: 50
Exam	: 3 Hrs.	Credits	: 2

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

Students be able to

1. Understand the importance of professional communication.
2. Learn language skills and vocabulary in order to improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self-attributes which require improvement and motivation.
5. Build their confidence and overcome their inhibitions.
6. Improve their strategies in reading skills.

Course Outcomes:

1. Students enhance their vocabulary and use it in the relevant contexts .
2. They improve speaking skills.
3. They learn and practice the skills of composition writing.
4. They enhance their reading and understanding of different texts.
5. They enrich their communication both in formal and informal contexts.
6. They strengthen their confidence in presentation skills.

SYLLABUS

Speaking Skills

PPT

Describing event/place/thing

Picture Description

Extempore

Debate

Telephonic Skills

Analyzing Proverbs

Vocabulary

Affixes

Pairs of Words

Reading Skills

Reading Comprehension

Reading/Summarizing News Paper Artic

Writing Skills

Designing Posters

Essay writing

Resume Writing

Reference Books:

1. Interchange (4th edition) Student's books 1&2 by Jack C. Richards, CUP.
2. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
3. English and Communication Skills for Students of Science and Engineering, by S.P.
4. Dhanavel, Orient Blackswan Ltd. 2009
5. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
6. The Oxford Guide to Writing and Speaking by John Seely OUP

(***Note: Sessional Marks will be evaluated based on Continuous Comprehensive Evaluation of the students' Performance - 40M, Attendance – 10M and External Marks will be evaluated based on Presentation Skills – 30M, Project 20M)

INDUSTRY ORIENTED TRAINING
(WEB Development)
(Common to CSE & IT)

Lab: 2 Periods**Exam : 3 Hrs.****Sessionals : 50****Credits : 1****Course Objective:**

Web development technologies such as HTML, CSS, JavaScript and others are at the core of all modern online systems. The objective of this course is to expose & train the students on web programming concepts and develop the ability to build web sites with dynamic presentation.

Course Outcomes:

Upon completion of this web development training course, able to:

1. Design and develop basic web pages using HTML.
2. Apply cascading style sheets to web pages in order to separate form from content.
3. Understand & Apply basic control of elements with JavaScript.
4. Understand the basic concepts of PHP scripting
5. Able to design & complete a project by applying above all the concepts.

Syllabus: Industry Oriented Applications on following topics.

HTML:- HTML Introduction, HTML Basic Tags, HTML Lists, HTML Tables, HTML Images, HTML Links & Navigation, HTML Forms.

CSS:-CSS Introduction, CSS Properties - Controlling Fonts, CSS Properties - Text Formatting, Selectors - id and class, Pseudoclasses, CSS for Links, CSS for Lists, CSS for Tables.

JAVA SCRIPT:-JavaScript Introduction, Empty Field Validation Example, Name & Numbers Only Validation Example, Email Validation Example, innerHTML Error Display Example.

PHP:-Installation of Wamp Server, PHP Introduction, Creating PHP Script, Running PHP Script, PHP Numeric Variables, Sample PHP Programs

MINI PROJECT

(Note: Total Marks will be evaluated based on Continuous evaluation - 25 Marks, Mini Project- 25 Marks)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

II/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

COMPUTER SCIENCE AND ENGINEERING

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 CS 2201	Operating Systems	4	3	1	--	4	30	70	100
B16 CS 2202	Computer Organization	4	3	1	--	4	30	70	100
B16 CS 2203	Microprocessors	4	3	1	--	4	30	70	100
B16 CS 2204	Data Communications	4	3	1	--	4	30	70	100
B16 CS 2205	Advanced Data Structures	4	3	1	--	4	30	70	100
B16 CS 2206	Computer Graphics	4	3	1	--	4	30	70	100
B16 CS 2207	Operating Systems & Unix programming Lab	2	--	--	3	3	50	50	100
B16 CS 2208	Digital Electronics & Microprocessors Lab	2	--	--	3	3	50	50	100
B16 CS 2209	Competitive Programming	1	1	--	1	2	50	--	50
B16 ENG 2203	Industry Oriented Training	1	--	--	2	2	50	--	50
Total		30	19	6	9	34	380	520	900

OPERATING SYSTEMS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To understand operating system as a layer of abstraction above physical hardware that facilitates usage convenience and efficient resource management.
2. To learn design and implementation of policies and mechanisms of OS.
3. To investigate case studies to understand the design paradigms for popular multiuser or single user operating system.

Course Outcomes:

1. The student understands OS evolution, its structure and services provided by it.
2. Learn process life cycle, process scheduling objectives, policies and mechanisms, process synchronization, inter process communication, deadlocks and other process subsystem related concepts.
3. Learn memory hierarchy, allocation, de-allocation policies and mechanism for main and auxiliary memory, file system design and implementation issues.
4. Investigate UNIX/ LINUX and Windows OS platforms w.r.t similarities and differences in design paradigms.

SYLLABUS

Introduction to Operating Systems:

Over View of Operating Systems, Types of Operating Systems, Operating System Structures, Operating System Services, System Calls, Virtual Machines, Operating System Design and Implementation.

Process Management:

Process Concepts, Operations on Processes, Co-operating Processes, Threads, Inter Process Communication, Process Scheduling, Scheduling Algorithms, Multiple - Processor Scheduling, Thread Scheduling.

Process Synchronization:

The Critical Section Problem, Peterson's Solution, Synchronization Hardware, Semaphores, Classical Problems of Synchronization, Critical Regions, Monitors.

Deadlocks:

System Model, Deadlock Characterization, Methods For Handling Deadlocks, Deadlock Prevention, Avoidance, Deadlock Detection, Recovery from Deadlocks

Memory Management:

Logical versus Physical Address, Swapping, contiguous memory allocation, paging, structure of the page table, segmentation, Virtual Memory, Demand Paging, Page Replacement, Allocation of Frames, Thrashing, Memory-Mapped files

File Systems, Implementation, and Secondary-storage Structure:

Concept of a file, Access Methods, Directory Structure, Protection, File System Structure, Allocation Methods, Free Space Management, Directory Management, Device Drivers, overview of Mass-storage structure, Disk structure, disk attachment, disk scheduling, swap-space management.

Case study: Overview of LINUX, Windows Operating systems

Text Book:

1. Operating Systems, Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne, John Wiley Publ., Seventh Edition.

Reference Books:

1. Modern Operating Systems, Andrew S. Tanenbaum, , 2nd edition, 1995, PHI.
2. Operating Systems, William Stallings 5th Edition - PHI
3. Operating Systems: A Design-Oriented Approach', Charles Crowley, _Tata Hill Co.,1998 edition.

COMPUTER ORGANIZATION

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To study about structure and functional components of a computer.
2. Understanding the hierarchical organization of a computer system which consists of Instruction set of commands.
3. Learn about the architecture of a computer.
4. To study about designing a balance system that minimizes performance and utilization of all elements.

Course Outcomes:

1. Apply the basic knowledge about Digital logic to the Functional components of computer.
2. Students will be able to Describe the major components of a computer.
3. Students will be able to classify different Computer Instructions.
4. Students will be able to Describe Instruction set architecture.
5. Recognize the importance of peripheral devices.
6. Students should be able to classify Computer memories

SYLLABUS**Register Transfer and Micro operations:**

Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro operations, Shift Micro operations, Arithmetic Logic Shift Unit.

Basic Computer Organization and Design:

Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory-Reference Instructions, Input- Output and Interrupt, Complete Computer Description, Design of Basic Computer, Design of Accumulator Logic.

Micro programmed Control:

Control Memory, Address Sequencing, Micro program Example, Design of Control Unit.

Central Processing Unit:

Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, Reduced Instruction Set Computer(RISC)

Pipeline and Vector Processing:

Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISK Pipeline, Vector Processing, Array Processors.

Input/output Organization:

Peripheral Devices, I/O interface, Asynchronous data transfer, Modes of transfer, priority Interrupt, Direct memory access, Input-Output Processor (IOP), Serial Communication.

Memory Organization:

Memory Hierarchy, Main memory, Auxiliary memory, Associate Memory, Cache Memory, and Virtual memory.

Text Book :

1. Computer System Architecture, M. Morris Mano, Prentice Hall of India Pvt. Ltd., Third Edition, Sept.2008.

Reference Books :

1. Computer Architecture and Organization, William Stallings, PHI Pvt. Ltd., Eastern Economy Edition, Sixth Edition, 2003.
(Computer Organization and Architecture-Designing of Performance)
2. Computer Organization and Architecture, Linda Null, Julia Lobur, Narosa Publications, Third Edition ,2003.
3. Computer Architecture and Organization, John. P. Hayes, Third Edition, Tata Mc Grawhill International Edition,1998.

MICROPROCESSORS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To discuss the architectures of 8085, 8086 microprocessors, their instruction sets and related ALP programs.
2. To discuss interfacing semiconductor memories, interfacing peripheral to Intel 8085.

Course Outcomes:

1. Understand the basic architectures of 8085 and 8086 microprocessors.
2. Ability to write ALP programs using instruction sets of 8085 & 8086.
3. Understand the various interfacing concepts.

SYLLABUS

Internal Architecture functional/signal description of 8085 microprocessor, Instruction set, Addressing modes and programming in 8085.

Timing diagram, counters and time delays, stacks and subroutines and Interrupts in 8085

Classification and interfacing semiconductor memories with 8085 MPU. Interfacing characteristics of IO devices, IO device addressing methods.

Interfacing peripherals to INTEL 8085: Parallel I/O interface-8255, Serial I/O Interface-8251, Timer Interface-8253.

Interfacing peripherals to INTEL 8085: Keyboard/Display Interface-8279, Interrupt controller Interface-8259.

The 8086 Microprocessor architecture: Internal Architecture & functional /signal description of 8086, segmented memory, Maximum & Minimum mode of 8086.

Instruction set and programming the 8086: Addressing modes, Instruction set and assembly language programming techniques with 8086.

Text Books

1. Microprocessor Architecture and Applications with the 8085 , Ramesh S. gaonkar, 4th Edition, Penram International, 1999
2. Advanced Microprocessors and Peripherals, A K RAY & K M Bhurchandi , 2nd Edition, The McGraw-Hill companies.

Reference Books

1. The 80X86 Family , Design, Programming and Interfacing, John E. Uffenbeck, 3rd Edition, Pearson Education Inc., 2002.
2. Walter A . tribal and Avatar Singh. The 8088 and 8086 Microprocessors, Programming interfaing, software, hardware and Applications, 4th Edition Pearson education Inc., 2003
3. Microprocessors and Interfacing. Programming and hardware, 2ne Edition, Douglass V. Hall. MH Edition , 1999.

DATA COMMUNICATIONS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To study basics of data communication systems.
2. To study the various types of transmission media.
3. To study the various hardware concepts related to data communications.
4. To discuss about multiplexing techniques.

Course Outcomes:

1. Students will have the ability to use Data Communications and Networking Protocols and protocol architectures
2. Students will have the ability to develop communication models for providing data transmission facility
3. Students will have the ability to outline Data Communication terminology
4. Students will have the ability to classify various transmission media
5. Students will have the ability to discriminate various types of signals for data transmission and ability to describe data encoding techniques
6. Students will have the ability to describe data communications interface
7. Students will have the ability to apply various flow control , error control techniques of data link control protocols
8. Students will have the ability to use various data communication terminals and processing hardware
9. Students will have the ability to demonstrate multiplexing techniques

SYLLABUS**Introduction to Data Communications:**

A Communications Model, Data Communications and Data Communications Networking, Protocols and Protocol Architecture, Characteristics of Data Transmission: Concepts and Terminology, Analog and Digital Data Transmission, Transmission Impairments.

Transmission Media:

Guided Transmission Media, Wireless Transmission. **Data Encoding:** Digital Data-Digital Signals, Digital Data-Analog Signals, Analog Data-Digital Signals, Analog Data-Analog Signals.

Data Communication Interface:

Asynchronous and Synchronous Transmission, Line Configurations, Interfacing. Data Link Control Flow Control, Error Detection, Error Control, High-Level Data Link Control (HDLC)

Data Communications Hardware:

Terminals: Introduction, Basic Terminal Components, Enhanced Terminal Components, General-Purpose Terminals, Remote Job Entry Terminals, Transaction Terminals, Clustering of Terminal Devices.

Communications Processing Hardware:

Introduction, Switching Processors, Multiplexers, Concentrators, Front-End Processors

Multiplexing:

Frequency-Division Multiplexing, Synchronous Time-Division Multiplexing: Characteristics, TDM Link Control, Digital Carrier Systems, Statistical Time-Division Multiplexing: Characteristics.

Text Books:

1. William Stallings, Data and Computer Communications, 7th Edition, PH/Pearson Edu. Inc.,
2. Mary E.S. Loomis, Data Communications, PHI-N.J., 1983 (Chapter 3, Chapter 5)

Reference Books:

1. Behrouz A. Forouzan, Data Communications and Networking, 3rd Edition TMH, 2004
2. William A. Shay, Understanding Data Communications & Networks, 2nd Edition Thomson-Brooks/Cole –Vikas Publishing House, 1999.
3. Michale A. Miller, Data & Network Communications, Thomson/Delmar –Vikas Pub. House, 2000

ADVANCED DATA STRUCTURES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To study the concepts related to trees such as binary trees, BST , AVL trees etc.
2. To discuss various hashing technique.
3. To study the various external sorting algorithms.
4. To discuss the concepts related to disjoint set ADT.
5. To study several graph algorithms and their time complexities.

Course Outcomes:

1. Student will be able to write programs to implement various trees.
2. Ability to understand various hashing techniques.
3. Ability to write programs to implement sorting techniques.
4. Ability to understand concepts related to graph theory.

SYLLABUS**Trees:**

Definition , operations and applications of Binary search trees, AVL trees, Red-Black Trees, Splay trees, Tries and B-Trees, B+ Trees

Priority Queues:

Heap model and implementations, Binary Heap, Applications of Priority Queues, d-Heaps, Leftist Heaps, Skew Heaps, Binomial Queues structure, operations and Implementation

Hashing & External sorting:

Hash Table Structure, Hash Function, Collision handling, Separate Chaining, Open Addressing, Rehashing, Extendible hashing, Difference between internal and external sorting, Model and simple algorithm for External sorting, Multi-way Merge, Poly-phase Merge, Replacement selection.

Graph algorithms:

Representation of graphs, Topological sort, Network flow problems, Applications of Depth first search for finding Bi-connectivity, Euler circuits, strong components, Introduction of NP-Completeness

Disjoint Set ADT & Amortized analysis:

Equivalence relations, Dynamic equivalence problem, Basic data structure, smart union algorithms, path compression, Analysis of union/find algorithm, applications of ADT Disjoint set, Introduction to amortized analysis, Basic approaches, binary queues, skew heaps, Aggregate analysis, The accounting method, The potential method and Dynamic tables.

Text Books:

1. Data Structures and Algorithm Analysis in C – Mark Allen Weiss, Pearson Edu Publishers.
2. Data Structures and Algorithms: Concepts, Techniques and Applications – G.A.V.Pai, Tata Mc Graw Hill Publishers

Reference Books:

1. Advanced Data Structures – Peter Brass, Cambridge University Press, 2008
2. Introduction to Algorithms by Thomas H. Cormen, 3rd Edition, PHI

COMPUTER GRAPHICS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Provide a comprehensive introduction to computer graphics leading to the ability to understand contemporary terminology, progress and trend.
2. Design algorithms for digitization of graphic primitives
3. Apply computer graphics techniques for two-dimensional and three-dimensional transformations
4. Visualize viewing transformations
5. Demonstrate working of I/O devices

Course Outcomes:

At the end of the course the student will be able to

1. Summarize the application areas of computer graphics
2. Implement algorithms for scan converting graphic primitives in a graphic package.
3. Apply direct and indirect methods for two-dimensional transformations using matrices.
4. Construct three-dimensional geometric transformations using matrices.
5. Visualize two-dimensional viewing transformations
6. Produce views of three-dimensional scenes.
7. Visualize the working of I/O devices

SYLLABUS**Overview of Graphics Systems:**

Applications of Computer Graphics-Graphical User Interfaces-Video Display Devices-Raster Scan Systems-Random Scan Systems-Graphics Monitors and Workstations -Input Devices-Logical Classification of Input Devices-Hard Copy Devices- Graphics Software-Overview of C-Graphics, Open GL and PHIGS.

Output Primitives:

Points and Lines-Line Drawing Algorithms- Circle Generating Algorithms- Parallel Line Algorithms-Functions in C-Graphics for Output Primitives.

Attributes of Output Primitives:

Line and Curve Attributes-Color and Gray Scale Levels-Area Fill Attributes-Character Attributes-Bundled Attributes-Inquiry Functions-Anti-aliasing Techniques -Functions in C-Graphics for Attributes of Output Primitives-Filled area primitives- Boundary Fill Algorithm-Flood Fill Algorithm.

Two-Dimensional Geometric Transformations:

Basic Transformations- Matrix Representations-Homogeneous Coordinates-Composite Transformations-Reflection-Shear-Transformations between Coordinate Systems-Affine Transformations- Raster Methods for Transformations.

Two-Dimensional Viewing:

The Viewing Pipeline-Viewing Coordinate Reference Frame-Window-to-Viewport Coordinate Transformation-Clipping Operations-Point Clipping-Line Clipping-Polygon Clipping-Curve Clipping- Text and Exterior Clipping

Three-Dimensional Object Representations and Viewing:

3D Display Methods- 3D Graphics-Polygon Surfaces- Curved Lines and Surfaces- Quadratic Surfaces-Super Quadrics-Blobby Objects-Spline Representations-Cubic Spline methods- Bézier Curves and Surfaces- 3D Viewing Pipeline- Viewing Coordinates- Projections- View Volumes- General Projection Transformations.

Three-Dimensional Geometric Transformations:

Translation- Rotation- Scaling- Reflection – Shear-Composite Transformations-Modeling and Coordinate Transformations.

Case studies- Implementation of algorithms in c-graphics**Text Book:**

1. Computer Graphics C Version, Donald Hearn& M. Pauline Baker, Pearson Education, New Delhi, 2004

Reference Books:

1. Procedural Elements for Computer Graphics, David F. Rogers, Tata McGraw Hill Book Company, New Delhi, 2003
2. Computer Graphics: Principles & Practice in C, J.D. Foley, S.K. Feiner, A. Van Dam. F.H John, Pearson Education, 2004
3. Computer Graphics with Open-GL, Donald Hearn, M. Pauline Baker& Warren Carithers,4th Edition, 2011
4. Computer Graphics, Zhigang Xiang and Roy A. Plastock, McGraw-Hill Education, 2nd Edition, ,2015
5. Mathematical and computer programming techniques for computer graphics, Peter Comninos, Springer -Verlag, 2006.

OPERATING SYSTEMS AND UNIX PROGRAMMING LAB**Lab : 3 Periods****Exam : 3 Hrs.****Sessionals : 50****Ext. Marks : 50****Credits : 2****Course Objectives:**

1. To learn about UNIX/LINUX operating system, its intervals.
2. To learn system programming for UNIX/LINUX Operating System.
3. To understand UNIX/LINUX shell programming.
4. To understand resource management policies, mechanisms and their performance evaluation.

Course Outcomes:

1. The student practices UNIX commands, Vi editor, shell commands.
2. The student develops skill in writing C programs using system calls for process management, inter process communication and memory management aspects.
3. The student learns shell programming and develops skill for writing scripts for batch level tasks.

SYLLABUS**Module I**

OS lab familiarization, Home Assignment on Unix commands, Vi editor
 Simple C programs using command line arguments, system calls, library function calls,
 make utility C programs using fork system call to create process and study parent, child
 process mechanism
 C programs to create process chaining, spawning
 C programs to handle errors using errno, perror() function
 C programs to use pipe system call for inter process communication

Module II

Familiarization of Unix shell programming
 Simple shell programming exercises
 Shell programming using decision making constructs
 Shell programming using loop constructs
 Shell programming for file and directory manipulation

Module III

C programs to study process scheduling (FCFS, Shortest Job First, and Round Robin) C
 programs to study page replacement (FIFO, Optimal, and LRU page replacement) C
 programs to study deadlock avoidance and detection

C Programs to simulate free space management

Reference Books:

1. Unix concepts and applications by Sumitabha Das, TMH Publications.
2. Unix programming by Stevens, Pearson Education.
3. Shell programming by YashwanthKanetkar.
4. Operating SystemConcepts bySilberschatz, and Peter Galvin

DIGITAL ELECTRONICS AND MICRO PROCESSORS LAB**Lab : 3 Periods****Exam : 3 Hrs.****Sessionals : 50****Ext. Marks : 50****Credits : 2****Course Objectives:**

1. To learn about logic gates, half adders, full adders and flip-flops.
2. To learn about the microprocessor programming.
3. To learn about the microprocessor interfacing with stepper motor, R-2R ladder.

Course Outcomes:

1. The student understands the logic gates, half adders, full adders and flip-flops to design a circuit.
2. The student develops the skill of writing microprocessor programming.
3. The student understands the interfacing of microprocessor with stepper motor, R-2R ladder.

SYLLABUS**DIGITAL EXPERIMENT**

Verification of Truth tables of OR, AND, NOT, NAND, NOR, EX-OR gates (by using 7400-series)

Construction of gates using NAND, NOR gates.

Construction of Half and Full adders and verifying their truth tables.

Operation and verifying truth tables of flip-flops-RS, D and JK using IC's

Up/Down counters using JK flip-flops.

4-bit shift right and left registers using JK flip-flops.

MICROPROCESSORS: 8085

Binary Addition of 'N' 8-bit numbers.

Binary to BCD conversion

Arranging –Ascending/descending order

To find the largest /smallest numbers in the array.

ASCII to HEXA & HEXA to ASCII conversion.

MICROPROCESSORS: 8086

Linear Search

Factorial of a given number

To copy string from S1 to S2

To find GCD and LCM

MICROPROCESSOR INTERFACING WITH 8085

Elevator

Traffic Light.

Analog to Digital & Digital to Analog Conversion

Interrupt controller

Stepper Motor controller.

Reference Books

1. Microprocessor Architecture and Applications with the 8085 , Ramesh S. Gaonkar, 4th-Edition, Penram International, 1999
2. Advanced Microprocessors and Peripherals, A K RAY & K M Bhurchandi , 2nd Edition, The Mcgraw-Hill companies.
3. The 80X86 Family , Design, Programming and Interfacing, John E. Uffenbeck, 3rd Edition, Pearson Education Inc., 2002.
4. Walter A . tribal and Avatar Singh. The 8088 and 8086 Microprocessors, Programming interfacing, software, hardware and Applications, 4th Edition Pearson education Inc., 2003
5. Microprocessors and Interfacing. Programming and hardware, 2ne Edition, Douglass V. Hall. MH Edition , 1999.

COMPETITIVE PROGRAMMING**Theory: 1 Period****Lab : 1 Period****Exam : 3 Hrs.****Sessionals : 50****Credits : 1**

Course Objectives:

1. To Learn basic concepts of Python Programming Language
2. To Learn various Object Oriented Programming Concepts
3. To Learn various advanced Data structures

Course Outcomes:

At the end of the course the student can able to

1. Write programs using python programming
2. Write algorithms
3. Implement various data Structures
4. To apply object oriented mechanisms
5. To Implement various Advance data Structures like AVL trees, B-Trees, Splay trees etc

SYLLABUS**Introduction to Python**

- The basic elements of python
- Branching Programs
- Control Structures
- Strings and Input
- Iteration

Functions, Scoping and Abstraction

- Functions and scoping
- Recursion
- Files

Classes and Object-Oriented Programming

- Abstract Data Types and Classes
- Inheritance
- Encapsulation and Information Hiding

Algorithms and Data structures

- Sequences
- Lists
- Item Ordering
- Two-Dimensional Sequences
- The Minmax

Sets and Maps

- Playing Sudoku
- Sets
- Hashing
- The HashSet Class
- Solving Sudoku

Maps

- Memorization
- Correlating Two Sources of Information

Membership Structures

- Bloom Filters
- The Trie Data type

Balanced Binary Search Trees

- Binary Search Trees
- AVL Trees
- Splay Trees
- Iterative Splaying
- Recursive Splaying

B-Trees

- B-Tree Implementation
- B-Tree Insert
- B-Tree Delete

PROJECT

Reference Books :

1. Kent D. Lee, Steve Hubbard, "Data Structures and Algorithms with Python", Springer Publications.
- .

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Coding Contest - 25 Marks)

INDUSTRY ORIENTED TRAINING
(Common to CSE & IT)

Lab : 2 Periods
Exam : 3 Hrs.

Sessionals : 50
Credits : 1

Course Objectives:

Students learn about

1. Linked lists, file operations and their applications.
2. Searching and Sorting algorithms.
3. Stack & Queue operations and their applications.
4. The concepts and applications of Trees and Graphs.

Course Outcomes:

Students will be able to

1. Implement the linked lists in real time applications.
2. Apply the file handling operations.
3. Apply the Searching & Sorting algorithms.
4. Implement Stack & Queue operations.
5. Implement the concepts and applications of Trees and Graphs.

Syllabus: Industry Oriented Applications on following topics.

Linear Linked Data: Singly linked list, operations on a linked list, circular linked list, double linked list, operations on double linked list

Standard Storage: Introduction to files, file types, file modes, file functions

Searching & Sorting: Linear search and Binary search, Bubble sort, Selection sort, Insertion sort, Quick sort, Heap sort, Merge sort: Worst and Average case analysis. Decision Tree Model and (worst case) Lower Bound on Sorting. Sorting in linear time- shell sort, radix sort, bucket sort, counting sort.

Stack & Queue: Stack structure, operations. Stack using linear list data. Stack using linear linked data. Queue structure, operations. Queue using linear list data. Queue using linear linked list. Circular queues.

Non Linear Data: Tree Structure and terminology, Binary Trees, Binary Tree traversals, Applications of Binary Tree, Binary Tree Operations.

Priority queues, union-find sets, (augmented) interval trees, (augmented) balanced BSTs and binary indexed trees, Binary Indexed Tree or Fenwick tree, Segment Tree (RMQ, Range Sum and Lazy Propagation), K-D tree, Union Find Disjoint Set, Tries, Interval Tree

Graphs: Graphs and their basic properties- degree, path, cycle, subgraphs, isomorphism, Eulerian and Hamilton walks, graph coloring, planar graphs, trees. Breadth first search and connected components. Depth first search in directed and undirected graphs.

More Trees: Binary search trees, Operations on BST, balanced binary search trees, AVL trees, Red-Black trees, skip lists, hashing. Priority queues, heaps, Fibonacci heap, union-find, splay trees Interval trees, tries.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Coding Contest- 25 Marks)



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

ESTD: 1980

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

III/IV B.TECH

(With effect from 2016-2017 Admitted Batch onwards)

Under Choice Based Credit System

COMPUTER SCIENCE AND ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 CS 3101	Computer Networks	4	3	1	--	4	30	70	100
B16 CS 3102	Web Technologies	4	3	1	--	4	30	70	100
B16 CS 3103	Formal Languages & Automata Theory	4	3	1	--	4	30	70	100
B16 CS 3104	Database Management Systems	4	3	1	--	4	30	70	100
#ELE-I	Elective-I	4	3	1	--	4	30	70	100
B16 CS 3110	Database Management Systems Lab.	2	--	--	3	3	50	50	100
B16 CS 3111	Application Development Lab.	2	--	--	3	3	50	50	100
B16 ENG 3102	Verbal & Quantitative Aptitude-I	2	5	--	--	5	100	--	100
#M-I	MOOCS-I	2	4	--	--	4	100	--	100
#M-II	MOOCS-II	2	4	--	--	4	100	--	100
Total		30	28	5	6	39	550	450	1000

#ELE-I	B16 CS 3105	Embedded Systems
	B16 CS 3106	Bio-Informatics
	B16 CS 3107	Image Processing
	B16 CS 3108	Application Development Using Java
#M-I	B16 ENG 3104	Advanced Coding
	B16 CS 3112A	Digital Image Processing
	B16 CS 3112B	E-Business
	B16 CS 3112C	Introduction to parallel Programming in Open MP
#M-II	B16 CS 3113A	Social Networks
	B16 CS 3113B	Introduction to Internet of Things
	B16 CS 3113C	Block chain Architecture Design and Use Cases
	B16 CS 3113D	Introduction to R Software

COMPUTER NETWORKS

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. Be familiar with basic Computer Network Terminology
2. Able to distinguish between Switching Techniques
3. To Know various types of Networks and Protocols
4. To Know the causes and effects of Congestion
5. Familiar with Internetwork operation.
6. In depth knowledge of various Transport Protocols.

Course Outcomes:

At the end of the course students can able to

1. Distinguish between Circuit Switching and Packet Switching approaches
2. Apply various concepts of ATM networks
3. Distinguish between various types of Networks
4. Apply various Congestion Control Techniques
5. Know Internetwork Operation.
6. Know various Connection Oriented Transport Control Mechanisms.

SYLLABUS

Switched Networks, Circuit-Switching Networks, Circuit Switching Concepts, Soft switch Architecture, Packet Switching Principles, X.25, Frame Relay

Asynchronous Transfer Mode: Protocol Architecture, ATM Logical Connections, ATM Cells, ATM Service Categories, Routing in Switched Networks

Congestion Control in Switched Data Networks: Effects of Congestion, Congestion Control, Traffic management, Congestion Control in Packet Switched networks, Principles of Cellular Networks

Local Area Network Overview: Background, Topologies and transmission media, LAN Protocol Architecture, Bridges, Layer 2 and Layer 3 Switches

High Speed LANs: The Emergence of High Speed LANs, Ethernet

Wireless LANs: Overview, Wireless LAN Technology, IEEE802.11 Architecture and Services. Internet Protocols: Basic protocol Functions, Principles of Internetworking, Connectionless Internetworking, Internet Protocol

Internet Operation: Multicasting, Routing Protocols: Autonomous Systems & Approaches to Routing

Transport protocols: Connection oriented Transport Protocol Mechanisms: Reliable Sequencing Network Service, TCP: TCP Services, TCP Header Format, TCP Mechanisms, UDP

Text Book:

1. Data and Computer Communications, William Stallings 7th Edition, Pearson Education, 2004

Reference Books:

1. Data Communications and Networking, Behrouz A. Forouzan, 3rd Edition, TMH, 2004.
2. Computer Networking: A Top-Down Approach Featuring the Internet, James F. Kurose and Keith W. Ross, 2nd Edition, Pearson Education, 2002.
3. Computer Networks, Andrew S. Tanenbaum, 4th Edition, Pearson Education, 2003.
4. An Engineering Approach to Computer Networking, S. Keshav, Pearson Education, 1997.
5. Computer Networks and Internets with Internet Applications, Douglas E. Comer, 4th Edition, Pearson Education, 2003.

WEB TECHNOLOGIES

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

On completing this course student will be able to

1. Understand the principles of Web based application development.
2. Design dynamic content in Web Pages using JavaScript.
3. Understanding the concepts of java Servlets, java Server Pages and design applications using them.
4. Understand the concepts of Component development and design applications by establishing connections to Databases

Course Outcomes :

1. Students will be able to construct web based applications and Identify where data structures are appearing in them.
2. Students will be able to connect java programs to different databases.
3. Students will be able to develop EJB programs

SYLLABUS

Introduction to HTML , Core Elements , Links and Addressing, Images , Text , Colors and Background, Lists, Tables and Layouts , Frames, Forms , Cascading Style Sheets.

Introduction to Java Scripts, Elements of Objects in Java Script, Dynamic HTML with Java Script.

Document type definition, XML Syntax, XML Schemas, Document Object model, Presenting XML, Using XML Processors .

Introduction to Servlet, Servlet Life Cycles, Servlet Basics, Tomcat Web Server, Configuring Apache Tomcat, Handling Client Request and Response, Handling Cookies, Session Tracking.

Introduction to PHP, Language Basics, Functions, Strings, Arrays.

Web Techniques, Data bases, Graphics, PDF, Dates and Times.

MYSQL Installation, Accessing MySQL Using PHP, Form Handling, Cookies, Sessions, and Authentication, Tables, Inserting Data into Tables, Selecting Data from a Table, Updating Table, Deleting data from Table, Webpage creation.

Text Books:

1. Web Programming, building internet applications, 2nd Ed., Chris Bates, Wiley Dreamtech
2. The complete Reference HTML and DHTML, Thomas A. Powey
3. Learning Php, Mysql, Robin Nixon
4. Programming Php, Kevin Tatroe, Peter MacIntyre & Rasmus Lerdorf foreword by Michael Bourque.

Reference Books:

1. Internet , World Wide Web , How to program, Dietel , Nieto, PHI/PEA Web Technologies, Godbole, kahate, 2nd Ed., TMH

FORMAL LANGUAGES AND AUTOMATA THEORY

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To introduce the concepts in automata theory to design grammars and recognizers for different formal languages.
2. To Employ finite state machines to solve problems in computing.
3. To introduce finite state machines ,context free grammars, Push Down Automata and Turing Machines and their properties as the basis for the formal expressivity of computer languages for solving linguistic decision problems.
4. To understand the concepts of tractability and decidability, the concepts of NP-completeness and NP-hard problem and also the challenges for Theoretical Computer Science and its contribution to other sciences.

Course Outcomes:

1. Ability to identify analytically and intuitively for problem- solving situations in related areas of theory in computer science by using Finite-State Machines, Deterministic Finite-State Automata, Nondeterministic Finite-State Automata.
2. Ability to describe the language accepted by an automata or generated by a regular expression or a context-free grammar.
3. Ability to describe the functioning of Pushdown Automata.
4. Ability to describe the functioning of Turing Machines.

SYLLABUS

Definitions of alphabet, strings, language, Definition of finite automata, Deterministic Finite Automata, Non-Deterministic Finite Automata, representations in mathematical diagram, tabular etc., id of finite automata, design of finite automata from the given description, Conversion of NFA to DFA, Non-Deterministic Finite Automata with ϵ -transitions, elimination of ϵ -transitions, Finite Automata with Output.

Definition of regular expression and Regular Sets, construction of regular expression from the given description, minimization of regular expressions, Conversion of regular Expression to finite automata, finite automata to regular Expression, construction of regular expression for the given finite state machine- a systematic way using Arden's theorem, discussion of pumping lemma on regular sets and Applications of pumping lemma for regular sets, closure properties and Decision Algorithms for Regular Sets, Applications of finite automata.

Definition of Context-Free Grammars, Construction of CFG from the given description, Derivation of Strings from CFG, Derivation Trees, Ambiguous Grammar, Simplification of CFG, CNF and GNF, Pumping Lemma for CFL, Applications of Pumping lemma for CFL, Closure Properties of CFL and Decision Algorithms.

Definition of push down automata, types of push down automata, push down automata to context free grammar, context free grammar to push down automata, design methodology of various push down automata's from the given language description, push down automata by empty stack, push down automata by final states.

Definition of Turing machine, ways of representing Turing machine's- tabular form, diagram, id of Turing machine, design of Turing machine from the given language description, types of Turing machine, types of grammar, types of automata, Relationship between different languages and automata, halting problem, church's thesis, universal Turing machine, Gödel number, definitions of recursive functions- prf, rf, decidability.

NOTE : Theorem proofs are eliminated

Text Book

1. Introduction to automata theory, languages and computation, John.E.H.P croft/ Rajeev Motwani & JD Ullman—pearson education- III edition

Reference Books

1. Theory of computation, K.L.P.Mishra and N.Chandrasekhar, PHI
2. Theory of computation, formal languages and automata theory, G P Saradhi Varma, B.Thirupathi Rao –Sci Tech publications.

DATABASE MANAGEMENT SYSTEMS

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To learn the evolution of DBMS Versus File systems, data models, and layers of abstraction.
2. To understand conceptual and Logical aspects of database design.
3. To learn formal language specifications.
4. To learn commercial query languages.
5. To understand concurrency control, recovery management, and other related issues

Course Outcomes:

1. Generalize the basic concepts of DBMS
2. Discover the relational model and formal query languages
3. Prepare SQL commands for definition, constructing and manipulation of databases
4. Apply conceptual and logical database design
5. Apply normalization on tables
6. Schedule concurrent transactions using locking protocols and protocols without locking
7. Recover transactions
8. Examine database connection techniques

SYLLABUS

Introduction: File System versus DBMS, Advantages of DBMS, Types of DBMS, The Relational model, Levels of abstraction, Data Independence, Introduction to Transaction management, Structure of a DBMS.

ER Model: Basics of Database Design, ER Diagrams, Entities, Attributes and Entity Sets, Relationships & Relationship Sets, Additional Features of the ER Model: IS A relationship, Aggregation

Database Design and the Relational Model: Conceptual Design with ER Model, Introduction to the Relational Model, Integrity Constraints over Relations, Enforcing Integrity Constraints, Logical Database Design: ER to Relational.

Relational Algebra and SQL: Relational Algebra, The form of a Basic SQL Query, Nested Queries, Joins, UNION, INTERSECT and EXCEPT, Aggregate Operators, Null Values, Complex Integrity Constraints in SQL, Introduction to Views, Destroying/ Altering Tables and Views, Triggers, Embedded SQL, Dynamic SQL, JDBC.

Normalization: Schema Refinement and Normal Forms, Introduction to Schema Refinement, Functional Dependencies, Reasoning about FD's, Normal Forms, Properties of Decomposition, Normalization, Other kinds of Dependencies: Multivalued and Join dependencies.

Transaction Management and Concurrency Control: The ACID Properties, Transactions & Schedules, Concurrent Execution of Transactions, Lock-Based Concurrency Control, 2PL, Serializability and Recoverability, Introduction to Lock Management, Lock Conversions, Dealing with Deadlocks, Specialized Locking Techniques, Concurrency Control without Locking.

Crash Recovery: Introduction to ARIES, The Log, Other Recovery-Related Structures, The Write-Ahead Log Protocol, Check pointing, Recovering from a System Crash, Media Recovery.

Text Book:

1. Database Management Systems, Raghu Ramakrishnan, Johannes Gehrke, 4th Edition, McGraw-Hill

Reference Books:

1. Database System Concepts, A. Silberschatz, H. Korth, 5th Edition, McGraw-Hill
2. Fundamentals of Database Systems 5th edition. Ramez Elmasri, Shamkant B.Navathe, Pearson Education,2008
3. Database Management System Oracle SQL and PL/SQL,P.K.Das Gupta, PHI, 2009.

EMBEDDED SYSTEMS (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs	Credits	: 4

Course Objectives:

1. To provide in-depth knowledge about Embedded Processor, its hardware and software.
2. To explain the programming concepts, Embedded programming in C & Assembly language.
3. To explain various architectures of Embedded Systems.
4. To explain Real Time Operating Systems.
5. To explain about Inter task communication.
6. To explain about Embedded software development tools.

Course Outcomes:

On successful completion of the course a student will able

1. To describe the differences between general computing system and Embedded System.
2. To recognize the classification of Embedded System.
3. To understand various architectures of Embedded System.
4. To design Real Time Embedded System using the concepts of RTOS.
5. To load embedded software on Host machine.
6. To test Host machine.

SYLLABUS

Examples of Embedded Systems – Typical Hardware – Memory – Microprocessors – Busses – Direct Memory Access – Introduction to 8051 Microcontroller – Architecture-Instruction set – Programming- Microprocessor Architecture – Interrupt Basics – The Shared-Data problem – Interrupt Latency.

Round-Robin Architecture - Round-Robin with Interrupts Architecture - Function-Queue-Scheduling Architecture – Real-Time Operating Systems Architecture – Selection of Architecture.

Tasks and Task States – Tasks and Data – Semaphores and Shared Data – Semaphore Problems – Semaphore variants, Message Queues – Mailboxes – Pipes – Timer Functions – Events – Memory Management – Interrupt Routines in RTOS Environment.

RTOS design – Principles – Encapsulation Semaphores and Queues – Hard Real-Time Scheduling Considerations – Saving Memory Space – Saving Power.

Host and Target Machines – Linker/Locator for Embedded Software- Getting Embedded Software into the Target System, Testing on your Host Machine – Instruction Set Simulators – Laboratory Tools used for Debugging.

Text Books:

1. David A. Simon, An Embedded Software Primer, Pearson Education, Inc., 1999.
2. Sriram V Iyer and Pankaj Gupta, Embedded Real Time Systems programming, TMH, 2004

Reference Books:

1. Frank Vahid/ Tony Givargis, Embedded Systems Design – A Unified Hardware/Software Introduction, John Wiley & Sons, 2006.
2. Raj Kamal, Embedded Systems, Architecture, Programming and Design, Tata McGraw-Hill Education, 2011

BIO-INFORMATICS
(Elective-I)

Theory :3 Periods
Tutorial :1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To Know the basic information related to Biological sequences, Genome Project, Pattern Recognition
2. To have an understanding of Protein Databases and Genome Databases
3. To perform DNA sequence Analysis
4. To use various sequence analysis packages

Course Outcomes:

At the end of the course the students can able to

1. Remember various bio-informatic terminology and Biological sequences.
2. Use various Genome and Protein databases.
3. Analyse various DNA sequences.
4. Use various tools for Sequence analysis.

SYLLABUS**Introduction:**

Definitions, Sequencing, Biological sequence/structure, Genome Projects, Pattern recognition, a prediction, Folding problem, Sequence Analysis, Homology and Analogy.

Protein Information Resources

Biological databases, Primary sequence databases, Protein Sequence databases, Secondary databases, Protein pattern databases, and Structure classification databases.

Genome Information Resources

DNA sequence databases, specialized genomic resources

DNA Sequence analysis

Importance of DNA analysis, Gene structure and DNA sequences, Features of DNA sequence analysis, EST (Expressed Sequence Tag) searches, Gene hunting, Profile of a cell, EST analysis, Effects of EST data on DNA databases

Pair wise alignment techniques

Database searching, Alphabets and complexity, Algorithm and programs, Comparing two sequences, sub-sequences, Identity and similarity, The Dotplot, Local and global similarity, different alignment techniques, Dynamic Programming, Pair wise database searching.

Multiple sequence alignment

Definition and Goal, The consensus, computational complexity, Manual methods, Simultaneous methods, Progressive methods, Databases of Multiple alignments and searching

Secondary database searching

Importance and need of secondary database searches, secondary database structure and building a sequence search protocol

Analysis packages

Analysis package structure, commercial databases, commercial software, comprehensive packages, packages specializing in DNA analysis, Intranet Packages, Internet Packages.

Text Books:

1. Introduction to Bioinformatics, T K Attwood & D J Parry-Smith Addition Wesley Longman.
2. Bioinformatics- A Beginner's Guide, Jean-Michel Claverie, Cerdric Notredame WILEY dreamlech India Pvt. Ltd

Reference Books:

1. Introduction to Bioinformatics, Arthur M.Lesk, OXFORD publishers (Indian Edition)

IMAGE PROCESSING (Elective-I)

Theory : 3 Periods

Tutorial : 1 Period

Exam : 3 Hrs.

Sessionals : 30

Ext. Marks : 70

Credits : 4

Course Objectives:

1. To explain fundamentals of Image processing concepts.
2. To provide mathematical foundation of image enhancement , image compression and image segmentation.
3. To explain the students about Morphology and its applications in image processing.

Course Outcomes:

1. Ability to develop algorithms for fundamental concepts in Image processing.
2. Ability to perform image enhancement , image compression and image segmentation using various methods.
3. Ability to implement Image transformation techniques

SYLLABUS

Fundamentals of Image Processing: Image Acquisition, Image Model, Sampling, Quantization, Relationship Between Pixels, Distance Measures, Connectivity , Image Geometry, Photographic Film. Histogram: Definition, Decision Of Contrast Basing On Histogram, Operations Basing on Histograms Like Image Stretching, Image Sliding, Image Classification. Definition and Algorithm of Histogram Equalization.

Image Enhancement in Spatial Domain : Arithmetic and Logical Operations, Pixel or Point Operations, Size Operations; Smoothing Filters-Mean, Median, Mode Filters – Comparative Study;

Edge enhancement in spatial domain: Edge enhancement filters, Directorial Filters, Sobel, Laplacian, Robert, KIRSCH Homogeneity & DIFF filters, PREWITT Filter, Contrast based edge enhancement techniques, Comparative study, Low pass filters, High pass filters, Sharpening filters, Comparative study, Color fundamentals and color model

Image Compression: Run Length Encoding, modified run length encoding, Contour Coding, Huffman Code, Compression Due to Change in Domain, Compression Due to Quantization Compression at the Time of Image Transmission. Brief Discussion on:- Image Compression Standards.

Image Segmentation: Definition of segmentation, Characteristics of Segmentation, Detection of Discontinuities, Thresholding. Pixel Based Segmentation Method. Region Based Segmentation Methods, Segmentation by Pixel Aggregation, Segmentation by Sub Region Aggregation, Histogram Based Segmentation, Spilt and Merge Technique, Segmentation of moving objects

Morphology: Dilation, Erosion, Opening, Closing, Hit-And-Miss Transform, Thinning, Thickening, Skeletons , Pruning Extensions to Gray – Scale Images Application of Morphology in I.P.

Image Transforms : A Detail Discussion On Fourier Transform, DFT,FFT, Properties of Fourier transform, WALSH Trans Form , WFT, HADAMARD Transform, DCT Image Enhancement in Frequency Domain: Design of Low Pass, High Pass, EDGE Enhancement, Smoothing Filters in Frequency Domain. Butter Worth Filter, Homomorphic Filters in Frequency Domain Advantages of Filters in Frequency Domain, Comparative Study of Filters in Frequency, Domain and Spatial Domain.

Text Book:

1. Digital Image Processing, Rafael C. Gonzalez And Richard E. Woods, Addison Wesley

Reference Books:

1. Fundamentals Of Electronic Image Processing By Arthyr – R – Weeks, Jr.(PHI).
2. Image Processing, Analysis, And Machine Vision By Milan Sonka Vaclan Halavac Roger Boyle, Vikas Publishing House.
3. Digital Image Processing, S. Jayaraman, S. Esakkirajan& T. Veera Kumar, TMH Fundamentals of Digital Image Processing, Chris Solomon, Tobi Breckon, Wiley-Blackwell

APPLICATION DEVELOPMENT USING JAVA (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Study of object oriented programming.
2. Learn about web based applications such as AWT components.
3. Study of multitasking by using multithreading concept.
4. Learn about network programming and applications development.
5. Students will have the ability to develop Web Applications using JSP.
6. Students will have knowledge about types of JDBC and they able to access database using JSP.
7. Getting the student to be well trained in Advanced Java Programming skills for an easy entry in the IT Industry.

Course Outcomes:

1. Able to do projects for web based and internet applications.
2. Understand multitasking and multiprogramming development
3. Able to do network programming.
4. Able to Construct Web application using Java Server Pages

SYLLABUS

Multithreading and AWT:

Thread, life cycle of Thread, Separate classes that implement Runnable, Inner classes that implement Runnable, Thread Priorities, Race conditions and synchronization

Basic **AWT Components**, container, Canvas, Panel, Frame, Processing events in GUI controls, Basic AWT user interface controls (Button, checkbox, CheckboxGroup, Scroll Bars , TextField, TextArea radio button, list box). **Event-handling:** Handling events with classes, Handling events by implementing interfaces, Organizing Windows with Layout Managers (Flow Layout, Border Layout, Card Layout, Grid Layout, GridBagLayout)

Java Swings

The Origins of Swing, Swing Is Built on the AWT, Two Key Swing Features, Components and Containers, the Swing Packages, Event Handling, Exploring Swing

Database Access

Structure of JDBC, Databases and Drivers, Connecting to the Database, The JDBC Classes for Creating a Connection, Basic Database Access, SQL Data types and Java Data types, Scrollable Result Sets, The JDBC Support Classes, Prepared Statements, JAVA Database connection program for MS Accesss, Oracle and MySQL

JSP

Introduction to JSP, Tomcat Server & Testing Tomcat. Displaying Values Using an Expression to Set an Attribute, Declaring Variables and Methods, Requests, and Users Passing Control and Data between Pages – Sharing Session and Application Data , JSP Program for Database Access

Input /Output , NIO and Network Programming:

The **I/O** Classes and Interfaces, I/O Exceptions, The Stream Classes, The Byte Streams, The Character Streams, **NIO** Fundamentals

The Networking Classes and Interfaces, InetAddress, TCP/IP Client Sockets, URLConnection, HttpURLConnection, TCP/IP Server Sockets, DatagramSockets , DatagramSocket ,DatagramPacket

Text Books:

1. The Complete Reference, Java Eighth Edition, Herbert Schildt, Oracle Press (Chapters: 11,19,20, 22,23,24,29,30)
2. Java Server Pages , Hans Bergstan, Oreilly, 2nd Edition, August 2002 (Chapters: 1-10)
3. Database Programming with JDBC and Java, George Reese, Second Edition, Published by O'Reilly & Associates (Chapters: 3,4)

Reference Books:

1. Internet and World wide web- How to program, Dietel and Nieto, Pearson.
2. Java server pages, Pekowsky, Pearson
3. Beginning Web Programming, Jon Duckett ,Wrox, Wiley

Code: B16 CS 3110

DATABASE MANAGEMENT SYSTEMS LAB

Lab : 3 Periods	Sessionals : 50
Exam : 3 Hrs.	Ext. Marks : 50
	Credits : 2

Course Objectives:

1. To introduce to a commercial DBMS such as ORACLE.
2. To learn and practice SQL commands for schema creation, data manipulation.
3. To learn conceptual database design based on a case study.
4. To learn Logical database design based on a case study.
5. To apply database design stages by a case study.

Course Outcomes:

1. The student is exposed to a commercial RDBMS environment such as ORACLE.
2. The student will learn SQL commands for data definition and manipulation.
3. The student applies conceptual design.
4. The student applies Logical data base design.
5. The student takes up a case study and applies the design steps.

SYLLABUS

Features of two commercial RDBMS package such as ORACLE/DB2, MS Access, MYSQL & Structured Query Language (SQL) used with the RDBMS.

- I. Laboratory Exercises Should Include
 - a Defining Schemas for Applications
 - b Creation of Database
 - c Writing SQL Queries
 - d Retrieve Information from Database
 - e Creating Views
 - f Creating Triggers
 - g Normalization up to Third Normal Form
 - h Use of Forms
 - i Report Writing
- II Some sample applications are given below:
 - a Accounting Package for Shops
 - b Database Manager for Magazine Agency or Newspaper Agency
 - c Ticket Booking for Performances
 - d Preparing Greeting Cards & Birthday Cards
 - e Personal Accounts - Insurance, Loans, Mortgage Payments, Etc.
 - f Doctor's Diary & Billing System
 - g Personal Bank Account
 - h Class Marks Management

- i. Hostel Accounting
- j. Video Tape Library
- k. History of Cricket Scores
- l. Cable TV Transmission Program Manager
- m. Personal Library
- n. Sailors Database
- o. Suppliers and Parts Database

Reference Books:

1. Oracle Database 11g The Complete Reference, McGraw Hill Professional, 2008.
2. Database Systems Using Oracle, Nilesh Shah, Pearson Education India, 2016.
3. Introduction to SQL, Rick F Vander Lans, Fourth Edition, Pearson Education, 2007

APPLICATION DEVELOPMENT LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. Practice object-oriented programs and build java applications.
2. Implement java programs for establishing interfaces.
3. Implement sample programs for developing reusable software components.
4. Create database connectivity in java and implement GUI applications.

Course Outcomes:

1. Students will be able to create sophisticated applications to Manipulate Window Interfaces Using Swing Objects, Graphics Objects and working with Streams and File Input /Output.
2. Develop Swing-based GUI.
3. Develop client/server applications and TCP/IP socket programming.
4. Update and retrieve the data from the databases using SQL.

SYLLABUS**CYCLE-1 :**

1. Write a java program that implements thread class methods.
2. Write a java program that handles all mouse events and shows the event name at the center of the window when a mouse event is fired. Use adapter classes.
3. Write a java program to demonstrate the key event handlers.
4. Write a java program to display the table using labels in grid layout.
5. Write a java program to display the Header information of the given URL
6. Write a java program to split the given URL.
7. Write a java program by using JDBC to execute a SQL query for a database and display the results.
8. Implementing one-one chat Application without threads
9. Write a java program for Datagram server and Client interaction as per given below.
 - i). A program to create Datagram server to send a message to client.
 - ii). A program to create Datagram client to receive the message sent by the server
10. Write JSP Program to validate user name and password on server side?
11. Write an online book purchase application using JSP. Consider a login validation page and one billing page for bill payment process. Assume any information if required?
12. Write JSP Program for Database Access.

CYCLE-2 :

Each batch should develop one project out of this list using **HTML,CSS,JS,PHP and MYSQL**

1. Design Airlines Ticket Reservation System
2. Online Shopping
3. Design Library Information system.
4. Design Gram Panchayat Information system for House tax, water tax, wealth tax, Library tax collection, phone bill, Electricity bill collection.
5. Design student information system portal which maintain attendance, marks etc.
6. Design online examination system.
7. Event management System.
8. Car Rental System.
9. Cinema Booking System.
10. Hotel Management System.
11. Complaint management System.
12. Online voting system.
13. Student Result System.
14. Car Comparison System Project.
15. Selling your old stuff.
16. Aquaculture Updates.
17. Timesheet using PHP
18. Online Help Desk using PHP
19. Online marriage beuro system
20. EAMCET web counseling

Reference Books:

1. P. J. Deitel, H. M. Deitel, “Java for Programmers”, Pearson Education, PHI, 4th Edition, 2007.
2. P. Radha Krishna, “Object Oriented Programming through Java”, Universities Press, 2nd Edition, 2007
3. Bruce Eckel, “Thinking in Java”, Pearson Education, 4th Edition, 2006.
4. Sachin Malhotra, Saurabh Chaudhary, “Programming in Java”, Oxford University Press, 5th Edition, 2010.

VERBAL & QUANTITATIVE APTITUDE – I
(Common to All Branches)

Theory	: 5 Periods(VA-3+QA-2)	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Part-A: Verbal and Soft Skills-I

Course objectives:

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS

Grammar: (VA)

Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)

Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)

Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).

Usage: (VA)

Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:

Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners’ Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal’s books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cmbridge University Press.
5. Soft Skills & Employability Skills by SaminaPillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from ‘IMS, Career Launcher and Time’ institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by RashmiBansal
8. I have a Dream by RashmiBansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion

Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance

Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions

Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms

Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1. www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

**ADVANCED CODING
(Common for CSE & IT)
(MOOCS-I)**

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objectives

1. To understand the basics of modular programming
2. To learn about ADT, Linked Lists and Templates
3. To investigate different methods to find time complexities
4. To learn about Java collections and Libraries

Course Outcomes

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of modular programming
2. Acquire Programming knowledge on linked lists
3. Acquire coding knowledge on ADT
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on Java libraries and Collections

SYLLABUS

UNIT I Review Coding essentials and modular programming

Introduction to Linear Data, Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding.

Introduction to modular programming: Formation of methods, Methods: Signature and definition, Inter-method communication, Data casting & storage classes, Recursions

UNIT II Linear Linked Data

Introduction to structure pointer, Creating Links Basic problems on Linked lists, Classical problems on linked lists. Circular Linked lists, Operations on CLL, Multiple links, Operations on Doubly linked lists

UNIT III Abstract Data-structures

Stack data-structure, Operations on stack, Infix/Prefix/Post fix expression evaluations, Implementation of stack using array, Implementation of stack using linked lists.

Queue data-structure: Operations on Queues, Formation of a circular queue, Implementation of queue using stack, Implementation of stack using array, Implementation of stack using linked lists

UNIT IV Running time analysis of code and organization of linear list data

Code evaluation w.r.t running time, Loop Complexities, Recursion complexities, Searching techniques: sequential Vs. binary searching.

Organizing the list data, Significance of sorting algorithms, Basic Sorting Techniques: Bubble sort, selection sort, Classical sorting techniques: Insertion sort, Quick sort, Merge sort.

UNIT V Standard Library templates and Java collections

Introduction to C++ language features, Working on STLs, Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.

Note: This course should focus on Problems

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by Robat Laphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>

**DIGITAL IMAGE PROCESSING
(MOOCS-I)****Theory : 4 Periods**
Exam : 3 Hrs.**Sessionals : 100**
Credits : 2

- Introduction and signal digitization
- Pixel relationship
- Camera models & imaging geometry
- Image interpolation
- Image transformation
- Image enhancement I
- Image enhancement II
- Image enhancement III
- Image restoration I
- Image restoration II & Image registration
- Colour image processing
- Image segmentation
- Morphological image processing
- Object representation, description and recognition

Suggested Reading

1. Digital Image Processing by Rafael C Gonzalez & Richard E Woods, 3rd Edition
2. Fundamentals of Digital Image Processing by Anil K Jain
3. Digital Image Processing by William K Pratt

WEB LINK:**https://onlinecourses.nptel.ac.in/noc18_ee40/preview**

**E-BUSINESS
(MOOCS-I)****Theory : 4 Periods**
Exam : 3 Hrs.**Sessionals : 100**
Credits : 2

Introduction to E-Business

Making Functional Areas E-Business Enabled : Value chain and supply chain, inter and intra organizational business processes, ERP

Making Functional Areas E-Business Enabled : E-Procurement

Making Functional Areas E-Business Enabled : E-marketing, E-Selling, E-Supply Chain Management

Technologies for E-Business: Internet and Web based system

Technologies for E-Business: Security and payment systems

Technologies for E-Business: Supply chain integration technologies (EDI, RFID, Sensors, IoT, GPS, GIS)

Technologies for E-Business: Supply chain integration technologies (Web services and cloud)

Decision Support in E-Business: Web analytics

Decision Support in E-Business: Customer behavior modeling

Decision Support in E-Business: Auctions

Decision Support in E-Business: Recommender systems

SUGGESTED READING

Management Information Systems: Managing the Digital Firm, Laudon and Laudon, Pearson

1. Scaling for E-Business, Menasce & Almeida, PHI
2. eBusiness & eCommerce – Managing the Digital Value Chain, Meier & Stormer, Springer
3. eBook is available in springerlink.com
4. Some reference books, Internet Resources, and Research Papers

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_mg35/preview

**INTRODUCION TO PARALLEL PROGRAMMING IN OPEN MP
(MOOCS-II)**

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

- Single Processor Architecture and Basic OpenMP constructs & functions
- More OpenMP constructs & functions
- Basic Linear Algebra using OpenMP and OpenMP tasks
 - (Assignment 1: Programming assignment to implement and evaluate blocked matrix multiply in OpenMP)
- Critical Sections, locks and Matrix Factorization using OpenMP
 - (Assignment 2: Programming assignment to implement and evaluate task based algorithm for a BLAS routine).

Suggested Reading

1. Introduction to Parallel Computing (Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar)

WEB LINK:

<https://computing.llnl.gov/tutorials/openMP/>

**SOCIAL NETWORKS
(MOOCS-II)**

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

SYLLABUS

- Introduction
- Handling Real-world Network Datasets
- Strength of Weak Ties
- Strong and Weak Relationships (Continued) & Homophily
- Homophily Continued and +Ve / -Ve Relationships
- Link Analysis
- Cascading Behaviour in Networks
- Link Analysis (Continued)
- Power Laws and Rich-Get-Richer Phenomena
- Power law (contd..) and Epidemics
- Small World Phenomenon
- Pseudocore (How to go viral on web)

SUGGESTED READING:

1. Networks, Crowds and Markets by David Easley and Jon Kleinberg, Cambridge University Press, 2010.
2. Social and Economic Networks by Matthew O. Jackson, Princeton University Press, 2010.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_cs56/preview

**INTRODUCTION TO INTERNET OF THINGS
(MOOCS-II)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

SYLLABUS

- Introduction to IoT: Part I, Part II, Sensing, Actuation, Basics of Networking: Part-I
- Basics of Networking: Part-II, Part III, Part IV, Communication Protocols: Part I, Part II
- Communication Protocols: Part III, Part IV, Part V, Sensor Networks: Part I, Part II
- Sensor Networks: Part III, Part IV, Part V, Part VI, Machine-to-Machine Communications
- Interoperability in IoT, Introduction to Arduino Programming: Part I, Part II, Integration of Sensors and Actuators with Arduino: Part I, Part II
- Introduction to Python programming: Part I, Part II, Introduction to Raspberry Pi: Part I, Part II
- Implementation of IoT with Raspberry Pi: Part I
- Implementation of IoT with Raspberry Pi: Part II, Part III, Introduction to SDN: Part I, Part II, SDN for IoT: Part I
- SDN for IoT: Part II, Data Handling and Analytics: Part I, Part II, Cloud Computing: Part I, Part II
- Cloud Computing: Part III, Part IV, Part V, Sensor-Cloud: Part I, Part II
- Fog Computing: Part I, Part II, Smart Cities and Smart Homes: Part I, Part II, Part III
- Connected Vehicles: Part I, Part II, Smart Grid: Part 1, Part II, Industrial IoT: Part I
- Industrial IoT: Part I, Case Study: Agriculture, Healthcare, Activity Monitoring: Part I, Part II

SUGGESTED READING

1. "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", by Pethuru Raj and Anupama C. Raman (CRC Press)
2. "Internet of Things: A Hands-on Approach", by Arshdeep Bahga and Vijay Madisetti

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_cs46/preview

BLOCKCHAIN ARCHITECTURE DESIGN AND USECASES (MOOCS-II)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

SYLLABUS

Chapter	Lecture	Topic	Concepts
1.	1	Introduction to Blockchain – I (Basics)	What is Blockchain Public Ledgers Blockchain as public ledgers
	2	Introduction to Blockchain – II (History)	Bitcoin Blockchain 2.0 Smart Contracts
	3	Introduction to Blockchain – III (Architecture)	Block in a Blockchain Transactions Distributed Consensus
	4	Introduction to Blockchain – IV (Conceptualization)	The Chain and the Longest Chain Cryptocurrency to Blockchain 2.0 Permissioned Model of Blockchain
	5	Basic Crypto Primitives – I	Cryptographic Hash Function Properties of a hash function Hash pointer and Merkle tree
2.	6	Basic Crypto Primitives – II	Digital Signature Public Key Cryptography A basic cryptocurrency
	7	Bitcoin Basics – I	Creation of coins Payments and double spending FORTH – the precursor for Bitcoin scripting
	8	Bitcoin Basics – II	Bitcoin Scripts Bitcoin P2P Network
	9	Bitcoin Basics – III	Transaction in Bitcoin Network Block Mining Block propagation and block relay
	10	Distributed Consensus	Why Consensus Distributed consensus in open environments Consensus in a Bitcoin network
3.	11	Consensus in Bitcoin – I (The Basics)	Bitcoin Consensus Proof of Work (PoW) – basic introduction Hashcash PoW
	12	Consensus in Bitcoin – II (PoW and Beyond)	Bitcoin PoW Attacks on PoW and the monopoly problem Proof of Stake, Proof of Burn and Proof of Elapsed Time
	13	Consensus in Bitcoin – III (The Miners)	The life of a Bitcoin Miner Mining Difficulty Mining Pool
	14	Permissioned Blockchain – I (Basics)	Permissioned model and use cases Design issues for Permissioned blockchains Execute contracts State machine replication
	15	Permissioned Blockchain – II (Consensus)	Consensus models for permissioned blockchain Distributed consensus in closed environment Paxos

Chapter	Lecture	Topic	Concepts
4.	16	Permissioned Blockchain – III (RAFT Consensus)	1) RAFT Consensus 2) Byzantine general problem
	17	Permissioned Blockchain – IV (Byzantine General Problem)	1) Byzantine fault tolerant system 2) Lamport-Shostak-Pease BFT Algorithm
	18	Permissioned Blockchain – V (Practical Byzantine Fault Tolerance)	1) BFT over Asynchronous systems 2) Practical Byzantine Fault Tolerance 3) Three phase commit 4) View Change
	19	Blockchain for Enterprise – Overview	1) Concepts and benefits of block chain for enterprise 2) The Hyperledger Project
	20	Blockchain Components and Concepts	1) Actors in a Blockchain 2) Components in Blockchain design 3) Ledger in Blockchain
5.	21	Hyperledger Fabric – Transaction Flow	1) Fabric Architecture 2) Transaction flow in Fabric
	22	Hyperledger Fabric Details	1) Ordering Services 2) Channels in Fabric 3) Fabric Peer and Certificate Authority
	23	Fabric – Membership and Identity Management	1) Organization and Consortium Network 2) Membership Service Provide 3) Transaction Signing
	24	Hyperledger Fabric Network Setup	1) Steps for network setup 2) Endorsement policies
	25	Fabric Demo on IBM Blockchain Cloud – I	1) Setup Blockchain networks 2) Experience blockchain network as different organizations 3) Deploy a simple application on IBM cloud
6.	26	Fabric Demo on IBM Blockchain Cloud – II	1) Deploy a simple application on IBM Cloud 2) Marbles (asset transfer) 3) Example smart contract code, client SDK code 4) Perform blockchain transactions using a cool UI!
	27	Fabric Demo, deploy from scratch – III	1) Setup a Fabric network on your laptop or VM 2) Install and instantiate chaincode 3) Run application on the network you created
	28	Hyperledger Composer – Application Development	1) Goals of Hyperledger Composer 2) Key concepts for the business service provide 3) Key development concepts – Model files, Access control lists, Transaction processors, business network definition
	29	Hyperledger Composer – Network Administration	1) Key concepts for administrators 2) How composer maps to Fabric chaincode
	30	Blockchain Use Cases	1) Sample use cases by Industry 2) Business Problems and Participants 3) Communities in Blockchain network

Chapter	Lecture	Topic	Concepts
7.	31	Blockchain in Financial Service – I (Payments and Secure Trading)	1) Cross border payments 2) Stellar and Ripple protocols 3) Project Ubin
	32	Blockchain in Financial Service – II (Compliance and Mortgage)	1) Know Your Customer (KYC) 2) Privacy Consents 3) Mortgage over Blockchain
	33	Blockchain in Financial Service – III (Financial Trade)	1) Blockchain enabled Trade 2) We.Trade – Trade Finance Network 3) Supply Chain Financing
	34	Revolutionizing Global Trade	1) Blockchain for Trade Logistics 2) Global Trade Digitization 3) Blockchain for Container Management
	35	Blockchain in Supply Chain – I	1) Food Safety and Food Traceability 2) Supply Chain Orchestration
8.	36	Blockchain in Supply Chain – II	1) Everledger 2) The Diamond Lifecycle 3) Addressing Supply Chain Fraud through Blockchain
	37	Blockchain in Other Industries	1) Blockchain in Healthcare 2) Blockchain in Energy Markets 3) Blockchain in Media
	38	Blockchain in Government – I (Advantages)	1) Blockchain and Government 2) Preventing Cyber Crime through blockchain 3) Government Use-cases
	39	Blockchain in Government – II (Use Cases)	1) Auditing and Compliance 2) Blockchain for Defense 3) e-Estonia: A Case Study
	40	Blockchain in Government – III (Digital Identity)	1) Digital Identity and Single Sign On (SSO) 2) Principles of Digital Identity Management 3) Why Blockchain
9.	41	Blockchain in Government – IV (Hyperledger Indy)	1) Indy for Digital Identity Management 2) How Indy Works
	42	Blockchain in Government – V (Tax Payments and Land Registry Records)	1) Blockchain for Tax Payments 2) Blockchain for Managing Land Registry Records
	43	Blockchain Security – I (Overview)	1) Security properties 2) Security considerations for Blockchain 3) Intel SGX
	44	Blockchain Security – II (Membership and Access control in Fabric)	1) Identities and Policies 2) Membership and Access Control 3) Blockchain Crypto Service Providers
	45	Blockchain Security – III (Privacy in Fabric)	1) Privacy in a Blockchain System 2) Privacy through Fabric Channels 3) Smart Contract Confidentiality

Chapter	Lecture	Topic	Concepts
10.	46	Blockchain Security – III (Fabric SideDB)	1) SideDB Motivation 2) SideDB Overview
	47	Research Aspects – I (Consensus Scalability)	1) PoW vs BFT Consensus 2) Consensus Finality 3) Consensus Scalability
	48	Research Aspects – II (Bitcoin-NG)	1) Fairness and Scalability in Nakamoto Consensus 2) Bitcoin-NG: Working Principles 3) Key Blocks and Microblocks
	49	Research Aspects –III (Collective Signing)	1) Authority and Digital Signature 2) Collective Signing (CoSi) 3) Shnorr Multisignature and BLS Signatures
	50	Research Aspects – IV (Byzcoin)	1) Strong non-probabilistic consistency 2) BFT over Bitcoin – increasing scalability 3) Byzcoin Design and Performance
11.	51	Research Aspects – V (Algorand)	1) Strong Synchrony vs Weak Synchrony 2) Avoiding Forks 3) Transaction Neutrality and Frictionless Evolution
	52	Research Aspects – VI (Cross Fault Tolerance)	1) Asynchronous networks as network fault 2) Cross fault Tolerant (XFT) architecture 3) XPaxos
	53	Research Aspects – VII (Secured Multi-Party Computation)	1) Multi-Party Computation (MPC) 2) Fairness in MPC 3) MPC over Blockchain – ensuring fairness
	54	Blockchain for Science – I (Blockchain for Big Data)	1) Big Data and Big Network 2) Why Blockchain for Big data – application aspects 3) BigChainDB – The Blockchain Database
	55	Blockchain for Science – II (Blockchain and AI)	1) Data analysis over Blockchain 2) Logic over Blockchain network 3) Inferring Decisions through AI
12.	56	Comparing Ecosystems – Ethereum	1) Architecture and concepts 2) Smart contracts and EVM 3) Additional/upcoming capabilities
	57	Comparing Ecosystems – Ethereum development tools and Quorum	1) Ethereum Development Tools 2) Motivation and concepts 3) Architecture 4) Transaction processing and consensus
	58	Comparing Ecosystems – Corda Part 1& Comparing Ecosystems – Corda Part 2	1) Key features, Notaries and Oracles 2) Transactions, Transaction Flows 3) Corda ledger and smart contracts, Consensus

Suggested Reading :

1. Mastering Bitcoin: Unlocking Digital Cryptocurrencies, by Andreas Antonopoulos
2. Blockchain by Melanie Swa, O'Reilly
3. Hyperledger Fabric - <https://www.hyperledger.org/projects/fabric>
4. Zero to Blockchain - An IBM Redbooks course, by Bob Dill, David Smits - <https://www.redbooks.ibm.com/Redbooks.nsf/RedbookAbstracts/crse0401.html>

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_cs47/course

INTRODUCTION TO R SOFTWARE (MOOCS-II)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

SYLLABUS

- Basic fundamentals, installation and use of software, data editing, use of R as a calculator, functions and assignments.
- Use of R as a calculator, functions and matrix operations, missing data and logical operators.
- conditional executions and loops, data management with sequences.
- Data management with repeats, sorting, ordering, and lists.
- Vector indexing, factors, Data management with strings, display and formatting.
- Data management with display paste, split, find and replacement, manipulations with alphabets, evaluation of strings, data frames.
- Data frames, import of external data in various file formats, statistical functions, compilation of data.
- Graphics and plots, statistical functions for central tendency, variation, skewness and kurtosis, handling of bivariate data through graphics, correlations, programming and illustration with examples.

Suggested Reading

- Introduction to Statistics and Data Analysis - With Exercises, Solutions and Applications in R By Christian Heumann, Michael Schomaker and Shalabh, Springer, 2016
- The R Software-Fundamentals of Programming and Statistical Analysis -Pierre Lafaye de Micheaux, Rémy Drouilhet, Benoit Lique, Springer 2013
- A Beginner's Guide to R (Use R) By Alain F. Zuur, Elena N. Ieno, Erik H.W.G. Meesters, Springer 2009

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_cs52

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

III/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

COMPUTER SCIENCE AND ENGINEERING
II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 CS 3201	Data Warehousing & Data Mining	4	3	1	--	4	30	70	100
B16 CS 3202	Object Oriented Software Engineering	4	3	1	--	4	30	70	100
B16 CS 3203	Design and Analysis of Algorithms	4	3	1	--	4	30	70	100
B16 CS 3204	Compiler Design	4	3	1	--	4	30	70	100
B16 CS 3205	Artificial Intelligence	4	3	1	--	4	30	70	100
#ELE-II	Elective-II	4	3	1	--	4	30	70	100
B16 CS 3213	Software Engineering Mini Project Lab	2	--	--	3	3	50	50	100
B16 CS 3214	Network Programming Lab	2	--	--	3	3	50	50	100
B16 ENG 3202	Verbal & Quantitative Aptitude-II	2	5	--	--	5	100	--	100
#M-III	MOOCS-III	2	4	--	--	4	100	--	100
Total		32	27	6	6	39	480	520	1000

#ELE-II	B16CS3206	Cloud Computing
	B16CS3207	Mobile Computing
	B16CS3208	Distributed Systems
	B16CS3209	Advanced Computer Architecture
#M-III	B16 ENG 3205	Competitive Coding
	B16CS3215A	AngularJS
	B16CS3215B	ASP.NET
	B16CS3215C	C#.NET & VB.NET

DATA WAREHOUSING & DATA MINING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To understand the evolution of data warehousing and data mining techniques.
2. To understand sourcing , cleaning and transformation of data into a warehouse.
3. To learn the principles of statistics, information theory, databases, machine learning and other areas for design and implementation of data mining techniques.
4. To understand pattern mining classification and clustering methods.

Course Outcomes:

1. The student understands the differences between OLTP and OLAP.
2. The student learns how data cube technology supports structuring and querying high dimensional data.
3. The student is introduced to similarity , distance, information gain and other performance and error metrics used for data mining.
4. The student is introduced to association rule mining , supervised and unsupervised learning and the corresponding classification and clustering approaches involving decision trees, Bayesian approaches, model based and agglomerative approaches.

SYLLABUS

Introduction to Data Mining: Evolution of I T into DBMS, Motivation and importance of Data Warehousing and Data Mining, Kinds of Patterns, Technologies, Basic Data Analytics: Data Objects and Attributes Types, Statistical Descriptions of Data, Data Visualization, Estimating Data Similarity and Dissimilarity, Major Issues in Data Mining, Data Mining Applications

Tasks involved in data processing :Data Cleaning, Data Integration, Data Reduction, Data Transformation, Discretization and Concept Hierarchy Generation.

Data Warehouse and OLAP Technology: Basic Concepts of Data warehouse, Data Modeling using Cubes and OLAP, DWH Design and usage, Implementation using Data Cubes and OLAPs, Data Generalization with AOI.

Mining Frequent Patterns Based on Associations and Correlations: Basic Concepts, Frequent Itemset Mining Methods: Apriori Algorithm, Association Rule Generation, Improvements to A Priori, FP- Growth Approach, Mining Frequent Patterns using Vertical Data Formats, Mining Closed and Max Patterns, Pattern Evaluation Methods, mining in multilevel, multi-dimensional space.

Classification & Prediction: Basic Concepts, Decision Tree Induction, BayesClassification, Rule-Based Classification, Model Evaluation and Selection, Techniques to Improve Classification Accuracy Advanced Methods: Classification by Back Propagation, SVM, Associative Classification, Lazy Learning

Cluster Analysis: Basic Concepts and issues in clustering, Types of Data in ClusterAnalysis, Partitioning Methods-K-Means, K-Medoids, Hierarchical Methods-Agglomerative versus Divisive, Distance Measures in Algorithmic Methods, Birch, Chameleon, Density Based Methods- DBSCAN, OPTICS, Grid Based Methods-STING, CLIQUE, Evaluation of Clustering Solutions

Text Book:

1. Data Mining- Concepts and Techniques by Jiawei Han, MichelineKamber and Jian Pei – Morgan Kaufmann publishers ---3rd edition

Reference Books:

1. Introduction to Data Mining, Adriaan, Addison Wesley Publication.
2. Data Mining Techniques, A.K.Pujari, University Press Data mining concepts by Tan, Steinbech, and Vipin Kumar - Pearson Edu publishers.
3. Data Mining –Introductory and Advanced by Margaret Dunham -- Pearson Education publishers.
4. Data Warehousing for Real –world by Sam Annahory-- Pearson Education publishers.

OBJECT ORIENTED SOFTWARE ENGINEERING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To explain the importance of OOSE in Software development.
2. To explain the students the importance of Requirements Engineering.
3. To explain the role of UML and Testing in Software Development.
4. To explain the entire Software Development Process with aid of case studies.

Course Outcomes:

1. Ability to define a problem and perform Requirements Engineering.
2. Ability to draw UML diagrams for the requirements gathered.
3. Ability to design various aspects of the system.
4. Ability to implement the designed problem in Object Oriented Programming Language and test whether all the requirements specified have been achieved or not.
5. Able to apply various testing approaches to test the system.
6. Able to use various Process management activities.

SYLLABUS

Introduction to Object Oriented Software Engineering: Nature of the Software, Types of Software , Software Engineering Projects, Software Engineering Activities, Software Quality, Introduction to Object Orientation, Software Process Models-Waterfall Model, Opportunistic Model , Phased Released Model, Spiral Model, Evolutionary Model, Concurrent Engineering Model.

Requirements Engineering: Domain Analysis, Problem Definition and Scope, Requirements Definition, Types of Requirements, Techniques for Gathering and Analyzing Requirements, Requirement Documents, Reviewing, Managing Change in Requirements.

Unified Modeling Language & Use Case Modeling: Introduction to UML, Modeling Concepts, Types of UML Diagrams with Examples; User-Centred Design, Characteristics of Users, Developing Use - Case Models of Systems, Use-Case Diagram, Use- Case Descriptions, Basics of User Interface Design, Usability Principles, User Interfaces.

Class Design and Class Diagrams: Essentials of UML Class Diagrams, Associations and Multiplicity, Other Relationships, Generalization, Instance Diagrams, Advanced Features of Class Diagrams, Interaction and Behavioural Diagrams: Interaction Diagrams, State Diagrams, Activity Diagrams, Component and Deployment Diagrams.

Software Design and Architecture: Process of Design, Principles Leading to Good Design, Techniques for Making Good Design Decisions, Good Design Document; Pattern Introduction, Design Patterns: Abstraction-Occurrence Pattern, General Hierarchical Pattern, Play-Role Pattern, Singleton Pattern, Observer Pattern, Delegation Pattern, Adaptor Pattern, Façade Pattern, Immutable Pattern, Read-Only Interface Pattern and The Proxy Pattern; Software Architecture Contents of Architecture Model, Architectural Patterns: Multilayer, Client-Server, Broker, Transaction Processing, Pipe & Filter and MVC Architectural Patterns

Software Testing: Overview of Testing, Testing Concepts, Testing Activities, Testing Strategies, Unit Testing, Integration Testing, Function Testing, Structural Testing, Class Based Testing Strategies, Use Case/Scenario Based Testing, Regression Testing, Performance Testing, System Testing, Acceptance Testing, Installation Testing, OO Test Design Issues, Test Case Design, Quality Assurance, Root Cause Analysis, Post-Mortem Analysis.

Software Process Management: Introduction to Software Project Management, Software Engineering Teams, Project Scheduling, Tracking and Monitoring.

CASE STUDY

1. Simple Chat Instant Messaging System
2. GPS Based Automobile Navigation System
3. Waste Management Inspection Tracking System (WMITS)
4. Geographical Information System

Text Books:

1. Object-Oriented Software Engineering Practical software development using UML and Java by Timothy C. Lethbridge & Robert, Langanieri McGraw-Hill
2. Object-Oriented Software Engineering: Using UML, Patterns and Java, Bernd Bruegge and Allen H. Dutoit, 2nd Edition, Pearson Education Asia.

Reference Books :

1. Software Engineering: A Practitioner's Approach, Roger S Pressman.
2. A Practical Guide to Testing Object-Oriented Software, John D. McGregor; David A Sykes, Addison-Wesley Professional.
3. Software Engineering, K.K. Agarwal, New Age Publications 2008

DESIGN AND ANALYSIS OF ALGORITHMS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Analyze the asymptotic performance of algorithms.
2. Write rigorous correctness proofs for algorithms.
3. Demonstrate a familiarity with major algorithms and data structures.
4. Synthesize efficient algorithms in common engineering design situations

Course Outcomes:

1. Students will be able to Argue the correctness of algorithms using inductive proofs and invariants and Analyze worst-case running times of algorithms using asymptotic analysis.
2. Describe the various paradigms of design when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm and synthesize them .
3. Students will be able to Compare between different data structures. Pick an appropriate data structure for a design situation.

SYLLABUS**Introduction**

Algorithm specification, Recursive Algorithms, Performance analysis, Space Complexity, Time Complexity, Asymptotic Notation, Practical Complexities Performance Measurement, Priority queues, Heaps, Heap Sort, Sets and Disjoint set union, Union and Find Operations

Divide and Conquer

General method, Binary search, Finding maximum and minimum, Mergesort, Quicksort, Performance Measurement , Selection Problem, A Worst-Case Optimal Algorithm , Implementation of Select2, Strassen's matrix multiplication, Convex hull Problem, QuickHull Algorithm

The Greedy Method

General method, Knapsack problem, Tree vertex splitting, Job sequencing with deadlines, Minimum cost spanning trees, Prim's algorithm, Kruskal's algorithm, Optimal storage on tapes, Optimal merge patterns , Huffman coding, Single source shortest paths

Dynamic Programming

General method, Multistage graphs, All pairs shortest paths, Single source shortest paths with general weights, Optimal binary search trees, String editing, 0/1Knapsack, Reliability design, The travelling salesperson problem

Backtracking

General method, 8-Queens problem, Sum of subsets, Graph coloring, Hamiltonian cycles, Knapsack problem

Branch and Bound

The method, Least Cost(LC) Search, The15-puzzle problem, Control abstractions for LC-Search, FIFO Branch-and-Bound, LC Branch-and-Bound, 0/1 Knapsack problem, Travelling sales person problem

Algebraic Problems and Lower Bound Theory

The method, Evaluation and interpolation, Fast Fourier Transform, Modular arithmetic, Comparison trees, Ordered Searching, Sorting, Selection, Oracles and adversary arguments, Merging, Largest and Second largest

NP-Hard And NP-Complete Problems

Basic concepts, Nondeterministic Algorithms, The Classes NP-hard and NP-complete

Text Book:

1. Fundamentals of Computer Algorithms 2nd edition by Ellis Horowitz, Sartaj Sahni, S. Rajasekharan

Reference Book:

1. Introduction to Algorithms 3rd edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein

COMPILER DESIGN

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To explain the basic understanding of grammars and language definition and Introducing various phases of designing a compiler.
2. To make the student to understand the concepts underlying the design and implementation of language processors and its mechanisms.
3. To extend the knowledge of parser by parsing LL parser and LR parser.
4. To enrich the knowledge in various phases of compiler and its use, code optimization techniques, loop optimization techniques, machine code generation, use of symbol table, and Runtime-Environments

Course Outcomes:

1. Ability to describe the lexical analyzer, and different types of parsers.
2. Ability to describe the Intermediate Code generation in compiler.
3. Ability to explain code optimization techniques to improve the performance of a program in terms of speed & space.
4. Ability to explain code generation, symbol table and runtime storage administration.

SYLLABUS**Introduction**

Introduction to Language processors, The Structure of a Compiler, Bootstrapping & Cross compiler.

Lexical Analysis

The Role of Lexical Analyzer, Input Buffering, Specification of Tokens, Recognition of tokens, Lexical-Analyzer generator Lex, Design of Lexical-Analyzer generator.

Syntax Analysis

Introduction, Context-Free Grammar, Writing a Grammar, Top-Down parsing: Recursive-Descent Parsing and Nonrecursive Predictive parsing, Error recovery in Predictive Parsing, Bottom-Up parsing: Shift Reduce Parsing.

More Powerful parsers

Introduction to LR Parsing: Simple LR, More Powerful LR parsers, Using Ambiguous grammars, Parser Generators.

Syntax-Directed Translations

Syntax-Directed definitions, Evaluation Orders for SDD's, Applications of Syntax-Directed Translation, Syntax-Directed Translation schemes.

Intermediate-Code Generation

Variants of Syntax Trees, Three-Address code, Types and Declarations, Translation of Expressions, Type Checking.

Code Optimization

Basic Blocks and Flow Graphs, Optimization of Basic Blocks, The Principle Sources of Optimization, Introduction to Data-Flow Analysis.

Code Generation

Issues in the Design of Code Generator, The Target Language, Addresses in the Target Code, A Simple Code Generator, Peephole Optimization, Register Allocation and Assignment.

Symbol Tables and Runtime-Environments

Symbol Table Per Scope, The Use of Symbol Tables, Storage Organization, Stack Allocation of Space, Heap Management.

Text Book:

1. Compilers Principles, Techniques and Tools by Alfred V Aho, Monica S.Lam, Ravi Sethi, Jeffrey D. Ullman, Pearson Education Second Edition

Reference Books:

1. Compiler Construction by Kenneth. C. Loudon, Vikas Pub.House.
2. Compiler Design, A.A. Pentambekar, Technical Publications
3. Modern Compiler Design, Grune.D, Van Reeuwijk K, Bal H.E, Jacobs C J H, Langendoen K, Springer
4. Advanced Compiler Design and Implementation, Steven Muchnic, Elsevier Publications.

ARTIFICIAL INTELLIGENCE

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To learn about AI problem, Production Systems and their characteristics.
2. To understand the importance of search and the corresponding search strategies for solving AI problem.
3. To introduce to Planning, Natural Language Processing and Expert Systems.

Course Outcomes:

1. The Student understands AI problem characteristics, state space approach for solving AI problem, Production System framework.
2. The student learns several optimal search strategies and the use of heuristics.
3. The student learns relational, inferential, inheritable and procedural knowledge and the corresponding knowledge representation approaches.
4. The student is introduced to applying AI problem solving approaches to natural language processing, planning and expert systems

SYLLABUS**Introduction to Artificial Intelligence:**

Artificial Intelligence, AI Problems, AI Techniques, Defining the Problem as a State Space Search, Problem Characteristics, Production Systems.

Search Techniques:

Issues in The Design of Search Programs, Un-Informed Search, BFS, DFS; Heuristic Search Techniques: Generate-And- Test, Hill Climbing, Best-First Search, A* Algorithm, Problem Reduction, AO*Algorithm, Constraint Satisfaction, Means-Ends Analysis.

Knowledge Representation using Rules:

Procedural Vs Declarative Knowledge, Logic programming, Forward Vs Backward Reasoning, Matching Techniques, Partial Matching, RETE Matching Algorithm AI Programming languages: Overview of LISP and PROLOG, Production System in Prolog.

Symbolic Logic:

Propositional Logic, First Order Predicate Logic: Representing Instance and is-a Relationships, Computable Functions and Predicates, Unification & Resolution, Natural Deduction; **Structured Representations of Knowledge:** Semantic Nets, Partitioned Semantic Nets, Frames, Conceptual Dependency, Conceptual Graphs, Scripts.

Reasoning under Uncertainty:

Introduction to Non-Monotonic Reasoning, Truth Maintenance Systems, Logics for Non-Monotonic Reasoning, **Statistical Reasoning:** Bayes Theorem, Certainty Factors and Rule-Based Systems, Bayesian Probabilistic Inference, Bayesian Networks, Dempster- Shafer Theory, Fuzzy Logic: Crisp Sets ,Fuzzy Sets, Fuzzy Logic Control, Fuzzy Inferences & Fuzzy Systems.

Natural Language Processing: Steps in The Natural Language Processing, Syntactic Processing and Augmented Transition Nets, Semantic Analysis, NLP Understanding Systems.

Planning: Components of a Planning System, Goal Stack Planning, Non-linear Planning using Constraint Posting, Hierarchical Planning, Reactive Systems.

Experts Systems:

Overview of an Expert System, Architecture of an Expert Systems, Different Types of Expert Systems- Rule Based, Frame Based, Decision Tree based, Case Based, Neural Network based, Black Board Architectures, Knowledge Acquisition and Validation Techniques, , Knowledge System Building Tools, Expert System Shells.

Text Books:

1. Artificial Intelligence, Elaine Rich and Kevin Knight, Tata Mcgraw-Hill Publications.
2. Introduction To Artificial Intelligence & Expert Systems, Patterson, PHI publications

Reference Books:

1. Artificial Intelligence, George F Luger, Pearson Education Publications .
2. Artificial Intelligence : A modern Approach, Russell and Norvig, Printice Hall.
3. Artificial Intelligence, Robert Schalkoff, Mcgraw-Hill Publications .
4. Artificial Intelligence and Machine Learning, Vinod Chandra S.S., Anand Hareendran S.

CLOUD COMPUTING (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. This course is designed to examine the system models for cloud computing.
2. Understand the concepts of virtualization, hardware and storage
3. Identify cloud platform architecture and programming.
4. Develop cloud applications

Course Outcomes:

1. Define basic networking concepts for distributed and cloud computing. **(K1)**
2. Identify Virtual machine concept. **(K1)**
3. Explain the architecture of Cloud platform. **(K2)**
4. **Practice** cloud programming. **(K3)**
5. Explain the concepts regarding accessing the cloud and cloud storage. **(K2)**
6. Develop cloud applications. **(K5)**

SYLLABUS

Systems modeling, Clustering and virtualization: Scalable Computing over the Internet, Technologies for Network based systems, System models for Distributed and Cloud Computing, Software environments for distributed systems and clouds, Performance, Security And Energy Efficiency.

Virtual Machines and Virtualization of Clusters and Data Centers: Implementation Levels of Virtualization, Virtualization Structures/ Tools and mechanisms, Virtualization of CPU, Memory and I/O Devices, Virtual Clusters and Resource Management, Virtualization for Data Center Automation.

Cloud Platform Architecture: Cloud Computing and service Models, Architectural Design of Compute and Storage Clouds, Public Cloud Platforms, Inter Cloud Resource Management, Cloud Security and Trust Management. Service Oriented Architecture, Message Oriented Middleware.

Cloud Programming and Software Environments: Features of Cloud and Grid Platforms, Parallel & Distributed Programming Paradigms, Programming Support of Google App Engine, Programming on Amazon AWS and Microsoft Azure, Emerging Cloud Software Environments.

Hardware and Infrastructure: Clients, Security, Network, Services. **Accessing the Cloud:** Platforms, Web Applications, Web APIs, Web Browsers. **Cloud Storage:** Overview, Cloud Storage Providers.

Developing Applications: Google, Microsoft, Intuit QuickBase, Cast Iron Cloud, Bungee Connect, Development, Troubleshooting, Application Management.

Text Books:

1. Distributed and Cloud Computing, Kai Hwang, Geoffrey C. Fox, Jack J. Dongarra MK Elsevier.
2. Cloud Computing, A Practical Approach, Anthony T Velte, Toby J Velte, Robert Elsenpeter, TMH.

Reference Books:

1. Mastering Cloud Computing, Foundations and Application Programming, Raj Kumar Buyya, Christen vecctiola, S Tammaraiselvi, TMH.
2. Cloud Computing, A Hands on approach, ArshadeepBahga, Vijay Madiseti, University Press.

MOBILE COMPUTING (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To provide knowledge on mobile services and mobile technologies .
2. To know the emerging technologies in cellular networks.
3. To know the data management issues in mobile networks.
4. To know the various types of communications in mobile networks.

Course Outcomes:

1. To understand the mobile Technologies.
2. To understand the various issues in mobile devices.
3. To study the various generations of cellular networks.
4. To study various applications like SMS,MMS,Mobile IP.

SYLLABUS

Introduction to Mobile Computing, Overview of Mobile Technologies, Limitations, The Ubiquitous Network, Architecture for Mobile Computing, Three-Tier Architecture, Design Considerations for Mobile Computing, Mobile Computing Through Internet, Mobile Devices and Mobile-Enabled Applications.

Introduction To Wireless Networking, Various Generations of Wireless Networks, Wireless LANs, Advantages and Disadvantages of WLANs, Fixed Network Transmission Hierarchy, Differences in Wireless and Fixed Telephone Networks, Traffic Routing in Wireless Networks, WAN Link Connection Technologies, Cellular Networks.

WLAN Topologies, WLAN Standard IEEE 802.11, Comparison Of IEEE 802.11a, B, G and N Standards, Wireless PANs, Hiper LAN, Wireless Local Loop, ATM, Virtual Private Networks, Wireless Data Services, Common Channel Signaling, Various Networks for Connecting to The Internet.

Emerging Technologies: Introduction - Bluetooth - Radio Frequency Identification (RFID), WIMAX -Mobile IP - Ipv6 - Java Card, TCP/IP in the Mobile Setting, GSM and GPS.

Data Management Issues, Data Replication For Mobile Computers, Adaptive Clustering for Mobile Wireless Networks, File System, Disconnected Operations, Data Services in GPRS - Applications for GPRS - Limitations - Billing and Charging.

Communications Asymmetry, Classification of New Data Delivery Mechanisms, Push-Based Mechanisms, Pull-Based Mechanisms, Hybrid Mechanisms, Selective Tuning (Indexing) Techniques. CDMA, GSM , Wireless Data, 3GNetworks and Applications.

Introduction to Mobile IP, Introduction To Wireless Application Protocol, Application Layer MMS - GPRS Applications, Short Message Service (SMS): Mobile Computing Over SMS - SMS - Value Added Services Through SMS -Accessing the SMS Bearer.

Text Books:

1. Mobile Computing - Technology Applications And Service Creation, Asoke K Talukder and Roopa R.Yavagal, TMH 2006.
2. Mobile Cellular Communication, Gottapu Sasibhushana Rao,, Pearson Education, First Edition, 2013.

Reference Books:

1. Principles Of Computing, Uwe Hansmann, Lothar Merk, Martin S.Nicklous, Thomas Staber, 2nd Ed., Springer International Edition.
2. Mobile Communications, J.Schiller, Addison-Wesley, 2003.
3. Stojmenovic and Cacute, "Handbook Of Wireless Networks And Mobile Computing", Wiley, 2002.

DISTRIBUTED SYSTEMS (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objective:

This course provides an introduction to the fundamentals of distributed computer systems, assuming the availability of facilities for data transmission.

Course Outcomes:

1. Scale as the number of entities in the system increase.
2. Can sustain failures and recover from them.
3. Work with distributed, fault tolerant file systems .
4. Can handle and process large data volumes.
5. Are secure and handle certain classes of distributed denial of service attacks .
6. Are Loosely coupled, transactional and eventually stable

SYLLABUS

Introduction to Distributed Systems, What is a Distributed System?, Hard ware concepts, Software concepts, Design issues.

Communication in Distributed Systems, Lay red Protocols, ATM networks, The Client – server model, Remote Procedure call, Group communication.

Synchronization in Distributed System, Clock Synchronization, Mutual Exclusion, Election algorithms, Atomic transactions, Deadlocks in Distributed Systems.

Process and processors in Distributed System threads, System Models, Processors allocation, Scheduling in Distributed System, Fault tolerance, Real time Distributed System.

Distributed File Systems, Distributed File System Design, Distributed File System implementation, Trends in Distributed File System.

Distributed Shared Memory, Introduction, What is Shared memory?, Consistency models, Page based Distributed Shared memory, Shared – variable Distributed Shared memory, Object based Distributed Shared Memory.

Text Book:

1. Distributed Operating Systems, Andrew S. Tanenbanm

Reference Book:

1. Advanced Concepts in Operating Systems, Makes Singhal and Niranjn G.Shivaratna

**ADVANCED COMPUTER ARCHITECTURE
(Elective-II)**

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objective:

1. Study about multiprocessors, shares memory, distributed computers, vector super computers.
2. Learn about advanced processing , pipelining and parallel and multi threaded architectures.
3. Study of parallel programming models, languages, code optimization pipelining.
4. Study about multiprocessor , multithreaded and multicomputer unix design.

Course Outcomes:

1. Detailed idea about parallel computing models.
2. Knowledge on hand about advanced processing, pipelining and super scalar techniques, Parallel, multi vector and multithreaded architectures.
3. Knowing about parallel programming softwares.

SYLLABUS

Parallel Computer Models: Elements of Modern Computer, Multi Processor and Multi Computers- Shared Memory Multi processors, Distributed Memory Multi Computers, Multi Vector and SIMD Computers-Vector Super Computers and SMD Super Computers, Architectural Development Tracks- Multiple Processor Tracks, Multi Vector and SIMD Tracks.

Advanced Processing: Advance Processing Technology-Design Space of Processor, Instruction Set Architectures, CISC Scalar Processors, RISC Scalar Processors; Super Scalar and Vector Processor- Super Scalar Processors, VLIW Architecture, Vector and Symbolic Processors.

Pipelining and Super scalar Techniques: Linear Pipeline Processors, Non-linear pipelining Processors, Instruction Pipeline Design, Arithmetic Pipeline Design, Superscalar and Super pipeline Design.

Parallel Architectures: Multiprocessor Interconnects- Hierarchical Bus Systems, Crossbar Switch and Multiport Memory, Multistage and Combining Networks; Cache Coherence and Synchronization Mechanisms,- The Cache Coherence Problem, Snoopy bus Protocols, Directory based protocols, Hardware Synchronization Mechanisms.

Multivector and SIMD Computer: Vector Processing Principles- Vector Instruction Types, vector- Access Memory Schemes, past and Present Supercomputers; Multi Vector Multiprocessors- Performance-Directed Design Rules, Cray Y-MP, C-90, and MPP, Mainframes and Mini- supercomputers; SIMD Computer Organizations- Implementation Models, the CM-2 Architecture, The Maspar MP-1 Architecture.

Multithreaded Architecture: Latency Hiding techniques- Shared Virtual Memory, Perfecting Techniques, Distributed Coherent Caches, Scalable Coherence Interface, relaxed Memory Consistency. Principles of Multithreading- multithreading Issues and Solutions, Multiple Context Processors, Multidimensional Architectures.

Software for Parallel Programming: Parallel programming Models, parallel languages and Compilers, Code optimization and Scheduling, Loop parallelization and Pipelining.

Case Study: Multiprocessor UNIX Design Goals, Master-Slave and Multithreaded UNIX, Multicomputer UNIX Extensions.

Textbook:

1. Kai Hwang, Advanced Computer Architecture, Tata McGraw Hill Edition.

Reference Books:

1. Kai Hwang and Briggs, Advanced Computer Architecture and Parallel Processing, Tata McGraw Hill Edition.
2. Advanced Computer Architectures: A Design Space Approach Sima, Terence FOUNTAIN, Péter KACSUK Pearson Education India.
3. Computer Architecture: A Quantitative Approach, 5th Edition. John L. Hennessy, David A. Patterson, Morgan Kaufmann Publications.

SOFTWARE ENGINEERING MINI PROJECT LAB**Lab : 3 Periods****Exam : 3 Hrs.****Sessionals : 50****Ext. Marks : 50****Credits : 2****Course Objectives :**

1. To familiarize the students with Software Engineering development activities
2. To get the knowledge of various Software Engineering Tools
3. Hands on experience in developing Software Engineering projects

Course Outcomes :

At the end of the course the students can able to

1. Identify various phases of Software Development
2. Use various Software engineering tools
3. Develop small projects

SYLLABUS

The purpose of the Software Engineering Lab course is to familiarize the students with modern software engineering methods and tools, **Rational Products**. The course is realized as a project-like assignment that can, in principle, by a team of three/four students working full time. Typically the assignments have been completed during the semester by each project team.

The goal of the Software Engineering Project is to have a walk through from the requirements, design to implementing and testing. An emphasis is put on proper documentation. Term projects are projects that a group student might take through from initial specification to implementation by giving equal importance to both design and implementation.

Cycle I: Practicing UML diagrams using IBM Rational Rose. 6*3 periods= 18periods

Before developing a mini-project, in this cycle, the student is acquainted with different UML diagrams using Rational Rose. The experiments should include drawing UML diagrams listed below for two demo/example applications assigned by the lab Instructor. The input for the following experiments is problem statement for any two demo projects supplied by the instructor.

1. Introduction to Rational Rose and Practicing the following diagrams
 - A. Activity diagrams for the overall business process of the projects
 - B. Use-case diagram for the demo projects along with Use-case descriptions and sub-diagrams for Use-cases.

2. Class diagram- Class diagrams including the features like classes, relationships, attributes and methods along with their visibilities.
3. Interaction diagrams- Sequence diagrams and Collaboration diagrams for different scenarios of the systems with all features like actors, objects and interactions.
4. Activity diagrams, State chart and other diagrams - Activity diagrams including the features like fork join and swim lanes. State diagrams including composite states and transitions. Component diagrams, Package diagrams and Deployment diagrams.
5. Forward and Reverser Engineering- Forward Engineering Class diagrams to classes in C++ and java and persistent classes to a database. Reverse Engineering C++ code, java code and a database.
6. Documentation using Rational Rose clear quest.

Cycle II: Mini-Project

8*3 periods= 24periods

The project deliverables include

- Problem statement
- Requirements Analysis
- Design
 - o A Software Design Description and a System Design.
 - o A test specification.
- Implementation

Implement the assigned project with one of the following web technologies

Front end: Java technologies/PHP/MS.NET Technologies

Backend: Oracle/My-SQL/SQL-Server

- Testing

Reference Books :

1. Project-based software engineering: An Object-oriented approach, Evelyn Stiller, Cathie LeBlanc, Pearson Education .
2. Visual Modeling with Rational Rose 2002 and UML, Terry Quatrini, Pearson Education

NETWORK PROGRAMMING LAB

Lab : 3 Periods

Exam : 3 Hrs.

Sessionals : 50

Ext. Marks : 50

Credits : 2

Course Objectives

1. To understand the use of client/server architecture, inter process communication and to explain the basic communication protocols.
2. To understand elementary socket system calls, advanced socket system calls and Java Socket API and to explain the basic concepts relating to TCP and UDP based sockets
3. To Understand e-mail Programming like SMTP and POP
4. To understand File transfer protocol, remote login using pseudo terminal and RPC.

Course Outcomes

1. Students will be able to write Socket based Network application programs
2. Students will be able to design and develop Client Server applications using Java
3. Students will be able to write network applications like One-One chat ,Broadcasting and Multicasting
4. Students will be able to understand e-mail programming (SMTP, POP).

SYLLABUS

1. Write a program to identify well known ports on a Local/Remote System.

By trying to listen to the various well-known ports by opening client connections.
If the exception does not occur then the remote port is active else the remote port is inactive.

2. Write a program to implement Chat Application.

- i. One-One Chat: By opening socket connection and displaying what is written by one System to other.
- ii. Many-Many Chat (Broad Casting): Each Client opens a socket connection to the chat server and writes to the socket. Whatever is written by one system can be seen by all other systems.

3. Write a program to retrieve data from a remote database using Java.

At the remote database a server listens for client connections. This server accepts SQL queries from the client, executes it on the database and sends the response to the client.

4. Write a program to implement Mail Client

- i. SMTP Client: Gives the server name, send email to the recipient using SMTP Commands
- ii. POP Client: Gives the server name, user name and password, retrieve the mails and allow manipulation of mailbox using POP commands.

5. Write a program to simulate Telnet Client which allows to connect to well known servers and send and receive information.

Provide a user interface to contact well known ports so that client server interaction can be seen by the user.

6. Write a program to implement IP multicasting.

7. Write a program to implement simple file transfer between two systems (without protocols)

By opening socket connection to our server on one system and sending a file from one system to another

8. Write a program to implement TFTP Client and TFTP Server for file transfer

9. Write a program to implement HTTP-Server and HTTP Client.

The Server has to process the following commands: GET, POST, HEAD, and DELETE.
The server must handle multiple clients.

10. Write a program to implement UDP Echo Server and UDP Echo Client.

11. Write a program to get the attributes and contents of a web page using URL Connection class.

12. Write a program to implement DNS.

Reference Books:

1. “Java Network Programming”, Elliotte Rusty Harold , Fourth Edition, Orielly 2013.
2. “An Introduction to Computer Networking”, Kenneth C. Mansfield Jr and James L. Antonakos Pearson Education Asia.
3. William Stallings, “Data and Computer Communication”, Sixth Edition, Pearson Education, 2000.
4. Ferouz.A.Ferouzan, “Computer Communication and Networking”, Fifth Edition, Pearson Education, 2001.
5. James F. Kuross, Keith W. Ross, “Computer Networking, A Top-Down Approach Featuring the Internet”, Third Edition, Addison Wesley, 2004.

VERBAL & QUANTITATIVE APTITUDE – II
(Common to All Branches)

Theory	: 5 Periods(VA-3+QA-2)	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing .

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)

Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a 'Statement of purpose', 'Letters of Recommendation', business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)

Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)

Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron's GRE guide.
3. Newspapers like 'The Hindu', 'Times of India', 'Economic Times'.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and PushpaLatha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by SaminaPillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by RashmiBansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and menstruation.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation

Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II :Puzzle test, blood Relations, permutations, Combinations and probability

Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III :Periods,Clocks, Calendars, Cubes and cuboids

Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles

Selective puzzles from previous year placement papers, sitting arrangement, problems-circular arrangement, linear arrangement, different puzzles.

UNIT V: Geometry and Menstruation

Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of menstruation-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

COMPETITIVE CODING
(Common to CSE & IT)
(MOOCS-III)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives:

1. To understand the essentials of competitive coding
2. To learn about competitive programming like time and space complexities
3. To investigate different methods like dynamic Arrays, Set & Map structures and sorting
4. To learn about String, Tree, Graph Theory algorithms

Course Outcomes:

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of competitive coding
2. Acquire Programming knowledge on time & space complexities
3. Acquire coding knowledge on dynamic Arrays, Set & Map structures and sorting
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on String, Tree, Graph Theory algorithms

SYLLABUS

UNIT-I Introduction to Competitive Coding

Introduction to Competitive coding and coding Platforms. Coding solution Vs. Efficient Coding solution. Types of solution approaches. Analyzing problem specific data requirement, Various data representations. Essential Data structures for fast coding. Various Syntactical I/O techniques comparison. Numbers, operations (including exponentiation). Integer properties(positive, negative, even, odd, divisible, prime, etc), Fractions, percentages and ratios. Point, vector, Cartesian coordinates(2D integer grid).

UNIT-II Essentials to Competitive coding

Basic data structures: Arrays, Strings, Stacks, Queues, Linked Lists . Asymptotic Notations – (Big-O), Evaluating the runtime complexity – Space Complexity - Towers-of-Brahma – Standard Template Libraries - Square root functions, primality testing and related techniques. Euclidean algorithms. Recursion techniques. Organizing data in $O(n \log n)$. Binary search techniques. Red-Black Trees. Fenwick tree, Segment Tree.

Basic Techniques

Dynamic Arrays, Set structures, Map structures, Iterators and ranges, Generating Subsets, Generating permutations, Backtracking techniques, Pruning the search. Bit masking. Disjoint set union.

UNIT-III Essential Coding Algorithms

Selection based algorithms: sorting, Coin change problem, Fractional selections, Schedules matching, Activity marking, heap sort, Huffman coding techniques, Spanning Trees, Minimizing sums, Data compression. Finding method count, Subsequence and related problems, paths in grid. DP with Bit mask

UNIT-IV String & Tree Algorithms

TRIE data structure, Naïve string searching, z-algorithm, Manacher's algorithm, Rabin-Karp, KMP Algorithm, Tree Traversals, Diameter, All longest paths, Binary trees, Applying search property to tree structures. Suffix arrays.

UNIT-V Graph Algorithms

Graph Algorithms – DFS, BFS. Depth First and Breadth First Traversals - Shortest paths: Dijkstra's algorithm Bellman-Ford Algorithm – Floyd Warshall Algorithm - Adjacency List Representation – Euler path, tour , cycle – Eulerian Graph - Johnson's Algorithm for All-pairs shortest path – Shortest path in Directed Acyclic Graph. Bridges and articulation points. Topological sorting, Strongly connected components in directed graphs. 2-SAT.

Note: Introduce C++ STL/Java Collections and let students solve problems using C++ STL/Java Collections

References:

1. Data Structures and Algorithm Analysis in C – Mark Allen Weiss, Pearson Edu Publishers.
2. Data Structures and Algorithms: Concepts, Techniques and Applications – G.A.V.Pai, Tata Mc Graw Hill Publishers.
3. Advanced Data Structures – Peter Brass, Cambridge University Press, 2008.
4. Introduction to Algorithms by Thomas H. Cormen, 3rd Edition, PHI.
5. <https://www.geeksforgeeks.org/>.
6. <https://www.tutorialspoint.com/>

**ANGULAR JS
(MOOCS-III)****Theory : 4 Periods**
Exam : 3 Hrs.**Sessionals : 100**
Credits : 2

Course objectives:

1. Reduce the amount of code you write to build rich user interface applications.
2. Increase the reliability and maintainability of UI by using data binding.
3. Retrieve data from back-end server, manipulate it and display it with ease.
4. Modularise your code with the custom services and directives.
5. Providing two way binding of data.
6. Create Single Page Applications (SPA).

Course Outcomes:

1. The main objective of AngularJS is to reduce the code to build user interface application
2. To create single page applications
3. To restore data from back-end server and manipulate it easily

Module 1: Introduction

Introduction to AngularJS, MVC Architecture, Conceptual Overview, Setting up the Environment, First Application, Understanding ng attributes.

Module 2: Expressions and Data Biding

Number and String Expressions , Object Binding and Expressions, Working with Arrays, Forgiving Behavior, Understanding Data binding.

Module 3: Working with Directives

Conditional Directives ,Styles Directives, Mouse and Keyboard Events Directives.

Module 4: Controllers

Understanding Controllers , Programming Controllers & \$scope object, Adding Behavior to a Scope Object, Passing Parameters to the Methods, Having Array as members in Controller Scope. Nested Controllers and Scope Inheritance, Multiple Controllers and their scopes.

Module 5: Filters

Built-In Filters , Uppercase and Lowercase Filters, Currency and Number Formatting Filters, OrderBy Filter. Filter Filter ,Creating Custom Filter.

Module 6: Forms

Using Simple Form, Working with Select and Options, Input Validations, Using CSS classes, Form Events, Custom Model update triggers, Custom Validations.

Module 7: Modules

Why Module? , Module Loading and Dependencies, Recommended Setup of Application, Creation vs Retrieval.

Module 8: Services

Understanding Services, Developing Creating Services, Using a Service, Injecting Dependencies in a Service.

Module 9: Ajax in AngularJS

\$http Service, \$q Service, Ajax Impl using \$http and \$q Service.

Module 10: Routing

Introduction to SPA , Creating HTML Templates, Configuring Route Provider

WEB LINKS:

<https://www.udemy.com/angularjs/>

<https://www.coursera.org/learn/single-page-web-apps-with-angularjs>

**ASP.NET
(MOOCS-III)****Theory : 4 Periods**
Exam : 3 Hrs.**Sessionals : 100**
Credits : 2

Course Objectives:

1. Introduction to web technologies
2. Web Forms Architecture
3. ASP.Net and HTTP
4. Web application developing using Visual Studio
5. State Management and Web Applications
6. ASP.Net Server-side controls
7. Caching in ASP.Net
8. ASP.Net application configuration
9. Debugging, Diagnostics of application
10. Personalization and Security

Course Outcomes:

1. To successfully build database-driven Web applications and Web Sites.
2. To build web-based enterprise applications using ASP.NET and Visual Studio.
3. It is easy to develop the Web Services using .Net framework in Service-oriented Architectures.

SYLLABUS**Chapter 1: Introduction to ASP.NET**, From ASP to ASP.NET, Web Forms, Web Services
ASP.NET Features**Chapter 2: Web Forms Architecture**, Page Class, Web Forms Life Cycle, Web Forms
Event Model, Code-Behind**Chapter 3: ASP.NET and HTTP**, Request/Response Programming, HttpRequest Class,
HTTP Collections, HttpResponse Class, Redirection, HttpUtility Class**Chapter 4: Web Applications Using Visual Studio**, Using Visual Web Developer, Visual
Studio Forms Designer, Using Components, Shadow Copying, Using the Global.asax File,
Data Binding**Chapter 5: State Management and Web Applications**, Session State, Application State,
Multithreading Issues, Cookies**Chapter 6: Server Controls**, HTML Server Controls, Web Forms Server Controls, Rich
Controls, Validation Controls, User Controls**Chapter 7: Caching in ASP.NET**, What Is Caching, Page-Level Caching, Page Fragment
Caching, Optimizing Your ASP.NET Application, Application Caching

Chapter 8: ASP.NET Configuration and Security Fundamentals, Configuration Overview, Authentication and Authorization, Forms Authentication, Windows Authentication, Security and ASP.NET

WEB LINKS:

<https://www.asp.net/freecourses>

https://mva.microsoft.com/en-us/training-courses/introduction-to-aspnet-mvc-8322?l=nKZwZ8Zy_3504984382

**C#.NET & VB.NET
(MOOCS-III)****Theory : 4 Periods**
Exam : 3 Hrs.**Sessionals : 100**
Credits : 2**Course Objectives:**

1. Learn the fundamentals of C# programming in Visual Studio.
2. Working with variables, data types
3. Exception Handling in C#
4. Object oriented programming in C#
5. Use anonymous types, lambda expressions, extension methods, object initializers, and implicit type declarations
6. To build desktop applications using Windows Forms in VB.NET
7. To interact with relational databases using ADO.NET

Course Outcomes:

1. Understand .NET Framework and describe some of the major enhancements to the new version of Visual Basic.
2. Describe the basic structure of a Visual Basic.NET project and use main features of the integrated development environment (IDE)
3. Create applications using Microsoft Windows Forms Create applications that use ADO.NET

SYLLABUS

Chapter 1: C# Overview, First C# Console Application, Namespaces, Data Types, Conversions, Control Structures, Subroutines and Functions, Parameter Passing, Strings, Arrays, Console I/O, Formatting, Exception Handling

Chapter 2 : Object-Oriented Programming in C#, Classes, Access Control, Methods and Properties, Asymmetric Accessor Accessibility, Static Data and Methods, Inheritance, Overriding Methods, Abstract Classes, Sealed Classes, Access Control and Assemblies, Auto-Implemented Properties, Implicitly Typed Variables, Object Initializers, Collection Initializers, Anonymous Types, Partial Methods, Extension Methods, Lambda Expressions, Language-Integrated Query (LINQ)

Chapter 3: VB.NET - Features, IDE- Menu System, Toolbars, Code Designer, Solution Explorer, Object Browser, Toolbox, Class View Window, Properties Window, Server Explorer, Task List, Output Window, Command Window

Chapter 4: Elements of VB.net - Properties, Events and Methods of Form, Label, TextBox, ListBox, Combo Box, RadioButton, Button, Check Box, Progress Bar, Date Time Picker, Calendar, Picture Box, HScrollbar, VScrollbar, Group Box, ToolTip, Timer.

Chapter 5: Programming in VB.net: Data Types, Keywords, Declaring Variables and Constants, Operators, Understanding Scope and accessibility of variables, Conditional Statements- If- Then, If-Then-Else, Nested If, Select Case, Looping Statement- Do loop, For Loop, For Each-Next Loop, While Loop, Arrays- Static and Dynamic.

Chapter 6: Functions, Built-In Dialog Boxes, Menus and Toolbar

Menus and toolbars- Menu Strip, Tool Strip, Status Strip, Built-In Dialog Boxes –Open File Dialogs, Save File Dialogs, Font Dialogs, Color Dialogs, Print Dialogs, InputBox, Msg Box, Interfacing With End user- Creating MDI Parent and Child, Functions and Procedures- Built-In Functions- Mathematical and String Functions, User Defined Functions and Procedures.

Chapter 7: Advanced Concepts in VB.Net

Object Oriented Programming- Creating Classes , Objects, Fields, Properties, Methods, Events , Constructors and destructors, Exception Handling- Models, Statements, File Handling- Using File Stream Class, File Mode, File Share, FileAccess Enumerations, Opening or Creating Files with File Stream Class, Reading and Writing Text using StreamReader and StreamWriter Classes, Data Access with ADO.Net – Data Access with Server Explorer, Data Adapter and DataSets, ADO.NET Objects and Basic SQL.

Text Books:

1. Visual Basic.Net Black Book by Steven Holzner Dreamtech Press
2. The Complete Reference Visual Basic .NET Jeffery R. Shapiro Tata McGraw Hills

Reference Books:

1. Murach's Beginning Visual basic .Net By Anne Bohem

WEB LINKS:

<https://www.coursera.org/learn/introduction-programming-unity>

<https://www.pluralsight.com/courses/dotnet-csharp-tutorial>

<https://www.udemy.com/learn-vbnet/>



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

IV/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

COMPUTER SCIENCE AND ENGINEERING

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Sessional Marks	External Marks	Total Marks
B16CS 4101	Machine Learning	4	3	1	--	4	30	70	100
B16CS 4102	Big Data Analytics	4	3	1	--	4	30	70	100
B16ENG 4101	Principles of Economics and Management	4	3	1	--	4	30	70	100
B16CS 4104	Knowledge Engineering Lab	2	--	--	3	3	50	50	100
B16CS 4105	Big Data Analytics Lab	2	--	--	3	3	50	50	100
B16 CS 4106	Project Phase-I	2			3	3	50	--	50
Total		18	9	3	9	21	240	310	550

MACHINE LEARNING

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. Familiarity with a set of well-known supervised, unsupervised machine learning algorithms.
2. The ability to apply some basic machine learning algorithms to real world problems
3. Understand how machine learning algorithms are evaluated.

Course Outcomes:

1. Identify the characteristics of machine learning that make it useful to real-world problems.
2. Classify machine learning algorithms as supervised, unsupervised
3. Construct different tree models and rule models.
4. Be able to demonstrate linear and distance based models.
5. Be able to identify and construct features and ensemble models
6. Infer the concept of artificial neural networks, reinforcement learning

SYLLABUS

The ingredients of machine learning, Tasks: the problems that can be solved with machine learning, **Models:** the output of machine learning, **Features,** the workhorses of machine learning. (Peter Flach)

Binary classification and related tasks: Classification, Scoring and ranking **Beyond binary classification:** Handling more than two classes: *Multi-class classification*, Regression, Unsupervised and descriptive learning. (Peter Flach)

Concept Learning: The hypothesis space, Paths through the hypothesis space **Tree models:** Decision trees, Ranking and probability estimation trees, Tree learning as variance reduction **Rule models:** Learning ordered rule lists, Learning unordered rule sets (Peter Flach)

Linear models: The least-squares method: *Univariate linear regression*, The perceptron: a heuristic learning algorithm for linear classifiers **Distance Based Models:** Introduction, Neighbours and exemplars, Nearest Neighbours classification, Distance Based Clustering, Hierarchical Clustering. (Peter Flach)

Features: Kinds of feature, Feature transformations: *Thresholding and discretisation, Normalisation and Calibration, Incomplete Features*, Feature construction and selection. Model ensembles: Bagging and random forests, Boosting (Peter Flach)

Artificial Neural Networks: Introduction, Neural network representation, appropriate problems for neural network learning, Multilayer networks and the back propagation algorithm.

Reinforcement Learning: Introduction, The Learning Task, Q Learning: The Q function, the algorithm for learning Q, An illustrative example, Convergence (Tom M. Mitchell)

Text Books:

1. Machine Learning: The art and science of algorithms that make sense of data, Peter Flach, Cambridge.
2. Machine Learning, Tom M. Mitchell, MGH.

Reference Books:

1. Understanding Machine Learning: From Theory to Algorithms, Shai Shalev-Shwartz, Shai Ben-David, Cambridge.
2. Machine Learning in Action, Peter Harington, 2012, Cengage.

BIG DATA ANALYTICS

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives: On completing this course student will be able to

1. Understand big data and Apache Hadoop Eco system
2. Understand distributed , parallel, cloud computing and SQL concepts
3. Apply Hadoop concepts
4. Understand concepts of map and reduce and functional programming

Course Outcomes:

1. Gain conceptual understanding of analytics concepts, algorithms and statistical tests
2. Students will be able to look at the core projects used for both batch and real time data processing such as Hadoop
3. Students will be able to look at wider range of problems and data science based solutions

SYLLABUS

Introduction to Big Data: Big Data-definition, Characteristics of Big Data (Volume, Variety, Velocity, Veracity, Validity), Importance of Big Data , Patterns for Big Data Development, Data in the Warehouse and Data in Hadoop,

Introduction to Hadoop: Hadoop- definition, Understanding distributed systems and Hadoop, Comparing SQL databases and Hadoop, Understanding MapReduce, Counting words with Hadoop—running your first program, History of Hadoop, Starting Hadoop - The building blocks of Hadoop, NameNode, DataNode, Secondary NameNode, JobTracker and Task Tracker

MapReduce -A Weather Dataset, Analyzing the Data with Unix Tools, Analyzing the Data with Hadoop, Scaling Out, Hadoop Streaming, Hadoop Pipes, Developing a MapReduce Application - The Configuration API, Configuring the Development Environment, Running Locally on Test Data, Running on a Cluster, Tuning a Job, MapReduce Workflows

HDFS: Components of Hadoop -Working with files in HDFS, Anatomy of a MapReduce program, Reading and writing the Hadoop Distributed File system -The Design of HDFS, HDFS Concepts, The Command-Line Interface, Hadoop Filesystem, The Java Interface, Data Flow, Parallel Copying with distcp, Hadoop Archives

MapReduce Programming: Writing basic Map Reduce programs - Getting the patent data set, constructing the basic template of a Map Reduce program, Counting things, Adapting for Hadoop's API changes, Streaming in Hadoop, Improving performance with combiners.

MapReduce Advanced Programming: Advanced MapReduce - Chaining MapReducejobs, joining data from different sources, creating a Bloom filter, Passing job-specific parameters to your tasks, probing for task-specific information, Partitioning into multiple output files, Inputting from and outputting to a database, keeping all output in sorted order

Graph Representation in Map Reduce: Modeling data and solving problems with graphs, Shortest Path Algorithm, Friends-of-Friends Algorithm, PageRank Algorithm, Bloom Filter, Parallelized Bloom filter creation in MapReduce, Map-Reduce semi-join with Bloom filters

Text Books:

1. Dirk deRoos, Chris Eaton, George Lapis, Paul Zikopoulos, Tom Deutsch “Understanding Big Data Analytics for Enterprise Class Hadoop and Streaming Data”, 1st Edition, TMH,2012.
2. Hadoop: The Definitive Guide by Tom White, 3rd Edition, O’reilly

Reference Books:

1. Hadoop in Action by Chuck Lam, MANNING Publ.
2. Hadoop in Practice by Alex Holmes, MANNING Publishers
3. Mining of massive datasets, Anand Rajaraman, Jeffrey D Ullman, Wiley Publications.

PRINCIPLES OF ECONOMICS & MANAGEMENT
(Common to CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Apply economic reasoning to the analysis of selected contemporary economic problems.
2. Understand how households (demand) and businesses (supply) interact in various market structures to determine price and quantity of goods and services produced and consumed.
3. Analyze the efficiency and equity implications of government interference in markets.
4. Recognize and identify situations leading to market failures and government failures.

Course Outcomes:

1. Understand the links between production costs and the economic models of supply.
2. Represent supply, in graphical form, including the upward slope of the supply curve and what shifts the supply curve.
3. Understand the efficiency and equity implications of market interference, including government policy.
4. Understand how different degrees of competition in a market affect pricing and output.
5. Apply economic reasoning to individual and firm behavior.

SYLLABUS

Introduction to Managerial Economics: Wealth, Welfare and Scarce Definitions of Economics; micro and Macro Economics; Demand- Law of Demand, Elasticity of Demand, types of Elasticity and factors of determining price elasticity of Demand: utility- Law of Diminishing Marginal Utility and its limitations.

Conditions of Different Market Structures: Perfect Competition, Monopolistic Competition, Monopoly, Oligopoly, and Duopoly.

Forms of Business Organizations: Sole Proprietorship, Partnership, Joint Stock Company- Private Limited and Public Limited Companies, Public Enterprises and their types.

Introduction to Management: Functions of Management- Taylor's Scientific management; Henry Fayol's Principle of Management; Human Resource Management- basic Functions of HR Manager; Man Power Planning, Recruitment, Selection, Training, Development, Placement, Compensation and performance Appraisal(in brief).

Production Management: Production Planning and Control, plant Location, Break- Even Analysis, assumptions and applications.

Financial Management: Types of Capital: Fixed and Working Capital , and Methods of Raising Finance; Depreciation: Straight Line and Diminishing Balance Methods. Marketing Management: Functions of marketing and Distribution Channels.

Entrepreneurship: Entrepreneurial Functions, Entrepreneurial Development: Objectives, Training, Benefits: Phases of Installing a project

Text Books :

1. K.K.DEWETT, Modern Economic Theory, S.Chand and Company, New Delhi-55.
2. S.C. Sharma and Banga T. R., Industrial Organization & Engineering Economics, khanna Publications, Delhi-6.

Reference Books

1. A.R. AryaSri, Management Science, TMH publications, New Delhi-20.
2. A.R. AryaSri, Managerial Economics and Financial Analysis, TMH Publications, new Delhi-20.

KNOWLEDGE ENGINEERING LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals :50
Ext.Marks :50
Credits :2

Course Objectives:

1. To study the various data analysis techniques in R Programming language.
2. To apply the various data mining techniques available in WEKA for generating Knowledge such as Association Analysis, Classification and Clustering to various standard datasets and own datasets.
3. To build Rule based Knowledge Systems using forward chaining and Backward chaining using CLIPS and PROLOG respectively

Course Outcomes:

1. Student will be able to write R programs to perform several data analytics operations on datasets
2. Ability to extract patterns by applying appropriate data mining techniques from different types of datasets using WEKA.
3. Ability to apply knowledge represented in the form of rules to draw conclusions using either forward or backward chaining using CLIPS /PROLOG.

SYLLABUS**1. Exploratory data analysis using R**

1. Load the 'iris. CSV' file and display the names and type of each column. Find statistics such as min, max, range, mean, median, variance, standard deviation for each column of data.
2. Write R program to normalize the variables into 0 to 1 scale using min-max normalisation
3. Generate histograms for any one variable (sepal length/ sepal width/ petal length/ petal width) and generate scatter plots for every pair of variables showing each species in different color .
4. Generate box plots for each of the numerical attributes. Identify the attribute with the highest variance.
5. Study of homogeneous and heterogeneous data structures such as vector, matrix, array, list, data frame in R.
6. Write R Program using 'apply' group of functions to create and apply normalization function on each of the numeric variables/columns of iris dataset to transform them into a value around 0 with z-score normalization.

7. a) Use R to apply linear regression to predict evaporation coefficient in terms of air velocity using the data given below:

Air Velocity (cm/sec)	20,60,100,140,180,220,260,300,340,380
Evaporation Coefficient (sqmm/sec)	0.18, 0.37, 0.35, 0.78, 0.56, 0.75, 1.18, 1.36, 1.17, 1.65

- b) Analyze the significance of residual standard-error value, R-squared value, F- statistic. Find the correlation coefficient for this data and analyze the significance of the correlation value.
 - c) Perform a log transformation on the 'Air Velocity 'column, perform linear regression again, and analyze all the relevant values.
8. Write R Program using 'apply' group of functions to create and apply normalization function on each of the numeric variables/columns of iris dataset to transform them a value around 0 with z-score normalization.

2. WEKA Knowledge Extraction toolkit:

9. Create an ARFF (Attribute-Relation File Format) file and read it in WEKA. Explore the purpose of each button under the preprocess panel after loading the ARFF file. Also, try to interpret using a different ARFF file, *weather.arff*, provided with WEKA.
10. Performing data preprocessing in Weka Study **Unsupervised Attribute Filters** such as **Replace MissingValues** to replace missing values in the given dataset, **Add** to add the new attribute **Average**, **Discretize** to discretize the attributes into bins. Explore **Normalize** and **Standardize** options on a dataset with numerical attributes.
11. Classification using the WEKA toolkit Demonstration of classification process using id3 algorithm on categorical dataset(weather). Demonstration of classification process using naïve Bayes algorithm on categorical dataset ('vote'). Demonstration of classification process using Random Forest algorithm on datasets containing large number of attributes.
12. Classification using the WEKA toolkit – Part 2 Demonstration of classification process using J48 algorithm on mixed type of dataset after discretizing numeric attributes. Perform cross-validation strategy with various fold levels. Compare the accuracy of the results.
13. Performing clustering in WEKA Apply hierarchical clustering algorithm on numeric dataset and estimate cluster quality. Apply DBSCAN algorithm on numeric dataset and estimate cluster quality.
14. Association rule analysis in WEKA Demonstration of Association Rule Mining on supermarket dataset using Apriori Algorithm with different support and confidence thresholds. Demonstration of Association Rule Mining on supermarket dataset using FP- Growth Algorithm with different support and confidence thresholds.

3. Building Knowledge based Inference Systems:

14. Implement AI problem solving through Rule based forward chaining inference using public domain software tool like CLIPS.
15. Implement AI problem solving through Rule based Backward chaining inference using PROLOG

Reference Books:

1. Practical data science with R, Nina Zumel and John Mount- Dreamtech Press.

BIG DATA ANALYTICS LAB

Lab : 3 Periods
Exam : 3 Hours

Sessionals :50
Ext. Marks :50
Credits :2

Course Objectives:

1. Imparting the architectural concepts of Hadoop and introducing map reduce paradigm.
2. Introducing Java concepts required for developing map reduce programs.
3. To understand the applications using Map Reduce Concepts.
4. To acquire knowledge on variety of NoSQL databases.

Course Outcomes:

1. Applying data modeling techniques to large data sets.
2. Creating applications for Big Data analytics.
3. Building a complete business data analytic solution.

SYLLABUS**1. Getting Hadoop Up and Running in a cluster:**

- 1.Setting up Hadoop on standalone machine.
- 2.Wordcount Map Reduce program using standalone Hadoop.
- 3.Adding the combiner step to the Wordcount Map Reduce program.
- 4.Setting up HDFS.
- 5.Using HDFS monitoring UI
- 6.HDFS basic command-line file operations.
- 7.Setting Hadoop in a distributed cluster environment.
- 8.Running the WordCount program in a distributed cluster environment.
- 9.Using Map Reduce monitoring UI

2. Hadoop Map Reduce Applications:

10. Choosing appropriate Hadoop data types.
11. Implementing a custom Hadoop Writable data type.
12. Implementing a custom Hadoop key type.
13. Emitting data of different value types from a mapper.
14. Choosing a suitable Hadoop Input Format for your input data format.
15. Formatting the results of Map Reduce Computation – using Hadoop Output Formats.

3. Analytics

16. Simple analytics using Map Reduce.
17. Performing Group-By using Map Reduce.
18. Calculating frequency distributions and sorting using Map Reduce.
19. Plotting the Hadoop results using GNU plot.
20. Calculating histograms using Map Reduce.
21. Calculating scatter plots using Map Reduce.
22. Parsing a Complex dataset with Hadoop.
23. Joining two datasets using Map Reduce.

Reference Book:

1. Hadoop Map Reduce Cookbook, Srinath Perera & Thilina Gunarathne, 2013, PACKT PUBLISHING.

PROJECT PHASE-I**Lab : 3 Periods****Sessionals : 50****Credits : 2**

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Write report and present it effectively.

The phase-I of the project shall comprise of

- Problem identification in close collaboration with industry.
- Literature survey.
- Deriving work content and carry out of project requirement analysis.
- Submission of interim report.
- Presentation to an expert committee.

(Note: Sessionals 50 marks will be awarded based on
Continuous evaluation - 25 Marks
Seminar and Viva voce - 25 Marks.)

SCHEME OF INSTRUCTION & EXAMINATION
 (Regulation R16)
IV/IV B.TECH
 (With effect from **2016-2017** Admitted Batch onwards)
 Under Choice Based Credit System

COMPUTER SCIENCE AND ENGINEERING
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Sessional Marks	External Marks	Total Marks
B16CS 4201	Internet of Things	4	3	1	--	4	30	70	100
B16CS 4202	Cryptography and Network Security	4	3	1	--	4	30	70	100
B16CS 4203	Operations Research	4	3	1	--	4	30	70	100
B16CS 4204	Internet of Things Lab	2	--	--	3	3	50	50	100
B16CS 4205	Project Phase-II	12	--	--	9	9	50	100	150
Total		26	9	3	12	24	190	360	550

INTERNET OF THINGS

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course objectives:

1. Students will be explored to the interconnection and integration of the physical world and the cyber space.
2. They are also able to design & develop IOT Devices.

Course Outcomes: •

1. Able to understand the application areas of IOT •
2. Able to realize the revolution of Internet in Mobile Devices, Cloud & Sensor Networks •
3. Able to understand building blocks of Internet of Things and characteristics.

SYLLABUS

The Internet of Things: An Overview of Internet of things, Internet of Things Technology, behind IoTs Sources of the IoTs, M2M Communication, Examples of IoTs,

Design Principles For Connected Devices: IoT/M2M systems Layers and designs standardizations, Communication Technologies, Data Enrichment and Consolidation and Device Management Gateway, Ease of designing and affordability.

Design Principles for the Web Connectivity for connected-Devices: Web Communication protocols for Connected Devices, Message Communication protocols for Connected Devices, Web Connectivity for connected-Devices.

Internet Connectivity Principles: Internet connectivity, Internet-Based Communication, IP addressing in the IOT, Application Layer Protocols: MQTT, HTTP.

Data Acquiring and storage, organizing the Data and Analytics, Cloud Computing paradigm for Data Collection, Storage and computing.

Design Methodology-Embedded Computing logic- Microcontroller, System on chips-IoT system building blocks-ARDUINO-Board details, IDE programming - Raspberry Pi - Interfaces and Raspberry Pi with Python Programming.

Sensors, Participatory sensing, RFIDS, and Wireless sensor Networks: Sensor technology, Participatory sensing, Industrial IoT and Automotive IoT, Actuator, Sensor Data communication protocols, Radio frequency identification Technology and Wireless sensor Network technology.

Textbooks:

1. Internet of Things: Architecture, Design Principles And Applications, Rajkamal, McGraw Hill Higher Education Internet of Things,
2. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015, ISBN: 9788173719547

Reference Books:

1. Designing the Internet of Things, Adrian McEwen and Hakim Cassimally, Wiley
2. Getting Started with the Internet of Things CunoPfister , Oreilly
2. Arshdeep Bahga, Vijay Madisetti, —Internet of Things – A hands-on approach, Universities Press, 2015.
3. Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, Marina Ruggieri & Homayoun Nikookar, River Publishers Series in Communications.
4. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, —IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, Cisco Press, 2017.
5. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015, ISBN: 9788173719547
6. Internet of things : Srinivasa K.G., Siddesh G.M. and Hanumantha Raju R.- Cengage
7. Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD), 2014, ISBN: 9789350239759

CRYPTOGRAPHY AND NETWORK SECURITY

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Introduction of the issues in network security- its need and importance, taxonomy and terminology.
2. Discussion of various cryptographic techniques.
3. Exploration of different types of security threats and remedies.
4. Understanding of Internet security protocols and standards

Course Outcomes:

1. Realize the need and importance of network and data security in the Internet and in the distributed environments.
2. Identify the different types of network security issues and their remedies.
3. Application various cryptographic tools and techniques in different contexts and as per need of security levels.
4. Implementation of some Internet security protocols and standards

SYLLABUS

Overview: Computer Security Concepts, Threats, Attacks, and Assets, Security Functional Requirements, A Security Architecture for Open Systems, Computer Security Trends, Computer Security Strategy. Cryptographic Tools: Confidentiality with Symmetric Encryption, Message Authentication and Hash Functions, Public-Key Encryption, Digital Signatures and Key Management, Random and Pseudorandom Numbers, Practical Application: Encryption of Stored Data. User Authentication: Means of Authentication, Password-Based Authentication, Token-Based Authentication, Biometric Authentication, Remote User Authentication, Security Issues for User Authentication, Practical Application: An Iris Biometric System, Case Study: Security Problems for ATM Systems.

Access Control: Access Control Principles, Subjects, Objects, and Access Rights, Discretionary Access Control, Example: UNIX File Access Control, Role-Based Access Control, Case Study: RBAC System for a Bank. Database Security: The Need for Database Security, Database Management Systems, Relational Databases, Database Access Control, Inference, Statistical Databases, Database Encryption, Cloud Security.

Malicious Software: Types of Malicious Software (Malware), Propagation—Infected Content—Viruses, Propagation—Vulnerability Exploit—Worms, Propagation—Social Engineering—SPAM E-mail, Trojans, Payload—System Corruption, Payload—Attack Agent—Zombie, Bots, Payload—Information Theft—Key loggers, Phishing, Spyware, Payload—Stealth—Backdoors, Root kits, Countermeasures.

Denial-of-Service Attacks: Denial-of-Service Attacks, Flooding Attacks, Distributed Denial-of-Service Attacks, Application-Based Bandwidth Attacks, Reflector and Amplifier Attacks, Defenses Against Denial-of-Service Attacks, Responding to a Denial- of-Service Attack.

Intrusion Detection: Intruders, Intrusion Detection, Host-Based Intrusion Detection, Distributed Host-Based Intrusion Detection, Network-Based Intrusion Detection, Distributed Adaptive Intrusion Detection, Intrusion Detection Exchange Format, Honeypots, Example System: Snort. Firewalls and Intrusion Prevention Systems: The Need for Firewalls, Firewall Characteristics, Types of Firewalls, Firewall Basing, Firewall Location and Configurations, Intrusion Prevention Systems, Example: Unified Threat Management Products.

Buffer Overflow: Stack Overflows, Defending Against Buffer Overflows, Other Forms of Overflow Attacks, Software Security: Software Security Issues, Handling Program Input, Writing Safe Program Code, Interacting with the Operating System and Other Programs, Handling Program Output. Operating System Security: Introduction to Operating System Security, System Security Planning, Operating Systems Hardening, Application Security, Security Maintenance, Linux/Unix Security, Windows Security, Virtualization Security.

Symmetric Encryption and Message Confidentiality: Symmetric Encryption Principles, Data Encryption Standard, Advanced Encryption Standard, Stream Ciphers and RC4, Cipher Block Modes of Operation, Location of Symmetric Encryption Devices, Key Distribution. Public-Key Cryptography and Message Authentication: Secure Hash Function, HMAC, The RSA Public-Key Encryption Algorithm, Diffie- Hellman and Other Asymmetric Algorithms.

Internet Security Protocols and Standards: Secure E-mail and S/MIME, Domain Keys Identified Mail, Secure Socket Layer (SSL) and Transport Layer Security (TLS), HTTPS, IPv4 and IPv6 Security. Internet Authentication Applications: Kerberos, X.509, Public-Key Infrastructure, Federated Identity Management. Wireless Network Security: Wireless Security Overview, IEEE 802.11 Wireless LAN Overview, IEEE 802.11i Wireless LAN Security.

Text Book:

1. Computer Security - Principles and Practices (Except the Chapters 13, 14, 15, 16, 17, 18, 19), 2nd Edition by William Stallings, Pearson Education, Inc.

Reference Books:

1. Cryptography and Network Security by William Stallings, Pearson Education Asia, New Delhi.
2. Network Security Essentials Applications and Standards, by William Stallings, Pearson Education Asia, New Delhi.

OPERATIONS RESEARCH

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

This Course aims to:

1. Introduce students to use quantitative methods and techniques for effective Decisions-making; model formulation and applications that are used in solving business and industrial decision making problems.
2. Familiarize the students with the use of practice oriented mathematical applications for optimization functions in an organization.

Course Outcomes:

1. Describe the basic Operations Research models, and formulate and solve Linear Programming problems.
2. Formulate and solve engineering and managerial situations as Transportation and Assignment problems.
3. Determine the optimal solutions for various Job Sequencing and Replacement models
4. Understand Games theory and Apply them to various competitive situations.
5. Describe and illustrate various Inventory models. Discuss & schedule various project management problems by CPM & PERT.

SYLLABUS

Overview of Operations Research, Types of OR Models, Phases of Operations Research– OR Techniques, Introduction to Linear Programming, Formulation of Linear Programming Problem, Graphical Solution; Standard Form of LPP, Basic Feasible Solutions, Unrestricted Variables, Simplex Algorithm , Artificial Variables, Big M Method, Two Phase Simplex Method, Degeneracy, Alternative Optimal, Unbounded Solutions, Infeasible Solutions, Primal And Dual Problems and Their Relations, Dual Simplex Method.

Transportation Problem as LPP, Initial Solutions, North West Corner Rule, Lowest Cost Method, Vogel's Approximation Method, Optimum Solutions of TPP, Degeneracy in Transportation, Transportation Algorithms.

Assignment Problem, Assignment Problem as LPP, Hungarian Method, Travelling Salesman Problem, Solutions of TSP.

Sequencing Problems, N-Jobs Two Machine Problems, N-Jobs K Machines Problems, Two-Jobs M-Machine Problems, Replacement Problems-Individual and Group Replacement Policy, Reliability & System Failure Problems,

Network Representation of A Project, CPM and PERT, Critical Path Calculations, Time – Cost Optimizations, PERT Analysis and Probability Considerations, Resource Analysis in Network Scheduling.

Inventory-Factors Effecting Inventory-EOQ, Inventory Problems With and Without Shortages, Inventory Problems with Price Breakups.

Game Theory: Two Person Zero Sum Games, Mixed Strategy Games and Their Algorithms.

Text Books:

1. Operations Research, Kanti Swaroop, P.K. Gupta, Man Mohan, Sulthan Chand&Sons Education.
2. Publishers Operations Research – An Introduction, Handy A Taha – Pearson Education.

Reference Books:

1. Operations Research Panneer Selvan Prentice Hall of India.
2. Operations Research by S.D Sharma.
3. Introduction to Operations Research, F.S. Hiller, G.J. Liberman, TMH.
4. Operations Research by V. K. Kapoor.

INTERNET OF THINGS LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals :50
Ext Marks :50
Credits :2

Course objectives:

1. Students will be explored to the interconnection and integration of the physical world and the cyber space.
2. They are also able to design & develop IOT Devices.

Course Outcomes: ·

1. Able to understand and design the application areas of IOT ·
2. Able to realize and design the revolution of Internet in Mobile Devices, Cloud & Sensor Networks ·
3. Able to understand and design building blocks of Internet of Things and characteristics.

LIST OF EXPERIMENTS:

1. Study of various network protocols used in IOT.
2. Application of wifi in IOT systems.
3. Application of 6lowpan in IOT systems.
4. Application of bluetooth in IOT systems.
5. Application of 802.15.4 zigbee. In IOT systems.
6. Design a simple IOT system comprising sensors, wireless network connection, data analytics

Reference Books:

1. Internet of Things: Architecture, Design Principles And Applications, Rajkamal, McGraw Hill Higher Education Internet of Things,
2. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015, ISBN: 9788173719547
3. Designing the Internet of Things, Adrian McEwen and Hakim Cassimally, Wiley 2. Getting Started with the Internet of Things CunoPfister , Oreilly
4. Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, Marina Ruggieri & Homayoun Nikookar, River Publishers Series in Communications.
5. Internet of things : Srinivasa K.G., Siddesh G.M. and Hanumantha Raju R.- Cengage
6. Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD), 2014, ISBN: 9789350239759

PROJECT PHASE-II**LAB : 9 Periods**

Sessionals	: 50
Ext. Marks	: 100
Credits	: 12

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Analyze, design and develop a technology/ process.
3. Implement and evaluate the technology at the laboratory level.
4. Write report and present it effectively.

The phase-II of the project shall consists of

Implementing, Testing and validation.
Report Writing.

Sessionals (50 Marks) will be awarded by the Project Guide based on continuous evaluation.
External Evaluation (100 marks) of project report and viva voce will be conducted by a committee consisting of HOD, Guide and External Examiner.

May be carried out using in-house facilities or in an industry by specified number of students in a group.

Format for Preparation of Project Thesis for B. Tech:

1. Arrangement Of Contents: The sequence in which the project report material should be arranged and bound should be as follows:
 1. Cover Page & Title Page .
 2. Bonafide Certificate
 3. Abstract.
 4. Table of Contents
 5. List of Tables
 6. List of Figures
 7. List of Symbols, Abbreviations and Nomenclature
 8. Chapters
 9. Appendices
 10. References

*The table and figures shall be introduced in the appropriate places.